

TRANSPORTATION ENGINEERING

Complete Semester Examination Notes

Theory with diagrams • solved numericals • formula sheet • previous-year question bank

Unit	Topic	Hrs
1	Highway development and planning; classification of roads; road development in India; current projects; highway alignment and project preparation	8
2	Geometric design of highways — cross-section elements, sight distance, horizontal and vertical alignment, intersections	5
3	Traffic engineering and control — traffic characteristics, traffic engineering studies, traffic flow and capacity	6
4	Traffic regulation and control; design of road intersections; parking facilities; highway lighting	8
5	Pavement materials — soils, aggregates, bituminous binders and mixes, cement and cement concrete	5
6	Design of pavements — flexible and rigid pavements, stresses, IRC 37 and IRC 58 design	10

PREScribed TEXT / REFERENCE BOOKS

1. Khanna, S.K., Justo, C.E.G. and Veeraragavan, A. — *Highway Engineering*, Revised 10th Edition, Nem Chand & Bros, 2017. **[primary book for this paper]**
2. Kadiyali, L.R. — *Traffic Engineering and Transport Planning*, Khanna Publishers.
3. Partha Chakraborty and Animesh Das — *Principles of Transportation Engineering*, PHI Learning.
4. Mannering, F.L., Washburn, S.S. and Kilareski, W.P. — *Principles of Highway Engineering and Traffic Analysis*, 4th Edition, John Wiley.
5. Srinivasa Kumar, R. — *Textbook of Highway Engineering*, Universities Press, 2011.
6. Wright, P.H. and Dixon, K.K. — *Highway Engineering*, 7th Edition, Wiley Student Edition, 2009.
7. IRC codes referred throughout: IRC 37-2018, IRC 58-2015, IRC 73-1980, IRC 86-1983, IRC 65-2017, IRC 93-1985, IRC 106-1990, IRC SP 41, IRC 67, IRC 35, IRC 3-1983.

How to use these notes

These notes are written strictly to the BEU 101401 syllabus printed on the cover. Every unit follows the same fixed sequence so that you always know where to look:

Element	What it contains	How to use it in the exam
Numbered sections	The theory, in the order of the syllabus, with every definition and classification spelled out.	This is your answer material for the long (14-mark) questions.
Figures	47 labelled engineering diagrams — cross-sections, sight-distance diagrams, curve geometry, apparatus sketches, stress diagrams.	Reproduce the labelled sketch first, then write the theory around it. A diagram alone often carries 3–5 marks.
Blue formula boxes	The working formula with every symbol and its unit defined.	Quote the formula, then substitute. Never substitute without writing the formula first.
Orange solved numericals	Fully worked problems of exactly the type asked, solved step by step with the answer boxed.	Practise these with the book closed. The numerical part of this paper is highly repetitive.
Gold PYQ flags	The questions on that topic which recur in BEU / AKU papers.	Use them to decide what to revise first when time is short.
Teal exam tips / IRC boxes	Standard IRC values, common mistakes, and shortcuts.	The IRC values are frequently asked as 2-mark questions.
Formula sheet	Every formula in the paper condensed on to facing pages.	This is the last thing to read before you enter the hall.

The BEU question-paper pattern for this subject

The end-semester paper of Bihar Engineering University for a theory subject such as this one is normally set as follows:

Question	Nature	Choice	Marks
Q.1	Short-answer / objective questions covering the whole syllabus (ten parts, roughly two per unit)	Answer any seven of ten	$7 \times 2 = 14$
Q.2 to Q.8	Long questions. Each is usually split into two or three parts — commonly one theory part plus one numerical or one diagram part	Answer any four of seven	$4 \times 14 = 56$
		Total	70 (3 hours)

Table: the usual BEU / AKU B.Tech end-semester pattern. Always re-read the instructions printed on your own question paper — the choice pattern is occasionally varied.

STRATEGY THAT FOLLOWS FROM THE PATTERN

Because you answer only four long questions out of seven, you can leave roughly one-third of the syllabus thin — but Q.1 is compulsory and it samples the WHOLE syllabus. So: study four or five units in depth for the long questions, and make sure you know the definitions, IRC values and one-line facts from every unit for Q.1.

Unit-wise weightage and priority

The syllabus allots 42 contact hours. The share of each unit in the timetable is a good proxy for its share in the paper, and it matches what the past papers show:

Unit	Topic	Hrs	Share	Numericals?	Priority
6	Design of pavements	10	24 %	Very heavy — ESWL, MSA, Westergaard stresses, joints, dowel and tie bars	HIGHEST
1	Highway development and planning	8	19 %	Light — road density, saturation system	High (theory-scoring)
4	Traffic regulation, intersections, parking, lighting	8	19 %	Heavy — Webster signal design, rotary capacity, parking, lamp spacing	HIGHEST
3	Traffic engineering and control	6	14 %	Medium — moving observer, PCU, capacity, speed studies	High

Unit	Topic	Hrs	Share	Numericals?	Priority
2	Geometric design	5	12 %	Very heavy — SSD/OSD, superelevation, transition and vertical curves	HIGHEST (marks per hour of study are the best here)
5	Pavement materials	5	12 %	Light — CBR, aggregate test values, Marshall	Medium

EXAM TIP

Units 2, 4 and 6 carry almost all the numerical marks and together account for over half the paper. If you have only three days, do Unit 2, Unit 6 and Unit 4 properly, then skim Units 1, 3 and 5 for the short-answer questions.

► Reference: Syllabus: BEU B.Tech Civil Engineering, Course Code 101401, Transportation Engineering (6 units, 42 hours).

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UNIT 1 | Highway Development and Planning

Syllabus allotment: 8 hours • about 19 % of the paper

1.1 Modes of transportation and the place of roads

Transportation is the movement of men and materials from one place to another. The four modes are **roadways, railways, waterways and airways**. Road transport is the only mode that offers complete **door-to-door service**, and every other mode ultimately depends on it for collection and distribution — which is why roads carry about 87 % of passenger traffic and 60 % of freight traffic in India.

Point of comparison	Road transport	Rail transport
Initial cost of the facility	Low	Very high
Flexibility of route and time	Complete — can be changed at will	Fixed route, fixed timetable
Door-to-door service	Possible	Not possible
Economical lead (distance)	Short and medium leads	Long leads
Suitability for hilly terrain	Can negotiate steep gradients (up to 1 in 15)	Needs very flat gradients (about 1 in 100)
Energy per tonne-km	Higher	Lower
Speed over long distance	Lower	Higher
Accident rate	High	Low

Table 1.1 Road transport compared with rail transport.

BEU / AKU REPEAT QUESTION

- State the different modes of transportation and bring out the advantages of road transport over rail transport. [2 marks / part of a 14-mark question]

1.2 History of road construction

1.2.1 Roman roads (about 300 B.C. onwards)

- Built primarily for the movement of armies, therefore laid out as **straight lines** regardless of the gradient encountered.
- Very thick, 0.75 m to 1.2 m, built with heavy stone blocks in several courses.
- Constructed **below the natural ground level**, so surface water and sub-soil water were not drained — the great defect of Roman roads.
- Nevertheless extremely durable, because of the sheer thickness and the quality of the workmanship.

1.2.2 Tresaguet construction, France, 1764

Pierre Tresaguet was the first engineer to develop a **cheaper method** of construction by using a smaller thickness of material, and the first to appreciate the importance of the **moisture content of the subgrade**. His cross-section was 30 cm thick in three layers with a cross slope of about 1 in 45, the camber being obtained by **sloping the subgrade itself**.

(a) Tresaguet construction (1764) — total thickness 30 cm

Cross slope of subgrade AND of surface ≈ 1 in 45 — camber obtained by sloping the subgrade central carriageway 5.4 m

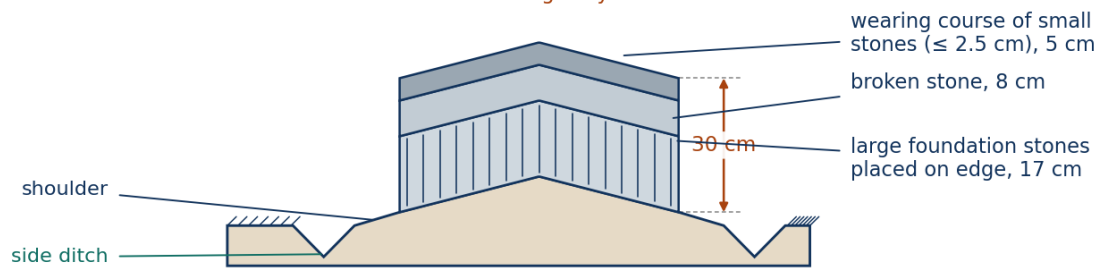


Fig. 1.1 Tresaguet's cross-section (1764) — total thickness 30 cm. Note that the camber of the finished surface is produced by giving the subgrade the same cross slope.

1.2.3 Metcalf construction, England, about 1815

John Metcalf, a blind engineer, built about 290 km of road in England. His work was similar in principle to Tresaguet's and he insisted on good **drainage** — he provided side ditches and a cross slope to the surface.

1.2.4 Telford construction, England, 1824

Thomas Telford, founder of the Institution of Civil Engineers, insisted on a **level subgrade** and obtained the camber by **varying the depth of the stone foundation** — 17 cm at the edges increasing to 22 cm at the centre. He also specified a maximum gradient of 1 in 20 and fixed the maximum size and shape of the stones to be used.

(b) Telford construction (1824)

Subgrade kept FLAT; camber produced by varying the depth of the stone foundation — surface slope ≈ 1 in 45 central carriageway 5.5 m

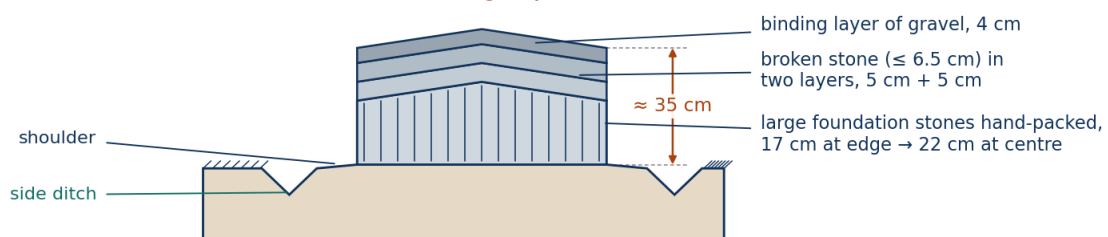


Fig. 1.2 Telford's cross-section (1824). The subgrade is FLAT; the camber comes from the varying depth of the hand-packed stone foundation.

1.2.5 Macadam construction, England, 1827

John Macadam's method was the **first scientific method** of road construction and is the basis of the modern water-bound macadam. His two great contributions were:

- He recognised that it is the **subgrade soil that ultimately carries the wheel load**, and that it must therefore be kept dry and well compacted; the function of the stone layers is only to spread the load and protect the subgrade.
- He recognised that **broken stone of a proper size, angular in shape, will interlock** under traffic and form a strong, load-spreading mass. He therefore fixed the size of the stone in each layer (5 cm, 3.75 cm and 2 cm) and kept the total thickness down to only 25 cm.
- He gave the subgrade and the surface the same cross slope of 1 in 36 so that water would drain out of the pavement.

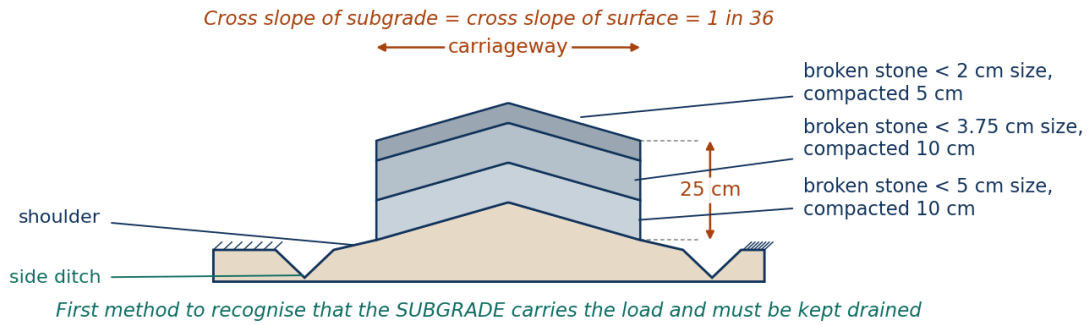
(c) Macadam construction (1827) — the basis of modern WBM

Fig. 1.3 Macadam's cross-section (1827) — total thickness only 25 cm, in three layers of decreasing stone size.

Feature	Roman	Tresaguet	Telford	Macadam
Year	300 B.C.	1764	1824	1827
Total thickness	0.75–1.2 m	30 cm	about 35 cm	25 cm
Subgrade	Below ground level, undrained	Sloped (gives the camber)	Flat / level	Sloped 1 in 36
Camber obtained from	Not provided	Sloping the subgrade	Varying the depth of the foundation	Sloping the subgrade and surface
Cross slope of surface	Nil	1 in 45	1 in 45	1 in 36
Large foundation stones	Yes, heavy blocks	Yes, 17 cm, set on edge	Yes, 17–22 cm, hand packed	NO — the key difference
Basic idea of load transfer	Thickness alone	Thickness + subgrade drainage	Stone foundation carries the load	The SUBGRADE carries the load; stones interlock and spread it

Table 1.2 Comparison of the historic methods of road construction — a very frequently asked question.

BEU / AKU REPEAT QUESTION

- Describe, with neat sketches, the Tresaguet, Telford and Macadam methods of road construction and compare them. [14 marks — asked almost every year in some form]
- What was the most important contribution of Macadam to highway engineering? [2 marks]

► Reference: Khanna, Justo & Veeraragavan — Highway Engineering, chapter 'Highway Development and Planning'.

1.3 Highway development in India — the milestones**1.3.1 Jayakar Committee, 1927**

After the First World War motor traffic increased sharply and the existing roads proved quite inadequate. The Government of India appointed the **Indian Road Development Committee** in 1927 under the chairmanship of **Mr. M.R. Jayakar**, which submitted its report in 1928. Its four principal recommendations, and what each of them produced, are the most examined facts in this unit:

Recommendation of the Jayakar Committee	Outcome	Year
Road development in the country should be considered as a matter of national interest , because the provincial and local governments do not have the resources or the technical knowledge for it	Led to the whole idea of a central road organisation and of national highways	—
An extra tax should be levied on petrol from the road users to create a fund dedicated to road development	Central Road Fund (CRF)	1929
A semi-official technical body should be formed to pool technical know-how and to act as an advisory body	Indian Roads Congress (IRC)	1934
A research organisation should be instituted to carry out road research	Central Road Research Institute (CRRI) , New Delhi	1950

Table 1.3 The Jayakar Committee (1927) and its outcomes.

1.3.2 Other important milestones

Year	Milestone
1929	Central Road Fund created (from the extra duty on petrol).
1934	Indian Roads Congress (IRC) formed — it frames the IRC standards and specifications used throughout this subject.
1939	Motor Vehicles Act passed (comprehensively revised in 1988) — governs registration, licensing, traffic regulation and liability.
1943	Nagpur Road Plan — the first twenty-year road plan, framed at the conference of chief engineers held at Nagpur.
1950	Central Road Research Institute (CRRRI), New Delhi, established.
1956	National Highways Act — declared certain roads to be national highways and vested their development in the Central Government.
1961	Second twenty-year plan (Bombay Road Plan) begins.
1973	National Transport Policy Committee.
1981	Third twenty-year plan (Lucknow Road Plan) begins.
1988	National Highways Authority of India (NHAI) constituted by an Act of Parliament; it became operational in 1995. It develops, maintains and manages the national highways.
1998	NHDP — National Highways Development Project launched.
2000	PMGSY launched (25 December 2000) and the Central Road Fund Act passed, giving the fund statutory backing.
2017	Bharatmala Pariyojana approved.

Table 1.4 Chronology of highway development in India.

BEU / AKU REPEAT QUESTION

- ▶ Discuss the recommendations of the Jayakar Committee and their outcome. [7 marks]
- ▶ Write short notes on: (i) Central Road Fund (ii) IRC (iii) CRRRI (iv) NHAI. [2 marks each]

1.4 The twenty-year road development plans

1.4.1 First plan — Nagpur Road Plan (1943–1963)

- Framed at the conference of chief engineers at **Nagpur in 1943**; it was the first attempt at a co-ordinated national road programme.
- Roads were classified for the first time into **five categories** — National Highways, State Highways, Major District Roads, Other District Roads and Village Roads.
- The road lengths were worked out on the assumption of a **star-and-grid pattern** of roads (a rectangular grid of roads with radial roads superimposed, connecting the market centres).
- Target **road density = 16 km per 100 km²** of area, giving a total road length of about 5,32,700 km by 1963. The target was in fact achieved ahead of time.
- A further aim was accessibility — no village in a developed agricultural area should be more than about **3.2 km** from a metalled road.

Star-and-grid road pattern assumed in the Nagpur Road Plan

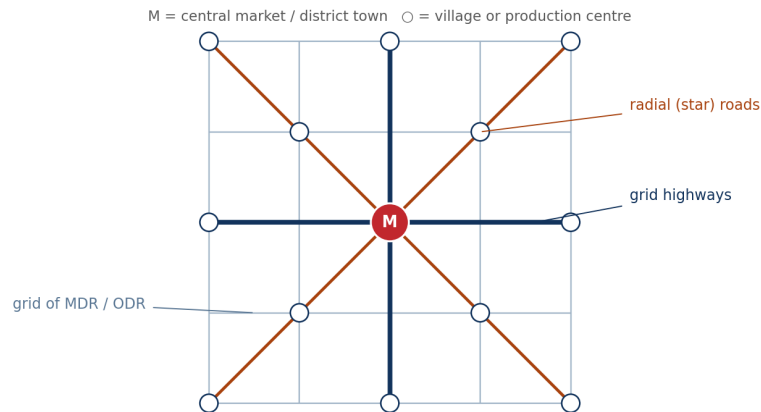


Fig. 1.4 The star-and-grid road pattern on which the Nagpur Road Plan formulae were based.

NOTE

Textbooks quote an empirical formula for the Nagpur plan of the form $L = A/8 + B/32 + N + 1.6 C + 8 D$, in which A is the developed (agricultural) area, B the undeveloped area, and N, C, D are counts of towns and villages in stated population ranges. **The symbols are defined differently in different books, so if such a formula is set in your paper the definitions will be printed with it — use those, and show the substitution clearly.** Numerical questions in BEU papers are far more often based simply on road density, as in Example 1.1 below.

1.4.2 Second plan — Bombay Road Plan (1961-1981)

- Target **road density** = **32 km per 100 km²** — double the first plan, giving a total of about 10,57,000 km.
- For the first time **expressways** were proposed as a separate category over and above the national highways.
- The plan also aimed at developing a nucleus of an all-India network of high-quality roads and at connecting all villages to the road system.

1.4.3 Third plan — Lucknow Road Plan (1981-2001)

- Target **road density** = **82 km per 100 km²**.
- Aimed that every village with a population above **500** be connected by an all-weather road, and that no part of the country be more than 50 km from a metalled road.
- Proposed an expressway network of about 2,000 km taken out of the national highway system.
- Introduced the idea of planning by the **saturation system** (maximum utility system) instead of by an arbitrary formula — see § 1.9.

1.4.4 Road Development Plan — Vision 2021 and beyond

The **Road Development Plan Vision 2021** (and the parallel **Rural Roads Development Plan Vision 2025**) replaced the twenty-year plans. Their thrust is no longer merely length but **capacity, riding quality, safety and connectivity** — four-laning and six-laning of corridors, bypasses, expressways, and all-weather connectivity to every habitation.

Plan	Period	Target road density	Special feature
First — Nagpur	1943-1963	16 km / 100 km ²	Star-and-grid pattern; five-fold classification of roads introduced
Second — Bombay	1961-1981	32 km / 100 km ²	Expressways proposed for the first time
Third — Lucknow	1981-2001	82 km / 100 km ²	Saturation system of planning; villages above population 500 to be connected
Vision 2021 / 2025	after 2001	Not length-based	Capacity, riding quality, safety, rural connectivity

Table 1.5 The road development plans at a glance — learn the three density figures 16, 32 and 82.

SOLVED NUMERICAL — Road density and the additional length required

Q. A district has a total area of 8,500 km² and an existing road length of 1,020 km. (a) Find the present road density. (b) Find the additional road length needed to achieve the target of the third twenty-year plan.

Solution:

Step 1 — present road density

Road density is defined as the road length per 100 km² of area.

$$\text{Density} = (1020 / 8500) \times 100 = \mathbf{12.0 \text{ km per } 100 \text{ km}^2}$$

Step 2 — target length for the third (Lucknow) plan

Target density for the third plan = 82 km per 100 km².

$$\text{Required length} = (82/100) \times 8500 = 6970 \text{ km}$$

Step 3 — additional length

$$\text{Additional length} = 6970 - 1020 = 5950 \text{ km}$$

Present density = 12 km/100 km² ; additional road length required = 5950 km

BEU / AKU REPEAT QUESTION

- Explain the three twenty-year road development plans of India and state the target road density of each. [7 marks]
- Define road density. The area of a state is ... and the road length is ... ; find the additional length needed to reach the third-plan target. [4 marks]

1.5 Classification of roads

1.5.1 Classification based on traffic volume and on the load carried

- **By traffic volume:** heavy, medium and light traffic roads.
- **By load or tonnage:** Class I to Class IV, or class A, B, etc., depending on the tonnes per day carried.
- These divisions are arbitrary, because the limits of volume and tonnage are not standardised — which is why the functional classification below is used in practice.

1.5.2 Nagpur Plan classification (five classes) — the standard answer

Class	Function	Authority responsible
National Highways (NH)	Main highways running through the length and breadth of the country, connecting state capitals, ports, major cities and foreign highways; they are the arterial routes of the nation.	Central Government / MoRTH, executed through NHAI, state PWDs and BRO
State Highways (SH)	Arterial routes of a state, connecting the district headquarters and important cities of the state with the national highways and with the highways of the adjacent states.	State Government (PWD)
Major District Roads (MDR)	Important roads within a district, connecting the areas of production with the markets and connecting these with the state highways and national highways.	State Government / Zilla Parishad
Other District Roads (ODR)	Roads serving rural areas of production, giving them outlets to the market centres, taluka headquarters, block development headquarters or other main roads.	Zilla Parishad / district board
Village Roads (VR)	Roads connecting the villages, or a group of villages, with each other and with the nearest road of a higher category.	Panchayat; funded largely through PMGSY

Table 1.6 The five classes of the Nagpur Plan classification. ODR + VR together are called RURAL ROADS.

1.5.3 Modified (Third Plan) classification — three classes

Class	Consists of	Character
Primary system	Expressways and National Highways	Highest design standards; through traffic at high speed; access controlled

Class	Consists of	Character
Secondary system	State Highways and Major District Roads	Collects traffic from the tertiary system and feeds it to the primary system
Tertiary system (rural roads)	Other District Roads and Village Roads	Provides accessibility to the land; lowest standards, lowest volumes

Table 1.7 The modified three-fold classification.

1.5.4 Classification of urban roads (IRC)

- **Expressways** — divided highways with full or partial control of access, for through traffic only; no access to abutting property.
- **Arterial streets** — for through traffic on a continuous route; parking, loading and pedestrian crossings are strictly controlled.
- **Sub-arterial streets** — like arterial streets but with a somewhat lower level of travel mobility and closer spacing of intersections.
- **Collector streets** — collect traffic from the local streets and feed it to the arterial system; some parking permitted.
- **Local streets** — give direct access to residences, shops and offices; through traffic is discouraged; free parking and pedestrian movement.

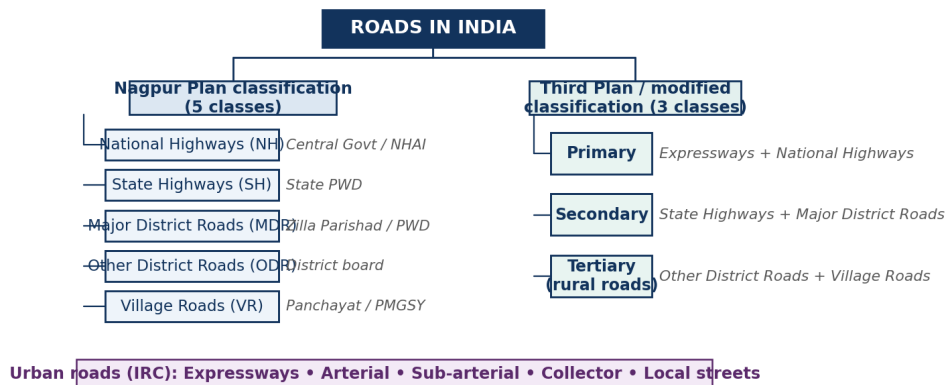


Fig. 1.5 The two systems of classifying roads in India, and the IRC classification of urban roads.

BEU / AKU REPEAT QUESTION

- ▶ Classify the roads as per the Nagpur Road Plan and state the function and the authority responsible for each class. [7 marks]
- ▶ Differentiate between an arterial street and a collector street. [2 marks]
- ▶ What are rural roads? Which classes constitute them? [2 marks]

1.6 Current road projects in India

Project	What it covers
NHDP — National Highways Development Project (1998, NHAI)	Phase I: the Golden Quadrilateral, about 5,846 km of four-laning connecting Delhi – Mumbai – Chennai – Kolkata. Phase II: the North-South corridor (Srinagar to Kanyakumari) and the East-West corridor (Silchar to Porbandar), about 7,300 km. Phase III: four-laning of about 12,100 km of high-density NH connecting state capitals and places of tourist and economic importance. Phase IV: two-laning with paved shoulders of about 20,000 km. Phase V: six-laning of about 6,500 km (mainly the GQ). Phase VI: about 1,000 km of expressways. Phase VII: ring roads, bypasses, flyovers, grade separators and other improvements on the NH network.

Project	What it covers
Bharatmala Pariyojana (2017)	The umbrella programme that succeeded the NHDP. Phase I covers about 34,800 km made up of economic corridors, inter-corridor and feeder routes, national corridor efficiency improvement, border and international connectivity roads, coastal and port connectivity roads, expressways , and the residual NHDP works.
PMGSY — Pradhan Mantri Gram Sadak Yojana (25 December 2000)	A rural connectivity programme: an all-weather road to every unconnected habitation with a population of 500 and above in the plains and 250 and above in hill, tribal and desert areas. Later phases (PMGSY-II, III) deal with upgradation and consolidation of the existing rural network.
Setu Bharatam (2016)	Replacement of level crossings on national highways by road over-bridges and under-bridges, and the inventory, rating and rehabilitation of old bridges, to make the NH network free of level crossings.
Chardham Pariyojana	All-weather improvement of the roads to the four pilgrimage centres in Uttarakhand.
SARDP-NE	Special Accelerated Road Development Programme for the North-Eastern region — connecting all state capitals and district headquarters of the north-east by at least two-lane roads.
Sagarmala	Port-led development, including the road and rail connectivity of ports (coastal and port connectivity roads overlap with Bharatmala).

Table 1.8 Major road projects currently in progress in India.

EXAM TIP

For a 2-mark question, remember three numbers: Golden Quadrilateral $\approx$ 5,846 km; NS-EW corridors $\approx$ 7,300 km; PMGSY threshold population 500 (250 in hill, tribal and desert areas).

BEU / AKU REPEAT QUESTION

- Write short notes on NHDP and its phases. [7 marks]
- What is Bharatmala Pariyojana? [2 marks]
- State the objective and the eligibility criterion of PMGSY. [2 marks]

1.7 Highway alignment

Alignment — the position or the layout of the centre line of the highway on the ground. It has a **horizontal alignment** (the straights and the curves in plan) and a **vertical alignment** (the gradients and the vertical curves in profile).

NOTE

A wrong alignment cannot be corrected later without heavy expenditure on land acquisition and earthwork. Hence great care is needed at the alignment stage — this sentence is worth writing in any alignment answer.

1.7.1 Requirements of an ideal alignment

The alignment between two terminal stations should be **short, easy, safe and economical**:

- **Short** — the shortest alignment is a straight line, but this is seldom possible; the aim is to keep the length as near to it as practicable.
- **Easy** — easy to construct and to maintain with the available resources, and easy for the vehicles to negotiate: easy gradients and easy curves.
- **Safe** — safe for traffic (adequate sight distance, gentle curves) and safe against the failure of embankment and cut slopes.
- **Economical** — the total cost (initial cost + maintenance cost + vehicle operation cost) taken together over the design life must be a minimum, not merely the initial cost.

1.7.2 Factors controlling the alignment

Factor	How it controls the alignment
Obligatory points	Points through which the alignment must pass — an intermediate town, a bridge site, a mountain pass, an existing bridge; and points through which it must not pass — religious places, costly structures, a lake or pond, marshy land, unstable hill features. Obligatory points may increase the length considerably.

Factor	How it controls the alignment
Traffic	The alignment should be decided on the basis of the origin-and-destination study, so that it serves the desire lines of the traffic it is meant to carry.
Geometric design	Gradient, radius of curve and sight distance govern the final alignment; for example the alignment must be changed if the ruling gradient cannot be maintained, and a bridge must be approached on a straight and level stretch.
Economy	The alignment should follow the natural ground as far as possible to balance cutting and filling, and should avoid deep cuts, high banks and long bridges.
Other considerations	Drainage and hydrological factors, the availability of construction materials along the route, the height of the water table, the possibility of subsidence or landslides, political and administrative boundaries, and the avoidance of long monotonous straights (which cause driver fatigue).

Table 1.9 Factors controlling highway alignment.

Factors that force a highway alignment away from the straight line

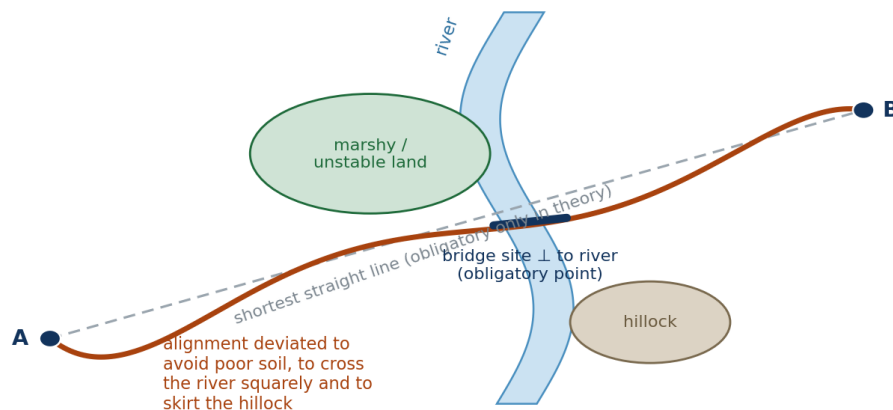


Fig. 1.6 Typical reasons for which an alignment departs from the straight line — obligatory points, poor soil, a square river crossing, and a hill feature to be skirted.

1.7.3 Special considerations for hill roads

- **Stability of the slopes** — the alignment should be taken along the side of the hill that is stable; the hill side rather than the valley side.
- **Hill drainage** — the number of cross-drainage structures should be kept to a minimum; the alignment should cross streams at the narrowest point.
- **Geometric standards** — the special standards of gradient, radius and hairpin bends for hill roads must be observed.
- **Resisting length** — the alignment should be so chosen that the total resisting length (which is the sum of the actual length and the equivalent length of the rise and of the curves) is a minimum.
- Avoid **excessive rise and fall**, which wastes the effort of climbing.

BEU / AKU REPEAT QUESTION

- ▶ What are the requirements of an ideal alignment? Discuss the factors that control the alignment of a highway. [14 marks — very frequently asked]
- ▶ What are obligatory points? Give two examples of each kind. [2 marks]
- ▶ State the special points to be considered in the alignment of hill roads. [7 marks]

1.8 Engineering surveys for highway location

The location of a new highway is fixed by four stages of survey, each more detailed and more expensive than the last, so that unsuitable routes are eliminated as early and as cheaply as possible.

Stage	What is done	Instruments / output
1. Map study	Alternative routes are marked on the available topographic maps (1:50,000 Survey of India sheets). Contours reveal valleys, passes, saddles, ponds and permanent structures, so that alignments avoiding excessive rise and fall can be picked out <i>on paper</i> .	Topo sheets; a set of alternative alignments

Stage	What is done	Instruments / output
2. Reconnaissance	A rapid field inspection of the whole area between the terminals, to verify on the ground the alternatives chosen on the map and to collect information not available on maps — the nature of the soil and rock, the sources of materials, the number of cross-drainage works, the highest flood level, unusual features.	Abney level / hand level, barometer, compass, pedometer, clinometer; unsuitable routes are rejected
3. Preliminary survey	A detailed survey of the two or three surviving alternatives: a rapid traverse of each, levelling along the centre line, cross-sections, soil survey, drainage and hydrological data, traffic survey. The alternatives are then compared economically and the best one chosen.	Theodolite / total station, levelling instrument; plans and profiles of each alternative, and the economic comparison
4. Final location and detailed survey	The centre line of the chosen alignment is transferred from the paper to the ground and staked out . All the data needed for the working drawings and estimates are then collected — longitudinal section, cross-sections at close intervals, details of every drainage structure, soil profile, and the land to be acquired.	Total station / theodolite, level; the working drawings and the estimate

Table 1.10 The four stages of engineering survey for highway location.

Engineering surveys for highway location and project preparation

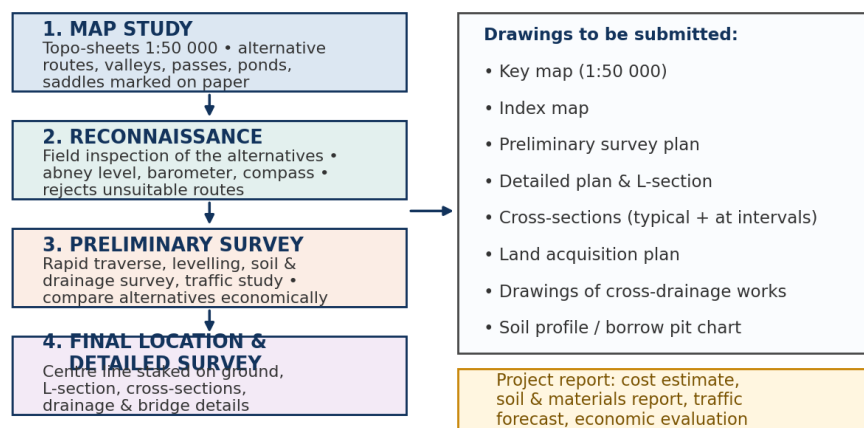


Fig. 1.7 The four stages of engineering survey, and the drawings and reports that make up the highway project.

1.9 Highway project preparation

1.9.1 Drawings to be prepared

- Key map** — shows the proposed road and the existing important roads, railways, towns and rivers; scale not less than 1:50,000.
- Index map** — shows the general topography, the alternative alignments considered and the one adopted.
- Preliminary survey plans** — showing the details of all the alternatives investigated.
- Detailed plan and longitudinal section** — the plan shows the alignment, the drainage works, the land width and the buildings; the L-section shows the ground line, the formation line, the gradients and the vertical curves (the vertical scale is exaggerated).
- Detailed cross-sections** — typical cross-sections plus cross-sections at every 50 m to 100 m and at every change of the section.
- Land acquisition plans** — giving the property lines and the areas to be acquired, for the revenue authorities.
- Drawings of cross-drainage and other structures** — culverts, bridges, retaining walls, breast walls.
- Drawings of road intersections.**
- A land plan showing the quarries** from which the materials will be obtained, with the lead.

1.9.2 Reports and estimates

- The **project report** containing the necessity and the justification of the project, the alternatives considered and the reasons for the choice, the traffic forecast, the design standards adopted, the soil and materials report, the construction programme and the maintenance proposals.

- The **detailed estimate** of cost, item by item, with the rate analysis.
- The **economic evaluation** — benefit-cost ratio, net present value or internal rate of return, to prove that the project is worth doing.

BEU / AKU REPEAT QUESTION

- ▶ Enumerate the drawings and reports to be prepared for a new highway project. [7 marks]
- ▶ Explain the different engineering surveys carried out for a new highway alignment. [14 marks — very frequently asked]

1.10 Planning surveys and the saturation system

1.10.1 Planning surveys

Before a road plan can be prepared, four studies are needed — an **economic study** (population, produce, industry, existing facilities), a **financial study** (sources of income and the funds available), a **traffic or road-use study** (volume, origin and destination, growth) and an **engineering study** (topography, soil, materials, drainage and the existing road condition).

1.10.2 Saturation system (maximum utility system)

Introduced in the third twenty-year plan for deciding the **priority** of new road links. The principle is simple: the road network that gives the **maximum utility per unit length** of road is the optimum. Utility is measured by the population served and the productivity (agricultural and industrial) served. Utility units are assigned to each, the total for an alignment is divided by its length, and the alignments are then ranked.

SATURATION SYSTEM

$$\text{Utility per unit length} = (\text{total utility units served}) / (\text{length of the road})$$

- Utility units are assigned for the population served and for the tonnage of agricultural and industrial produce served
- The rates of assignment are given in the question; they differ from book to book
- The alignment with the HIGHEST utility per km gets the highest priority

SOLVED NUMERICAL — Choosing between two alignments by the saturation system

Q. Two alternative alignments A and B are proposed to connect the same pair of towns. Utility units are to be assigned at the rate of **1 unit per 1,000 of population served, 1 unit per 1,000 tonnes of agricultural produce and 1 unit per 1,000 tonnes of industrial produce**. The data are:

Alignment A — length 20 km; villages served of population 2,000, 1,500, 1,000 and 500; agricultural produce 6,000 t; industrial produce 2,000 t.

Alignment B — length 15 km; villages served of population 1,200, 800 and 1,000; agricultural produce 4,500 t; industrial produce 1,500 t.

Which alignment should be given priority?

Solution:

Step 1 — utility units of alignment A

Population served = 2000 + 1500 + 1000 + 500 = 5000 $\Rightarrow$ 5000/1000 = 5.0 units

Agricultural produce = 6000 t $\Rightarrow$ 6000/1000 = 6.0 units

Industrial produce = 2000 t $\Rightarrow$ 2000/1000 = 2.0 units

Total utility of A = 5.0 + 6.0 + 2.0 = 13.0 units

Utility per km = 13.0 / 20 = **0.65 unit/km**

Step 2 — utility units of alignment B

Population served = 1200 + 800 + 1000 = 3000 $\Rightarrow$ 3.0 units

Agricultural produce = 4500 t $\Rightarrow$ 4.5 units

Industrial produce = 1500 t $\Rightarrow$ 1.5 units

Total utility of B = 3.0 + 4.5 + 1.5 = 9.0 units

Utility per km = 9.0 / 15 = **0.60 unit/km**

Step 3 — decision

Alignment A gives 0.65 unit/km against 0.60 unit/km for B.

Alignment A has the greater utility per unit length and should be given priority. (Note that B has the greater total length efficiency in no other respect — always compare utility PER KILOMETRE, not the total.)

BEU / AKU REPEAT QUESTION

- Explain the saturation system of road planning. Two alignments have the following data ... decide the priority. [7 + 7 marks]
- What are the four planning surveys required before preparing a road plan? [2 marks]

► *Reference: Khanna, Justo & Veeraragavan, chapters 'Highway Development and Planning' and 'Highway Alignment and Engineering Surveys'; Kadiyali — Traffic Engineering and Transport Planning for the planning surveys.*

UNIT 2 | Geometric Design of Highways

Syllabus allotment: 5 hours • about 12 % of the paper, but the largest share of the numericals

2.1 Introduction — what geometric design covers

Geometric design — the design of the **visible dimensions** of a highway — the cross-section, the horizontal alignment, the vertical alignment and the intersections — so that the road gives the optimum efficiency in operation with maximum safety at reasonable cost.

Geometric design must be done **once and for all** at the planning stage, because it is very costly to improve later. The elements are:

- **Cross-section elements** — pavement surface characteristics, camber, width of pavement, shoulders, medians, kerbs, side slopes, right of way.
- **Sight distance** — stopping, overtaking and intermediate sight distance, and sight distance at intersections.
- **Horizontal alignment** — radius of curve, superelevation, extra widening, transition curve, set-back distance.
- **Vertical alignment** — gradients, grade compensation, summit curves, valley curves.
- **Intersection elements** — dealt with in Unit 4.

2.1.1 Design speed and terrain — the starting point of everything

Design speed is the single most important factor in geometric design: sight distance, radius, superelevation, transition length and vertical curve length are all functions of it. IRC fixes the design speed from the **class of the road** and the **terrain**.

Terrain classification	Cross slope of the country
Plain	0 to 10 %
Rolling	10 to 25 %
Mountainous	25 to 60 %
Steep	greater than 60 %

Table 2.1 Terrain is classified by the general cross slope of the country, not by the road gradient.

Road classification	Plain ruling / min	Rolling ruling / min	Mountainous ruling / min	Steep ruling / min
National and State Highways	100 / 80	80 / 65	50 / 40	40 / 30
Major District Roads	80 / 65	65 / 50	40 / 30	30 / 20
Other District Roads	65 / 50	50 / 40	30 / 25	25 / 20
Village Roads	50 / 40	40 / 35	25 / 20	25 / 20

Table 2.2 IRC design speeds in km/h (ruling speed / minimum speed). Design for the RULING speed; the minimum speed is used only where site conditions make the ruling value impossible.

BEU / AKU REPEAT QUESTION

- How is terrain classified for the purpose of geometric design? [2 marks]
- What is design speed? On what factors does it depend? [2 marks]

2.2 Cross-section elements

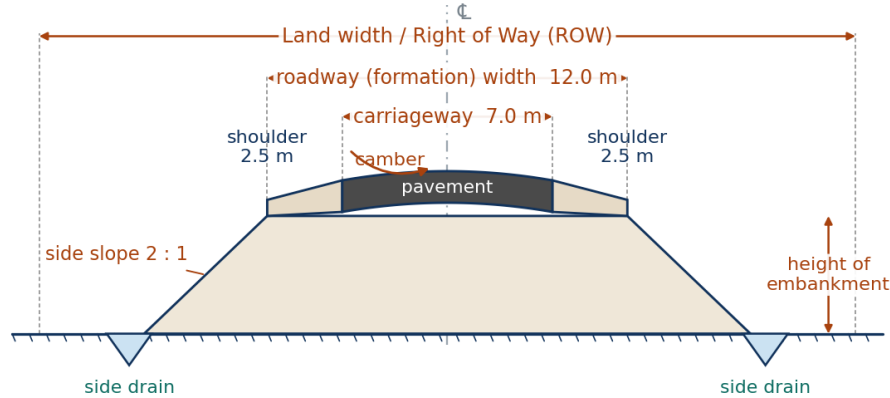
Typical cross-section of a two-lane highway in EMBANKMENT (vertical scale exaggerated)

Fig. 2.1 Typical cross-section of a two-lane highway in embankment, showing carriageway, shoulders, camber, side slopes, side drains, roadway width and right of way. **Learn to draw and label this from memory.**

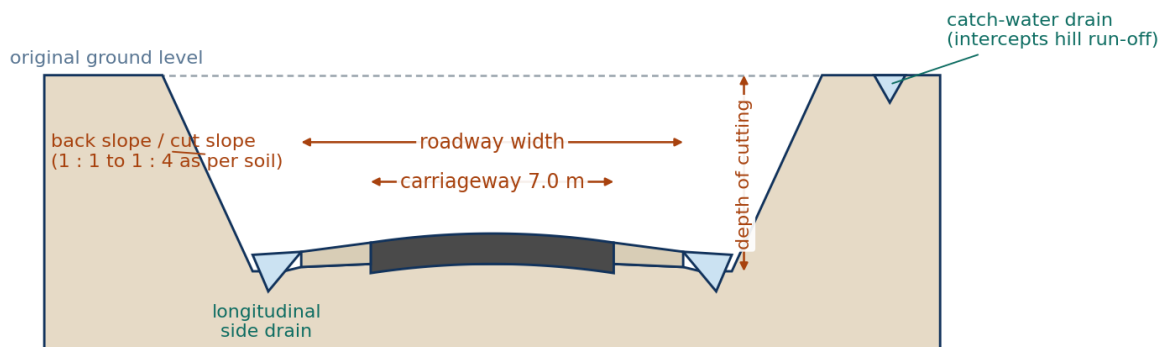
Cross-section of a highway in CUTTING

Fig. 2.2 Cross-section of a highway in cutting — note the side drain, the back (cut) slope and the catch-water drain.

2.2.1 Right of way, building line and control line

- **Right of way (land width)** — the area of land acquired for the road along its alignment. It should be adequate for the ultimate development, not merely the present one.
- **Building line** — a line on either side of the road beyond which the buildings must be set back, to give space for future widening and to keep the sight line clear.
- **Control line** — a line still further away, up to which the nature of the buildings is controlled by regulation.

Road class	Plain and rolling terrain Open area / Built-up area	Mountainous and steep terrain Open area / Built-up area
National and State Highways	45 m / 30 m	24 m / 20 m
Major District Roads	25 m / 20 m	18 m / 15 m
Other District Roads	15 m / 15 m	15 m / 12 m
Village Roads	12 m / 10 m	9 m / 9 m

Table 2.3 Recommended land width (right of way) as per IRC — normal values.

2.2.2 Camber (cross slope)

Camber — the transverse slope given to the road surface, to drain the rain water off the pavement quickly.

Objects of camber: (i) to drain off the rain water from the surface quickly, so that water does not enter the pavement layers and the subgrade; (ii) to prevent the entry of water into the bituminous layers, which would cause stripping; (iii) to keep the surface dry and thus preserve skid resistance.

Camber must be a compromise: **too little** camber leaves water standing on the surface, while **too much** camber makes the vehicles drift towards the edge, causes discomfort, wears the central portion of the tyres

and makes the surface prone to erosion of the shoulders.

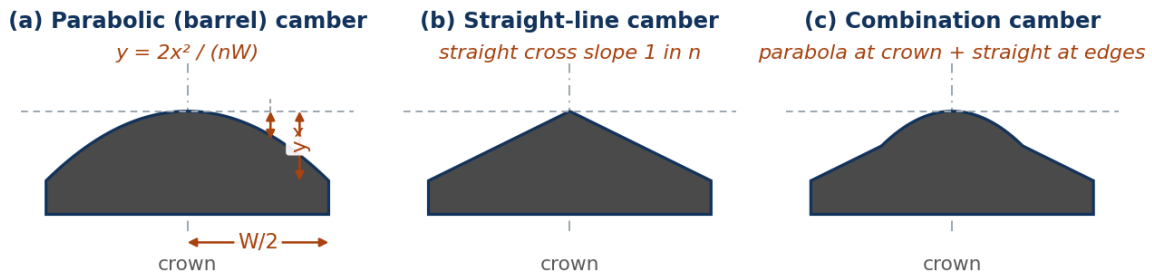


Fig. 2.3 Shapes of camber — parabolic (barrel), straight-line and combination. The parabolic shape is comfortable for fast traffic; the straight-line shape is easier to construct and maintain.

CAMBER

Parabolic (barrel) camber: $y = 2x^2 / (nW)$

Total rise of the crown above the edge = $W / (2n)$

- y = the drop of the surface below the crown, at a distance x from the crown (m)
- W = total width of the pavement (m)
- n = the camber expressed as 1 in n (so cross slope = $1/n$)
- Camber is stated either as 1 in n or as a percentage (1 in 50 = 2 %)

Type of road surface	Camber — range	Camber as 1 in n
Cement concrete and high-type bituminous surfaces	1.7 to 2.0 %	1 in 60 to 1 in 50
Thin bituminous surfaces	2.0 to 2.5 %	1 in 50 to 1 in 40
Water-bound macadam and gravel roads	2.5 to 3.0 %	1 in 40 to 1 in 33
Earth roads	3.0 to 4.0 %	1 in 33 to 1 in 25

Table 2.4 IRC recommended values of camber. Within each range the HIGHER value is adopted in areas of heavy rainfall and the LOWER value in areas of light rainfall.

IRC STANDARD VALUES

Learn these four camber values — they are asked as a 2-mark question almost every year. Cement concrete / high-type bituminous 1.7–2 %, thin bituminous 2–2.5 %, WBM and gravel 2.5–3 %, earth roads 3–4 %. Shoulders are given a cross slope 0.5 % steeper than the camber of the pavement, subject to a minimum of 3 %.

2.2.3 Width of pavement (carriageway)

Class of road	Width of carriageway
Single lane	3.75 m
Two lanes, without raised kerbs	7.0 m
Two lanes, with raised kerbs	7.5 m
Intermediate carriageway (between single and two lanes)	5.5 m
Multi-lane pavements	3.5 m per lane

Table 2.5 IRC recommended carriageway widths. Note the intermediate width 5.5 m — a width between 3.75 m and 5.5 m is NEVER provided, because two vehicles would try to cross and neither would have room.

2.2.4 Other cross-section elements

Element	Function and recommended value
Shoulder	Serves as an emergency lane and as a lateral support to the pavement layers. Minimum width 2.5 m for NH and SH. Cross slope 0.5 % steeper than the camber, minimum 3 %. It should be rougher than the pavement so that drivers do not use it as a running lane.

Element	Function and recommended value
Kerb	The boundary between the pavement and the shoulder or footpath. Low or mountable kerb (10 cm) — vehicles can climb it if needed; semi-barrier kerb (15 cm) — discourages encroachment; barrier kerb (20 cm) — in built-up areas, prevents vehicles from leaving the pavement; submerged kerb — used in rural roads to give lateral confinement to the pavement.
Median (traffic separator)	Separates the two directions of traffic and so prevents head-on collisions and headlight glare. Desirable width 5 m in rural areas; minimum 1.2 m , and where land is very restricted 0.6 m.
Footpath	Provided in urban roads where the pedestrian volume is significant; minimum width 1.5 m .
Cycle track	Minimum width 2.0 m , increased by 1 m for each additional cycle lane.
Side slopes	Embankment slopes usually 2 : 1 (H : V); cut slopes depend on the soil, from 1 : 1 in ordinary soil to nearly vertical in hard rock.
Guard rail / parapet	Provided at the edge of high embankments and on hill roads on the valley side, for safety.

Table 2.6 Other cross-section elements.

BEU / AKU REPEAT QUESTION

- ▶ What is camber? State its objects, the different shapes and the IRC recommended values for different surfaces. [7 marks]
- ▶ Draw a typical cross-section of a two-lane highway in embankment and label all the elements. [7 marks]
- ▶ Why is a carriageway width between 3.75 m and 5.5 m never provided? [2 marks]
- ▶ Differentiate between mountable, semi-barrier and barrier kerbs. [2 marks]

▶ Reference: Khanna, Justo & Veeraragavan, chapter 'Highway Geometric Design'; IRC 73-1980 (rural highways) and IRC 86-1983 (urban roads).

2.3 Sight distance

Sight distance — the length of road visible ahead to the driver at any instant.

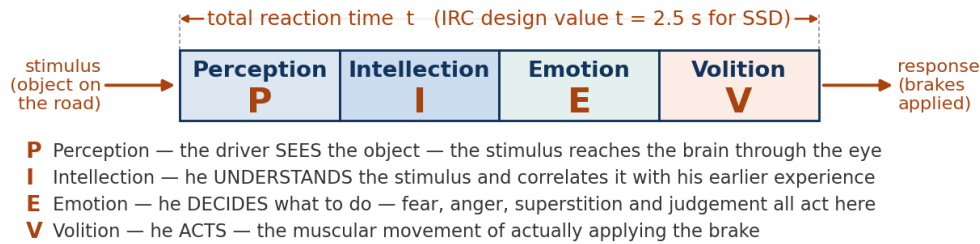
Three sight distances are considered in design:

- **Stopping (absolute minimum) sight distance, SSD** — the distance required to bring a vehicle to a stop after the driver sights an object on the carriageway. It must be available on **every part** of the road.
- **Overtaking (safe passing) sight distance, OSD** — the distance required to overtake a slow vehicle safely against the traffic in the opposite direction.
- **Intermediate sight distance, ISD** — taken as **twice the SSD**; provided where the OSD cannot be economically provided, so that at least a partial overtaking opportunity exists.
- In addition, **sight distance at intersections** and **head-light sight distance** (used in the design of valley curves) are considered.

2.3.1 Factors on which sight distance depends

- The **total reaction time** of the driver — explained by the **PIEV theory**; IRC takes $t = 2.5 \text{ s}$ for SSD and 2.0 s for OSD.
- The **speed** of the vehicle.
- The **efficiency of the brakes** — 100 % braking efficiency means the wheels are on the point of skidding.
- The **frictional resistance** between the tyre and the road, f (design value 0.35 to 0.40, decreasing with speed).
- The **gradient** of the road.

PIEV theory — the four components of driver reaction time



Lag (reaction) distance = $0.278 V t$ — the vehicle covers this distance BEFORE the brakes are applied

Fig. 2.4 PIEV theory — Perception, Intellection, Emotion, Volition. The sum of the four is the total reaction time t .

2.3.2 Stopping sight distance

SSD is the sum of two distances — the **lag (reaction) distance** covered during the reaction time, and the **braking distance** covered after the brakes are applied.

Stopping sight distance (SSD) — plan and sight-line elevation

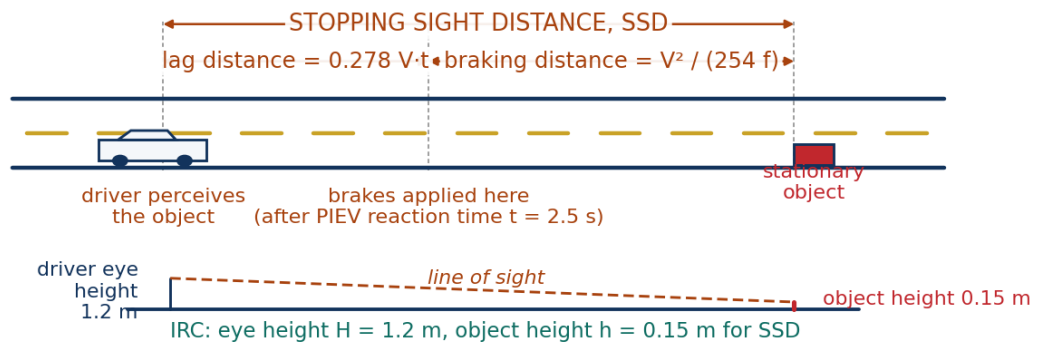


Fig. 2.5 Stopping sight distance = lag distance + braking distance. IRC criteria for measurement: driver's eye height $H = 1.2$ m, height of the object $h = 0.15$ m.

DERIVATION ASKED IN EXAM

Braking distance. The kinetic energy of the vehicle is destroyed by the work done against the skid resistance. If the vehicle of weight W moving at v m/s stops in a distance l , then

$$\text{work done against friction} = f \cdot W \cdot l = \text{kinetic energy} = \frac{1}{2} (W/g) v^2$$

$$\Rightarrow l = \frac{v^2}{(2 g f)}.$$

Adding the lag distance $v \cdot t$ gives $SSD = v t + v^2 / (2 g f)$. Converting v (m/s) to V (km/h) using $v = V/3.6$ gives the working formula.

STOPPING SIGHT DISTANCE

$$SSD = 0.278 V t + V^2 / (254 f) \quad [V \text{ in km/h}]$$

$$SSD = v t + v^2 / (2 g f) \quad [v \text{ in m/s}]$$

$$\text{On a gradient: } SSD = 0.278 V t + V^2 / [254 (f \pm 0.01 n)]$$

$$\text{Two-way traffic on a single-lane road: } SSD = 2 \times (\text{SSD for one way})$$

- V = design speed in km/h; v = design speed in m/s
- t = total reaction time = 2.5 s (IRC)
- f = design coefficient of longitudinal friction (0.40 at 20–30 km/h, 0.38 at 40, 0.37 at 50, 0.36 at 60, 0.35 at 65 km/h and above)
- n = gradient in per cent; use + for an ASCENDING gradient and – for a DESCENDING gradient
- $g = 9.8 \text{ m/s}^2$

EXAM TIP

On a descending gradient the friction available is reduced, so the braking distance and hence the SSD is LONGER. Therefore always design for the **descending** gradient. This is the single commonest slip in this topic.

SOLVED NUMERICAL — SSD on a level road and on a single-lane two-way road

Q. Calculate the stopping sight distance for a design speed of 80 km/h for (a) two-lane two-way traffic and (b) two-way traffic on a single-lane road. Take $t = 2.5$ s and $f = 0.35$.

Solution:

Step 1 — lag (reaction) distance

$$= 0.278 V t = 0.278 \times 80 \times 2.5 = 55.6 \text{ m}$$

Step 2 — braking distance

$$= V^2 / (254 f) = 80^2 / (254 \times 0.35) = 6400 / 88.9 = 72.0 \text{ m}$$

Step 3 — SSD

$$\text{SSD} = 55.6 + 72.0 = 127.6 \text{ m} \approx 128 \text{ m}$$

Step 4 — two-way traffic on a single lane

Both vehicles must stop, so twice the SSD is required:

$$= 2 \times 127.6 = 255.2 \text{ m} \approx 255 \text{ m}$$

(a) SSD for a two-lane road = 128 m (b) for a single-lane two-way road = 255 m

SOLVED NUMERICAL — SSD on a gradient

Q. Calculate the SSD on a highway at a descending gradient of 3 % and at an ascending gradient of 3 %, for a design speed of 65 km/h. Assume $t = 2.5$ s and $f = 0.36$.

Solution:

Step 1 — lag distance (the same in both cases)

$$= 0.278 \times 65 \times 2.5 = 45.18 \text{ m}$$

Step 2 — descending gradient ($n = -3$ %)

$$\text{Braking distance} = 65^2 / [254(0.36 - 0.03)] = 4225 / (254 \times 0.33) = 4225 / 83.82 = 50.41 \text{ m}$$

$$\text{SSD} = 45.18 + 50.41 = 95.6 \text{ m}$$

Step 3 — ascending gradient ($n = +3$ %)

$$\text{Braking distance} = 65^2 / [254(0.36 + 0.03)] = 4225 / 99.06 = 42.65 \text{ m}$$

$$\text{SSD} = 45.18 + 42.65 = 87.8 \text{ m}$$

SSD on the descending gradient = 95.6 m (governs the design); on the ascending gradient = 87.8 m

2.3.3 Overtaking sight distance

OSD is the minimum distance open to the vision of the driver of a vehicle intending to overtake a slow vehicle ahead, with safety against the traffic coming in the opposite direction. It is measured along the centre line of the road, with the height of the eye of the driver and the height of the object both taken as **1.2 m**.

Overtaking sight distance (OSD) on a two-lane two-way road

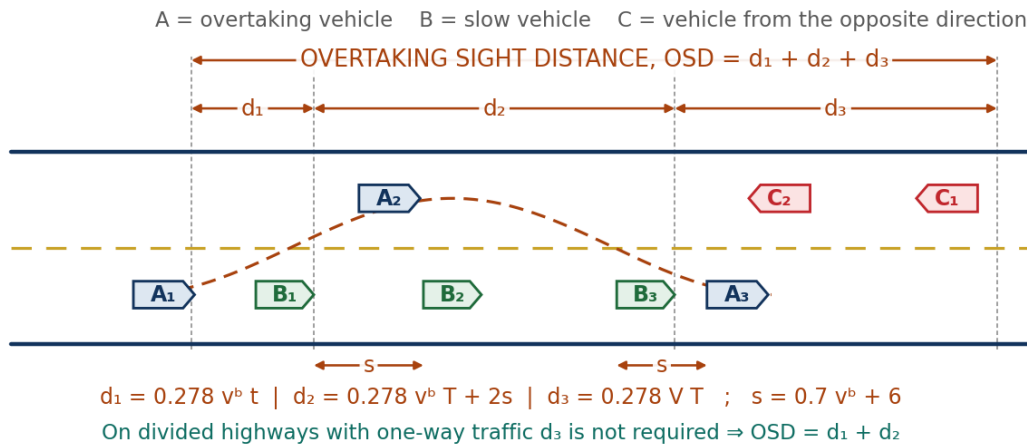


Fig. 2.6 Overtaking sight distance — $\text{OSD} = d_1 + d_2 + d_3$. A is the overtaking vehicle, B the slow vehicle and C the vehicle coming from the opposite direction.

The manoeuvre is in three parts. **d_1** is the distance travelled by the overtaking vehicle A during the reaction time t , while it closes up on B and decides to overtake. **d_2** is the distance travelled by A during the actual overtaking operation, which takes time T . **d_3** is the distance travelled in the same time T by the vehicle C coming from the opposite direction.

OVERTAKING SIGHT DISTANCE

$$\text{OSD} = d_1 + d_2 + d_3$$

$$d_1 = 0.278 V^b t \quad (t = 2 \text{ s})$$

$$d_2 = 0.278 V^b T + 2s$$

$$d_3 = 0.278 V T$$

$$s = 0.7 v^b + 6 \quad (v^b \text{ in m/s, } s \text{ in m})$$

$$T = \sqrt{(4s/a)} \quad [a \text{ in m/s}^2] \text{ or } T = \sqrt{(14.4s/A)} \quad [A \text{ in km/h/s}]$$

- V = design speed of the overtaking vehicle in km/h
- V^b = speed of the overtaken (slow) vehicle; if not given, take $V^b = V - 16 \text{ km/h}$
- s = the spacing of the vehicles, taken as $0.7 v^b + 6$ metres
- a = acceleration of the overtaking vehicle
- On a DIVIDED highway or a one-way road, d_3 is not required, so $\text{OSD} = d_1 + d_2$

SOLVED NUMERICAL — Overtaking sight distance and the overtaking zone

Q. The design speed of a two-lane two-way highway is 80 km/h. Calculate the overtaking sight distance and the minimum and desirable lengths of the overtaking zone. Take the acceleration of the overtaking vehicle as 0.92 m/s^2 and the reaction time as 2 s.

Solution:

Step 1 — speed of the slow vehicle

$$V^b = V - 16 = 80 - 16 = 64 \text{ km/h} = 64/3.6 = 17.78 \text{ m/s}$$

Step 2 — spacing s

$$s = 0.7 v^b + 6 = 0.7 \times 17.78 + 6 = 12.44 + 6 = 18.44 \text{ m}$$

Step 3 — time of the overtaking operation T

$$T = \sqrt{(4s/a)} = \sqrt{(4 \times 18.44 / 0.92)} = \sqrt{80.17} = 8.95 \text{ s}$$

Step 4 — the three distances

$$d_1 = 0.278 V^b t = 0.278 \times 64 \times 2 = 35.58 \text{ m}$$

$$d_2 = 0.278 V^b T + 2s = 0.278 \times 64 \times 8.95 + 2 \times 18.44 = 159.27 + 36.88 = 196.15 \text{ m}$$

$$d_3 = 0.278 V T = 0.278 \times 80 \times 8.95 = 199.05 \text{ m}$$

Step 5 — OSD

$$\text{OSD} = 35.58 + 196.15 + 199.05 = 430.8 \text{ m} \approx 431 \text{ m}$$

Step 6 — length of the overtaking zone

$$\text{Minimum length} = 3 \times \text{OSD} = 3 \times 431 = 1293 \text{ m}$$

$$\text{Desirable length} = 5 \times \text{OSD} = 5 \times 431 = 2155 \text{ m}$$

OSD = 431 m ; overtaking zone — minimum 1293 m, desirable 2155 m

NOTE

Overtaking zone. Where the OSD cannot be provided continuously, overtaking zones are provided at intervals. The zone is marked by sign posts SP1 and SP2 placed at a distance equal to the OSD *before* the beginning and *after* the end of the zone. Minimum length of the zone = $3 \times \text{OSD}$, desirable = $5 \times \text{OSD}$.

Sight distance	Height of the driver's eye	Height of the object
Stopping sight distance (SSD)	1.2 m	0.15 m
Overtaking sight distance (OSD)	1.2 m	1.2 m
Intermediate sight distance (ISD) = $2 \times \text{SSD}$	1.2 m	1.2 m
Head-light sight distance	0.75 m (head-light)	0.15 m, with a beam divergence of 1° upwards

Table 2.7 IRC criteria for measuring sight distance — a favourite 2-mark question.

BEU / AKU REPEAT QUESTION

- ▶ Derive an expression for the stopping sight distance. Explain the effect of gradient on it. [7 marks — very frequently asked]
- ▶ Explain, with a neat sketch, the overtaking sight distance and derive the expression for it. [7 marks — very frequently asked]
- ▶ What is an overtaking zone? What are its minimum and desirable lengths? [2 marks]
- ▶ State the criteria for measuring SSD and OSD. [2 marks]

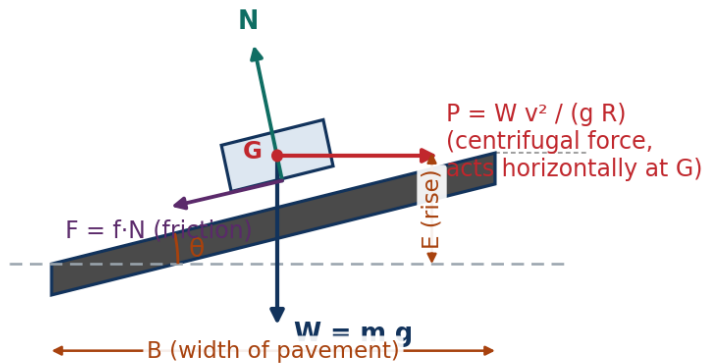
2.4 Design of horizontal alignment

2.4.1 Centrifugal force and superelevation

A vehicle moving on a horizontal curve of radius R at speed v is acted upon by an outward **centrifugal force** $P = W v^2/(gR)$, acting at the centre of gravity. The ratio $P/W = v^2/(gR)$ is called the **centrifugal ratio**. This force has two effects — it tends to **overturn** the vehicle about the outer wheels, and it tends to make the

vehicle **skid** sideways. To counteract it, the outer edge of the pavement is raised with respect to the inner edge; this is called **superelevation** or **banking**, $e = \tan \theta = E/B$.

Forces on a vehicle on a superelevated circular curve



$$e = \tan \theta = E / B ; \text{ for small } \theta, \tan \theta \approx \sin \theta$$

$$\Rightarrow e + f = v^2 / (g R) = V^2 / (127 R)$$

Fig. 2.7 Forces acting on a vehicle on a superelevated curve. Resolving along and perpendicular to the inclined surface gives $e + f = V^2 / (127 R)$.

DERIVATION ASKED IN EXAM

Resolving the forces along the inclined pavement surface, for equilibrium at the point of impending skidding:

$$P \cos \theta = W \sin \theta + F_A + F_B = W \sin \theta + f (R_A + R_B)$$

$$\text{and } R_A + R_B = W \cos \theta + P \sin \theta$$

$$\Rightarrow P \cos \theta - W \sin \theta = f (W \cos \theta + P \sin \theta)$$

Dividing throughout by $W \cos \theta$ and putting $\tan \theta = e$, $P/W = v^2 / (gR)$:

$$(v^2 / gR) - e = f (1 + e \cdot v^2 / gR). \text{ The term } e \cdot v^2 / gR \text{ is small and is neglected, giving}$$

$$e + f = v^2 / (g R) = V^2 / (127 R).$$

SUPERELEVATION

$$e + f = v^2 / (g R) = V^2 / (127 R)$$

Superelevation for mixed traffic (75 % of design speed, f neglected):

$$e = (0.75 V)^2 / (127 R) = V^2 / (225 R)$$

$$\text{Total rise of the outer edge: } E = e \times B$$

$$\text{Ruling minimum radius: } R = V^2 / [127 (e + f)]$$

- e = rate of superelevation (a ratio, or $\times 100$ for per cent)
- f = coefficient of lateral friction = 0.15 for design
- V = design speed in km/h, R = radius in m
- B = width of the pavement (including extra widening) in m
- Maximum e = 7 % (1 in 15) for plain and rolling terrain; 10 % for hill roads not bound by snow; 4 % for urban roads
- Minimum e = the camber of the road, provided for drainage even on very flat curves

The IRC four-step procedure for designing superelevation

1. Calculate **e for 75 % of the design speed**, neglecting friction: $e = V^2 / (225 R)$. If this is less than or equal to the maximum (7 %), **provide it** (but not less than the camber).
2. If the value from step 1 exceeds 7 %, **provide $e = 7$ %** and go to step 3.
3. Check the coefficient of friction actually required at the full design speed with $e = 0.07$: $f = V^2 / (127 R) - 0.07$. If $f \leq 0.15$, the curve is **safe at the full design speed** — the design is finished.
4. If $f > 0.15$, the curve is not safe at the design speed. Find the restricted (allowable) speed from $V_a = \sqrt{[127 R (0.07 + 0.15)]} = \sqrt{(27.94 R)}$ and either restrict the speed by regulatory signs or, preferably, **increase the radius**.

SOLVED NUMERICAL — Design of superelevation by the IRC method

Q. A two-lane highway of 7.0 m carriageway is to have a horizontal curve of radius 300 m. The design speed is 80 km/h. Design the superelevation and check whether the curve is safe at the design speed.

Solution:

Step 1 — e for 75 % of the design speed

$$e = V^2/(225 R) = 80^2/(225 \times 300) = 6400/67500 = 0.0948 = 9.48 \%$$

This exceeds the permissible maximum of 7 %.

Step 2 — limit the superelevation

$$\text{Provide } e = 7 \% = 0.07$$

Step 3 — check the friction required at the full design speed

$$f = V^2/(127 R) - e = 6400/(127 \times 300) - 0.07 \\ = 6400/38100 - 0.07 = 0.168 - 0.070 = 0.098$$

Since $f = 0.098 < 0.15$, the curve is safe at the full design speed of 80 km/h.

Step 4 — total rise of the outer edge

$$E = e \times B = 0.07 \times 7.0 = 0.49 \text{ m}$$

(If the extra widening of 0.61 m found in Example 2.6 is included, $B = 7.61 \text{ m}$ and $E = 0.07 \times 7.61 = 0.53 \text{ m}$.)

Provide $e = 7 \%$; the curve is safe at 80 km/h ; rise of the outer edge $E \approx 0.49 \text{ m}$ (0.53 m with extra widening)

SOLVED NUMERICAL — Ruling minimum radius of a horizontal curve

Q. Calculate the ruling minimum radius and the absolute minimum radius of a horizontal curve for a national highway in plain terrain. Take $e = 7 \%$ and $f = 0.15$.

Solution:

Step 1 — design speeds (Table 2.2)

For a NH in plain terrain: ruling design speed = 100 km/h, minimum design speed = 80 km/h.

Step 2 — ruling minimum radius (at the ruling design speed)

$$R = V^2/[127(e+f)] = 100^2/[127(0.07+0.15)] = 10000/27.94 = 357.9 \text{ m}$$

Step 3 — absolute minimum radius (at the minimum design speed)

$$R = 80^2/27.94 = 6400/27.94 = 229.1 \text{ m}$$

Ruling minimum radius $\approx 360 \text{ m}$; absolute minimum radius $\approx 230 \text{ m}$

2.4.2 Extra widening of the pavement on horizontal curves

The pavement is widened on a horizontal curve for two reasons, and the total extra widening is the sum of the two:

- **Mechanical widening** — because the rear wheels of a vehicle track on a smaller radius than the front wheels (**off-tracking**), so the vehicle occupies a greater width on a curve than on a straight.
- **Psychological widening** — extra width provided for psychological reasons: drivers tend to keep a greater clearance from other vehicles on a curve, and there is a greater difficulty in steering.

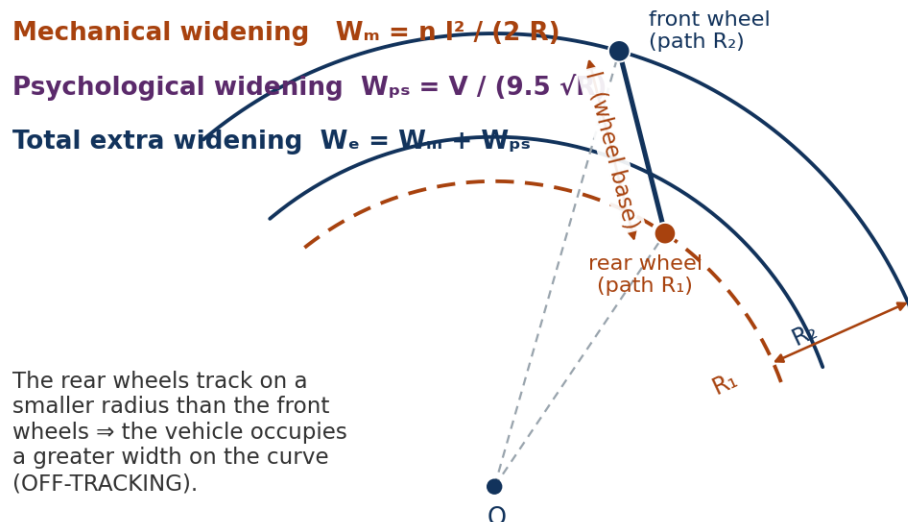


Fig. 2.8 Off-tracking of a vehicle on a horizontal curve, which makes mechanical widening necessary.

EXTRA WIDENING AT CURVES

Mechanical widening: $W_m = n l^2 / (2 R)$

Psychological widening: $W_{ps} = V / (9.5 \sqrt{V})$

Total extra widening: $W_e = W_m + W_{ps} = n l^2 / (2R) + V / (9.5 \sqrt{V})$

- n = number of traffic lanes
- l = length of the wheel base of the longest vehicle (6.1 m for a commercial vehicle as per IRC)
- R = radius of the curve in m, V = design speed in km/h
- The extra widening is provided half on the inside and half on the outside of the curve on a two-lane road (fully on the inside for hill roads)

SOLVED NUMERICAL — Extra widening of the pavement at a curve

Q. Calculate the extra widening required for a two-lane pavement on a horizontal curve of radius 300 m. The design speed is 80 km/h and the longest wheel base is 6.1 m. What total width should be provided at the curve?

Solution:

Step 1 — mechanical widening

$$W_m = n l^2 / (2R) = 2 \times (6.1)^2 / (2 \times 300) = 2 \times 37.21 / 600 = 0.124 \text{ m}$$

Step 2 — psychological widening

$$W_{ps} = V / (9.5 \sqrt{V}) = 80 / (9.5 \times \sqrt{300}) = 80 / (9.5 \times 17.32) = 80 / 164.6 = 0.486 \text{ m}$$

Step 3 — total extra widening

$$W_e = 0.124 + 0.486 = 0.61 \text{ m}$$

Step 4 — total width at the curve

$$= 7.0 + 0.61 = 7.61 \text{ m} \approx 7.6 \text{ m (0.305 m on each side)}$$

Extra widening = 0.61 m ; total width at the curve $\approx$ 7.6 m

2.4.3 Transition curves

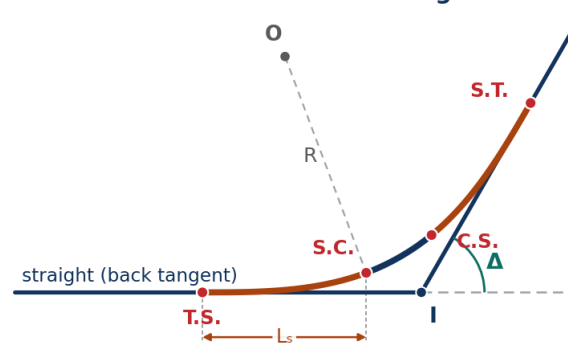
A transition curve has a radius that decreases gradually from infinity at the straight to the radius R of the circular curve. Its **functions** are:

- To introduce the **centrifugal force gradually**, so that the vehicle and the passengers are not subjected to a sudden jerk.
- To enable the driver to **turn the steering gradually**, which is both easier and safer.

- To provide a convenient length over which the **superelevation and the extra widening can be introduced gradually**.
- To improve the **aesthetic appearance** of the road.

The **ideal shape is the spiral (clothoid)**, because in a spiral the rate of change of curvature is constant, which is exactly what is wanted. IRC recommends the spiral. A cubic parabola or a lemniscate may also be used; the cubic parabola is easier to set out but is satisfactory only for small deflection angles.

Transition curve set out between the straight and the circular curve



Orange = transition (clothoid) curves, in which the radius falls from ∞ to R ; Blue = circular curve of radius R

T.S. tangent-spiral S.C. spiral-circle C.S. circle-spiral S.T. spiral-tangent Δ = deflection angle

Fig. 2.9 Transition curves inserted between the straight and the circular curve. T.S. = tangent-spiral, S.C. = spiral-circle, C.S. = circle-spiral, S.T. = spiral-tangent.

Shift of the circular curve on inserting a transition curve

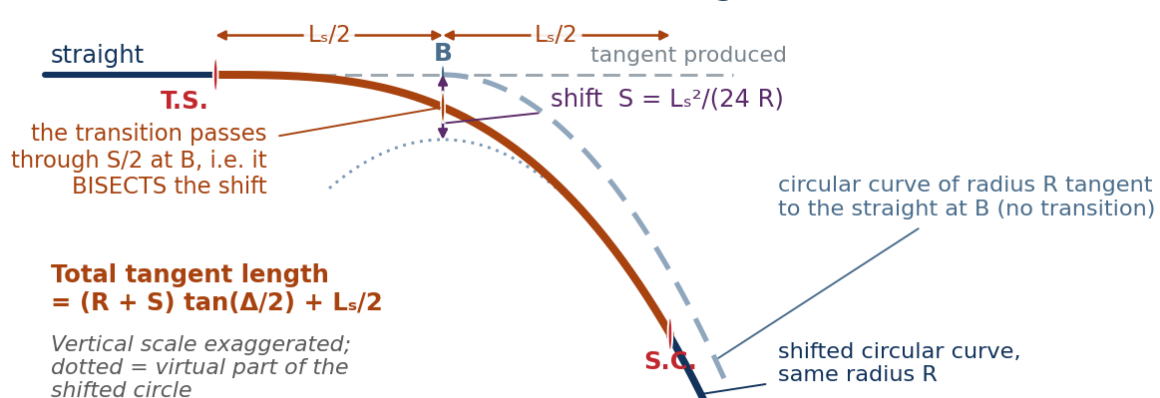


Fig. 2.10 Inserting a transition pushes the circular curve inward by the shift $S = L_s^2 / (24R)$, and the transition curve bisects the shift.

LENGTH OF TRANSITION CURVE

The length of the transition curve is the GREATEST of the three:

- (i) Rate of change of centrifugal acceleration: $L = 0.0215 V^3 / (C R)$, $C = 80/(75 + V)$
- (ii) Rate of introduction of superelevation: $L = e N (W + W_e)$ [rotated about the inner edge]
or $L = e N (W + W_e)/2$ [rotated about the centre line]

empirically $L = 2.7 V^2/R$ (plain, rolling) or $L = V^2/R$ (mountainous, steep)

- (iii) IRC minimum length as specified for the class of road

Shift: $S = L^2 / (24 R)$ Total tangent length = $(R + S) \tan(\Delta/2) + L/2$

- C = allowable rate of change of centrifugal acceleration in m/s^3 ; $C = 80/(75+V)$, subject to $0.5 \leq C \leq 0.8$
- N = rate of introduction of superelevation, expressed as 1 in N (1 in 150 for plain and rolling terrain, 1 in 60 for mountainous and steep terrain)
- W = normal width of the pavement, W_e = extra widening
- Δ = deflection angle of the curve

SOLVED NUMERICAL — Length of the transition curve and the shift

Q. A national highway in plain terrain has a horizontal curve of radius 300 m. The design speed is 80 km/h, the pavement is 7.0 m wide with an extra widening of 0.61 m and the superelevation is 7 %. The pavement is rotated about the centre line. Determine the length of the transition curve and the shift.

Solution:

Criterion (i) — rate of change of centrifugal acceleration

$C = 80/(75 + V) = 80/(75 + 80) = 80/155 = 0.516 \text{ m/s}^3$ (lies between 0.5 and 0.8, so acceptable)

$L = 0.0215 V^3/(C R) = 0.0215 \times 80^3/(0.516 \times 300)$
 $= 0.0215 \times 512000/154.8 = 11008/154.8 = 71.1 \text{ m}$

Criterion (ii) — rate of introduction of superelevation

Rotation about the centre line, so the outer edge is raised by $E/2$:

$E = e (W + W_e) = 0.07 \times (7.0 + 0.61) = 0.07 \times 7.61 = 0.533 \text{ m}$

Rise of the outer edge above the centre = $0.533/2 = 0.266 \text{ m}$

With a rate of 1 in 150, $L = 150 \times 0.266 = 39.9 \text{ m}$

(Check by the empirical formula: $L = 2.7 V^2/R = 2.7 \times 6400/300 = 57.6 \text{ m}$)

Criterion (iii) — IRC minimum

For a design speed of 80 km/h and $R = 300 \text{ m}$ the IRC minimum is of the order of 45–50 m, which is less than the value from criterion (i).

Choose the largest value

$L = \text{greatest of } (71.1, 57.6, 39.9) = 71.1 \text{ m} \rightarrow \text{adopt } L = 75 \text{ m}$ (rounded up to a convenient multiple of 5 m)

Shift

$S = L^2/(24 R) = 75^2/(24 \times 300) = 5625/7200 = 0.78 \text{ m}$

Length of the transition curve $L = 75 \text{ m}$; shift $S = 0.78 \text{ m}$

2.4.4 Set-back distance (clearance on horizontal curves)

On a horizontal curve the line of sight is a **chord**, while the road is an arc; obstructions such as buildings, trees or the cut slope on the inside of the curve therefore restrict the visibility. The **set-back distance m** is the clearance that must be kept free of obstruction, measured from the centre line of the road.

Set-back (clearance) distance on a horizontal curve

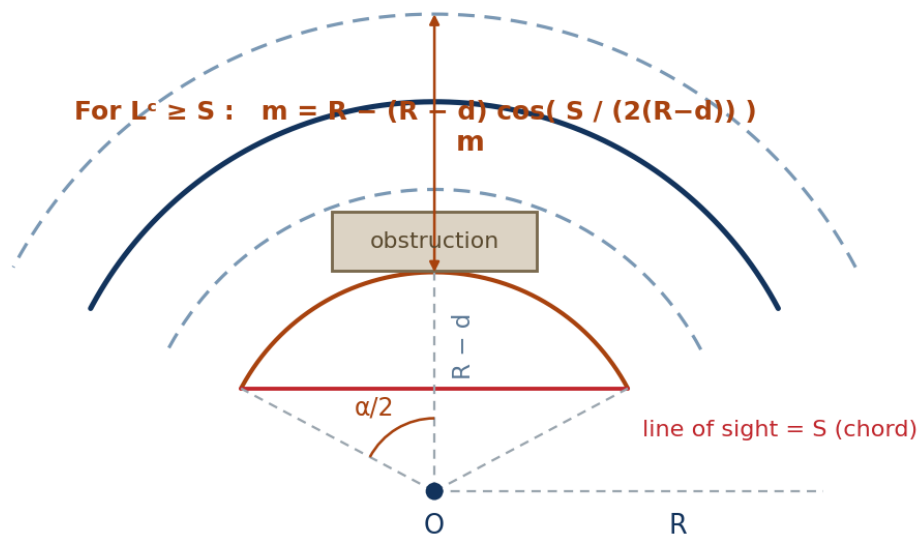


Fig. 2.11 Set-back distance on a horizontal curve. The sight line is the chord of the inner-lane centre line.

SET-BACK DISTANCE

Single-lane road: $m = R - R \cos(S / 2R)$

Multi-lane road: $m = R - (R - d) \cos[S / (2(R - d))]$

When the length of the curve L^c is LESS than the sight distance S :

$$m = R - (R - d) \cos(\alpha/2) + [(S - L^c)/2] \sin(\alpha/2), \alpha/2 = L^c / (2(R - d))$$

- m = set-back (clearance) from the centre line of the road, in m
- R = radius of the centre line of the road, in m
- d = distance between the centre line of the road and the centre line of the inner lane (= 1.75 m for a 7.0 m two-lane road)
- S = the sight distance to be provided (SSD, ISD or OSD)
- The angle inside the cosine is in RADIANS — the commonest mistake in this problem

SOLVED NUMERICAL — Set-back distance on a horizontal curve

Q. A two-lane road of 7.0 m width has a horizontal curve of radius 300 m and length 250 m. Calculate the set-back distance required to provide a stopping sight distance of 128 m.

Solution:

Step 1 — check which case applies

Length of the curve $L^c = 250$ m, which is greater than $S = 128$ m, so the first case ($L^c > S$) applies.

Step 2 — distance of the inner lane centre line

$d = 7.0/4 = 1.75$ m, so $R - d = 300 - 1.75 = 298.25$ m

Step 3 — the half angle

$$S/[2(R - d)] = 128/(2 \times 298.25) = 128/596.5 = 0.2146 \text{ radian} = 12.29^\circ$$

$$\cos(0.2146 \text{ rad}) = 0.9771$$

Step 4 — set-back distance

$$m = R - (R - d) \cos[S/(2(R-d))] = 300 - 298.25 \times 0.9771$$

$$= 300 - 291.42 = 8.58 \text{ m}$$

Set-back distance $m \approx 8.6$ m, measured from the centre line of the road. The area within this distance on the inside of the curve must be kept clear of obstructions.

BEU / AKU REPEAT QUESTION

- ▶ Derive the expression $e + f = V^2/127R$ for superelevation. [7 marks — asked very often]
- ▶ Explain the IRC procedure for the design of superelevation, and design the superelevation for $R = \dots$ and $V = \dots$ [14 marks]
- ▶ Why is extra widening provided at horizontal curves? Derive/state the expressions for mechanical and psychological widening. [7 marks]
- ▶ What is a transition curve? State its functions and the three criteria for determining its length. [7 marks — asked very often]
- ▶ Define shift. Prove that the transition curve bisects the shift. [2–4 marks]
- ▶ What is set-back distance? Calculate it for the given data. [7 marks]

2.5 Design of vertical alignment

2.5.1 Gradients

Gradient — the rate of rise or fall along the length of the road with respect to the horizontal, expressed as 1 in n or as a percentage.

Type of gradient	Meaning
Ruling gradient	The maximum gradient within which the designer attempts to design the vertical profile of the road. Also called the design gradient .
Limiting gradient	A steeper gradient which may be used where the topography compels it, and then only for short stretches.
Exceptional gradient	A very steep gradient used only in extraordinary situations, and for stretches not exceeding about 100 m at a time.
Minimum gradient	Needed from the point of view of DRAINAGE, not of traction: 1 in 500 for concrete drains and 1 in 200 for open soil drains, so that the water in the side drains flows away.
Floating gradient	A gradient on which a vehicle moving with a certain speed continues to move at the same speed without any tractive effort.

Table 2.8 Types of gradient.

Terrain	Ruling gradient	Limiting gradient	Exceptional gradient
Plain and rolling	3.3 % (1 in 30)	5 % (1 in 20)	6.7 % (1 in 15)
Mountainous and steep terrain, at elevations above 3,000 m	5 % (1 in 20)	6 % (1 in 16.7)	7 % (1 in 14.3)
Mountainous and steep terrain, up to 3,000 m elevation	6 % (1 in 16.7)	7 % (1 in 14.3)	8 % (1 in 12.5)

Table 2.9 IRC recommended gradients — learn the plain-terrain row (3.3 %, 5 %, 6.7 %) at least.

2.5.2 Grade compensation at horizontal curves

At a horizontal curve the vehicle has to overcome the extra tractive resistance caused by the curve in addition to that caused by the gradient. To avoid overloading the vehicle, the gradient is **eased (reduced)** at the curve; this reduction is the grade compensation.

GRADE COMPENSATION

$$\text{Grade compensation} = (30 + R) / R \text{ per cent}$$

subject to a maximum of 75 / R per cent

$$\text{Compensated gradient} = \text{ruling gradient} - \text{grade compensation}$$

- R = radius of the horizontal curve in m
- Grade compensation is NOT applied to gradients flatter than 4 %, because such gradients are easy in any case
- The compensated gradient should not be flatter than 4 %

SOLVED NUMERICAL — Grade compensation at a horizontal curve

Q. The gradient of a road in mountainous terrain is 1 in 20. A horizontal curve of radius 60 m is to be provided at this location. Determine the compensated gradient.

Solution:

Step 1 — the ruling gradient as a percentage

$$1 \text{ in } 20 = (1/20) \times 100 = 5 \% \text{ (greater than } 4 \% \text{, so compensation IS applicable)}$$

Step 2 — grade compensation

$$(30 + R)/R = (30 + 60)/60 = 90/60 = 1.5 \%$$

$$\text{Maximum permissible} = 75/R = 75/60 = 1.25 \%$$

Therefore adopt the lower value, 1.25 %

Step 3 — compensated gradient

$$= 5.0 - 1.25 = 3.75 \%$$

This is flatter than 4 %; hence limit the compensated gradient to 4 %.

Grade compensation = 1.25 % ; compensated gradient = 3.75 %, restricted to 4 % (= 1 in 25)

2.5.3 Vertical curves

A vertical curve is introduced at every change of grade to smooth out the vertical profile. A **summit curve** is one with convexity upwards and a **valley (sag) curve** one with convexity downwards. The **deviation angle N** is the algebraic difference of the two grades (an ascending grade is +, a descending grade is -).

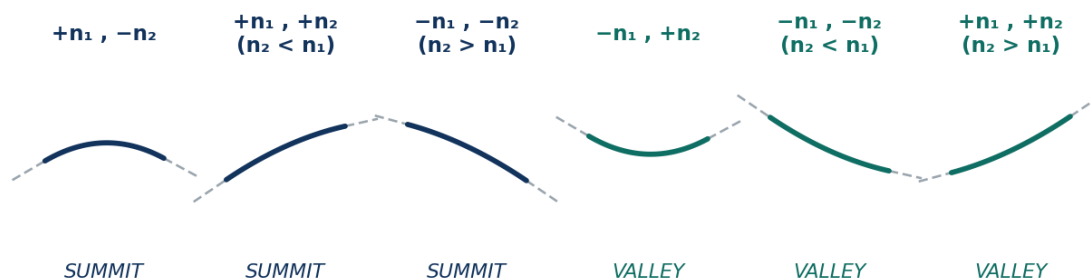
Types of summit and valley curves formed by different grade combinations

Fig. 2.12 Summit and valley curves produced by the different combinations of grades.

Summit curves

The **simple parabola** is used, because it gives the best riding comfort (the rate of change of grade is constant) and is easy to lay out. The design criterion is entirely **sight distance**; comfort is not a problem on a summit curve because the centrifugal force acts upwards, opposing gravity.

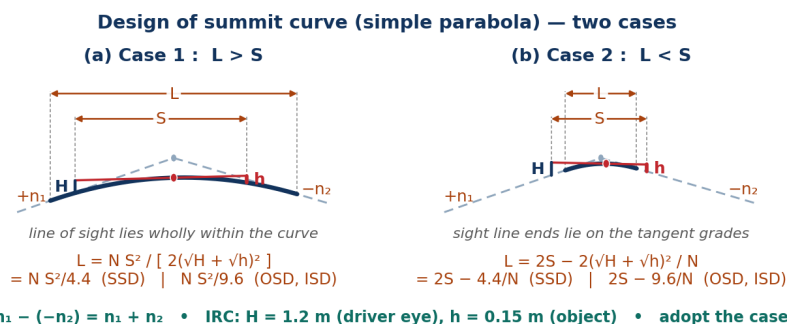


Fig. 2.13 Design of a summit curve, showing both cases. The formula to be used depends on whether the length of the curve is greater or less than the sight distance.

LENGTH OF A SUMMIT CURVE

Case 1 — the length of the curve is GREATER than the sight distance ($L > S$):

$$L = N S^2 / [2 (\sqrt{H} + \sqrt{h})^2]$$

$$= N S^2 / 4.4 \text{ for SSD (H = 1.2 m, h = 0.15 m)}$$

$$= N S^2 / 9.6 \text{ for OSD or ISD (H = h = 1.2 m)}$$

Case 2 — the length of the curve is LESS than the sight distance ($L < S$):

$$L = 2 S - 2 (\sqrt{H} + \sqrt{h})^2 / N$$

$$= 2 S - 4.4 / N \text{ for SSD} = 2 S - 9.6 / N \text{ for OSD or ISD}$$

EXAM TIP

How to handle the two cases. Assume Case 1 ($L > S$) first, because it usually governs. Compute L . If the L you get is indeed greater than S , the assumption was right and that is the answer. If it comes out less than S , discard it and recompute with the Case 2 formula, then check again.

SOLVED NUMERICAL — Length of a summit curve for stopping sight distance

Q. A vertical summit curve is formed by an ascending gradient of 1 in 30 meeting a descending gradient of 1 in 20. Design the length of the summit curve to provide a stopping sight distance of 128 m for a design speed of 80 km/h. Also find the length required for an intermediate sight distance.

Solution:

Step 1 — deviation angle

$$n_1 = +1/30 = +0.0333, n_2 = -1/20 = -0.05$$

$$N = n_1 - n_2 = 0.0333 - (-0.05) = 0.0833$$

Step 2 — assume $L > S$ and use the Case 1 formula for SSD

$$L = N S^2 / 4.4 = 0.0833 \times 128^2 / 4.4 = 0.0833 \times 16384 / 4.4$$

$$= 1364.8 / 4.4 = 310.2 \text{ m}$$

Step 3 — check the assumption

$L = 310.2 \text{ m}$ is greater than $S = 128 \text{ m}$, so the assumption $L > S$ is correct.

Adopt $L = 310 \text{ m}$

Step 4 — length for the intermediate sight distance

$$\text{ISD} = 2 \times \text{SSD} = 2 \times 128 = 256 \text{ m}$$

$$L = N S^2 / 9.6 = 0.0833 \times 256^2 / 9.6 = 0.0833 \times 65536 / 9.6$$

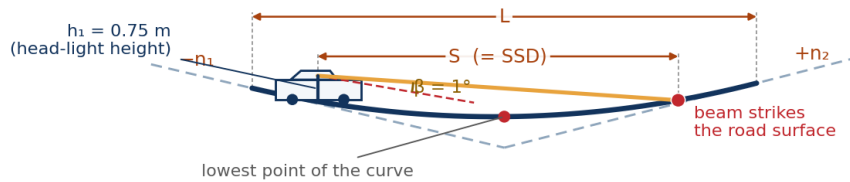
$$= 5459.2 / 9.6 = 568.7 \text{ m (again greater than } S = 256 \text{ m, so valid)}$$

Length of the summit curve for SSD = 310 m ; for ISD = 569 m. The SSD value is the absolute minimum; provide the ISD length if the site and the cost permit.

Valley (sag) curves

On a valley curve the centrifugal force acts **downwards**, in the same direction as gravity, so the total pressure on the springs is increased and **riding comfort** becomes a design criterion. Also, the critical condition is at **night**, when the visibility is limited to the length of road lit by the head-lights. Hence two criteria:

1. **Comfort criterion** — the allowable rate of change of centrifugal acceleration is taken as 0.6 m/s^3 .
2. **Head-light sight distance criterion** — the length of road lit by the head-lights must be at least equal to the SSD. Head-light height $h_1 = 0.75 \text{ m}$ and the beam is assumed to diverge 1° upwards ($\beta = 1^\circ$).

Valley (sag) curve — head-light sight distance criterion

Design for the GREATER of: (i) comfort $L = 0.38 (N V^3)^{1/2}$ (ii) head-light SD $L = N S^2 / (1.50 + 0.035 S)$ if $L > S$

Fig. 2.14 Valley curve — the head-light sight distance criterion.

LENGTH OF A VALLEY CURVE

Comfort criterion: $L = 0.38 (N V^3)^{1/2}$

Head-light sight distance, $L > S$: $L = N S^2 / [2 (h_1 + S \tan \beta)] = N S^2 / (1.50 + 0.035 S)$

Head-light sight distance, $L < S$: $L = 2 S - (1.50 + 0.035 S) / N$

DESIGN FOR THE GREATER OF THE TWO LENGTHS

- N = deviation angle, V = design speed in km/h, S = SSD in m
- $h_1 = 0.75$ m (height of the head-light), $\beta = 1^\circ$ so $\tan \beta = 0.0175$
- The valley curve is made of two similar transitions (a cubic parabola) of length $L/2$ each

SOLVED NUMERICAL — Length of a valley curve

Q. A valley curve is formed by a descending gradient of 1 in 25 meeting an ascending gradient of 1 in 30. Design the length of the valley curve for a design speed of 80 km/h. Take the SSD as 128 m.

Solution:

Step 1 — deviation angle

$$n_1 = -1/25 = -0.04, n_2 = +1/30 = +0.0333$$

$$N = n_2 - n_1 = 0.0333 + 0.04 = 0.0733$$

Step 2 — comfort criterion

$$\begin{aligned} L &= 0.38 (N V^3)^{1/2} = 0.38 \times (0.0733 \times 80^3)^{1/2} \\ &= 0.38 \times (0.0733 \times 512000)^{1/2} = 0.38 \times (37530)^{1/2} \\ &= 0.38 \times 193.7 = 73.6 \text{ m} \end{aligned}$$

Step 3 — head-light sight distance criterion (assume $L > S$)

$$\begin{aligned} L &= N S^2 / (1.50 + 0.035 S) = 0.0733 \times 16384 / (1.50 + 0.035 \times 128) \\ &= 1200.9 / (1.50 + 4.48) = 1200.9 / 5.98 = 200.8 \text{ m} \end{aligned}$$

Check: $L = 200.8 \text{ m} > S = 128 \text{ m}$, so the assumption is correct.

Step 4 — adopt the greater value

$$L = \text{greater of } (73.6, 200.8) = 200.8 \text{ m} \rightarrow \text{adopt } L = 200 \text{ m}$$

Length of the valley curve = 200 m, governed by the head-light sight distance criterion (as is almost always the case)

BEU / AKU REPEAT QUESTION

- ▶ Design the length of a summit curve for the given gradients and sight distance. [7 marks — appears in almost every paper]
- ▶ What are the two criteria for the design of a valley curve? Design the length for the given data. [7 marks — appears very often]
- ▶ Why is riding comfort a criterion for valley curves but not for summit curves? [2 marks]
- ▶ Define ruling, limiting and exceptional gradient. What is grade compensation? [7 marks]

2.6 Design of intersections — an outline

An **intersection** is the area where two or more roads join or cross. It is the point at which the capacity of a road system and its safety are usually decided. Intersections are of two broad kinds:

- **At-grade intersections** — all the roads meet at the same level: T, Y, skew, cross, staggered, multi-leg and rotary intersections. The conflicts are managed by **channelization, traffic islands, markings, signs, signals** or a **rotary**.
- **Grade-separated intersections (interchanges)** — the crossing roads are at different levels, connected by ramps: trumpet, diamond, cloverleaf and rotary interchanges. There is no crossing conflict at all, but the cost and the land requirement are high.

The **objectives** of intersection design are to minimise the number and the severity of the points of conflict, to fix the angle and the point of merging, to control the speed of the turning traffic, to provide adequate sight distance (the **sight triangle**) and to give clear guidance to the driver. The detailed design of intersections, including the design of a rotary and the design of traffic signals, is dealt with in **Unit 4, § 4.4 and § 4.5**.

► *Reference: Khanna, Justo & Veeraragavan, chapter 'Highway Geometric Design'; Kadiyali, chapters on geometric design and intersections; IRC 73-1980, IRC 86-1983, IRC SP 23 (vertical curves).*

UNIT 3 | Traffic Engineering and Control

Syllabus allotment: 6 hours • about 14 % of the paper

3.1 Scope of traffic engineering

Traffic engineering — that branch of engineering which deals with the improvement of traffic performance of road networks and terminals — the planning, geometric design and traffic operation of roads, streets and highways, their networks, terminals and abutting lands, and their relationship with other modes of transportation.

The scope of traffic engineering covers **traffic characteristics, traffic studies and analysis, traffic operation, regulation and control, planning and analysis of traffic facilities, geometric design, and accident analysis and prevention.**

3.2 Traffic characteristics

The three components of the traffic system are the **road user**, the **vehicle** and the **road**. Traffic characteristics are the characteristics of the first two.

3.2.1 Road user characteristics

Group of characteristics	What it includes
Physical	Vision (acuity, peripheral vision, glare recovery), hearing, strength, and the general state of health. Vision is the most important — the acute cone of vision is about 3° to 10° and the wider peripheral vision is about 120° to 180°.
Mental	Knowledge, skill, intelligence, experience and literacy — these decide how quickly and how correctly a driver responds.
Psychological	Emotions such as anger, fear, impatience, superstition and anxiety, which affect the reaction of the road user.
Environmental	The conditions in which the road user is placed — traffic stream characteristics, facilities to the traffic, atmospheric conditions and the surrounding land use.

Table 3.1 Road user characteristics.

Reaction time and the PIEV theory. The total reaction time of a driver is the time from the instant the object comes into his sight to the instant the brakes are actually applied. It is split into four parts — **Perception, Intellection, Emotion and Volition**. IRC takes the total reaction time as **2.5 s** for the design of stopping sight distance (see Fig. 2.4 in Unit 2).

3.2.2 Vehicle characteristics

Characteristic	Design significance	IRC / typical value
Width	Decides the lane width and the extra widening	Maximum 2.5 m
Height	Decides the clearance under structures and overhead cables	3.8 m (4.75 m for double-deck buses)
Length	Decides the extra widening (through the wheel base), the capacity, the parking bay size and the turning radius at intersections	Single unit truck 11 m; articulated / tractor-trailer 16 m; wheel base of a commercial vehicle 6.1 m
Weight and axle load	Decides the pavement thickness and the bridge loading	Legal maximum single axle load 10.2 tonnes; standard axle for pavement design 8.16 tonnes (80 kN)
Power and speed	Decides the gradient that can be negotiated, the overtaking sight distance and the capacity	Design speed as per Table 2.2
Braking and acceleration	Decides the stopping and overtaking sight distances	Design $f = 0.35$ – 0.40 longitudinal, 0.15 lateral

Table 3.2 Vehicular characteristics and their design significance (IRC 3-1983 for dimensions).

Because the traffic on Indian roads is **highly heterogeneous**, vehicles of very different sizes and speeds share the same carriageway. To add them up meaningfully they are converted into a common unit — the **Passenger Car Unit (PCU)**.

Type of vehicle	PCU	Type of vehicle	PCU
Motor cycle or scooter	0.50	Truck, bus or tractor-trailer	3.00
Passenger car, tempo, auto-rickshaw, tractor	1.00	Horse-drawn vehicle	4.00
Cycle	0.50	Small bullock cart or hand cart	6.00
Cycle rickshaw	2.00	Large bullock cart	8.00

Table 3.3 Commonly used PCU equivalency factors. IRC 106-1990 gives these as RANGES which depend on the percentage composition of the vehicle type in the stream; use the values given in the question if any are supplied.

SOLVED NUMERICAL — Converting a classified volume count into PCU

Q. The classified traffic volume count on an urban road for the peak hour is: cars 600, two-wheelers 800, buses 90, trucks 110, auto-rickshaws 150, cycles 200 and cycle rickshaws 40. Determine the traffic volume in vehicles per hour and in PCU per hour.

Solution:

Step 1 — total number of vehicles

$$600 + 800 + 90 + 110 + 150 + 200 + 40 = 1990 \text{ vehicles/hour}$$

Step 2 — convert each class to PCU (Table 3.3)

$$\text{Cars: } 600 \times 1.0 = 600 \text{ PCU}$$

$$\text{Two-wheelers: } 800 \times 0.5 = 400 \text{ PCU}$$

$$\text{Buses: } 90 \times 3.0 = 270 \text{ PCU}$$

$$\text{Trucks: } 110 \times 3.0 = 330 \text{ PCU}$$

$$\text{Auto-rickshaws: } 150 \times 1.0 = 150 \text{ PCU}$$

$$\text{Cycles: } 200 \times 0.5 = 100 \text{ PCU}$$

$$\text{Cycle rickshaws: } 40 \times 2.0 = 80 \text{ PCU}$$

Step 3 — total

$$\text{Total} = 600 + 400 + 270 + 330 + 150 + 100 + 80 = 1930 \text{ PCU/hour}$$

Traffic volume = 1990 vehicles/hour = 1930 PCU/hour. The design and capacity checks must be done in PCU, not in vehicles.

BEU / AKU REPEAT QUESTION

- What is PCU? Why is it necessary on Indian roads? Convert the given classified count into PCU. [7 marks]
- Discuss the road user and vehicular characteristics that influence geometric design and traffic operation. [7 marks]
- What is PIEV theory? [2 marks]

3.3 Traffic engineering studies

The five basic traffic studies are the **volume** study, the **speed** study (spot speed, and speed and delay), the **origin and destination** study, the **parking** study and the **accident** study.

3.3.1 Traffic volume study

Traffic volume — the number of vehicles crossing a section of the road per unit time at a given period. It is expressed in vehicles per day or vehicles per hour, or in PCU per hour.

Term	Definition
ADT (average daily traffic)	The average of 24-hour counts taken over a period shorter than a year.
AADT (annual average daily traffic)	The average of 24-hour counts taken over a full year , i.e. the total yearly volume divided by 365. Used for the economic analysis and for pavement design.

Term	Definition
Peak hour volume	The maximum number of vehicles that pass in one hour of the day.
Design hourly volume	The volume adopted for the geometric design. Designing for the absolute peak is uneconomical, so the 30th highest hourly volume of the year is commonly used for rural highways in India.
Peak hour factor, PHF	A measure of how uniformly the traffic flows within the peak hour: $PHF = (\text{peak hour volume}) / (4 \times \text{peak 15-minute volume})$. It approaches 1.0 for a uniform flow.
Traffic density	The number of vehicles occupying a unit length of the road, in veh/km.

Table 3.4 Volume terminology.

Methods of volume count

- **Manual count** — field enumerators with tally sheets. The only method that gives a **classified** count and turning movements, but it is labour-intensive and cannot be run continuously.
- **Mechanical / automatic count** — pneumatic tubes, inductive loop detectors, magnetic and photo-electric detectors, video image processing. Runs continuously but usually does not classify vehicles.
- **Moving observer (moving car) method** — gives the volume, the mean speed and the density from a single pair of test runs (see § 3.3.3).
- **Photographic and video methods** — a permanent record is obtained which can be analysed later at leisure.

Presentation of volume data

- **Traffic flow (volume) diagram** — a plan of the network with bands drawn along each link, the width of the band being proportional to the volume.
- **Traffic volume count / turning movement diagram** — an intersection diagram with the volume of each movement written on the arrow.
- **Trend charts** — volumes plotted against the year, used for forecasting future traffic.
- **Variation charts** — hourly, daily, monthly and seasonal variation, used to fix the design hour and to plan maintenance and enforcement.
- **The 30th highest hourly volume curve** — the hourly volumes of the year arranged in descending order and plotted, from which the design hourly volume is read.

SOLVED NUMERICAL — Peak hour factor and design hourly volume

Q. The volume counted in the peak hour on a road is 1,800 vehicles, and the highest 15-minute count within that hour is 550 vehicles. Determine the peak hour factor and comment on the uniformity of the flow.

Solution:

Step 1 — peak hour factor

$$\begin{aligned}
 PHF &= (\text{peak hour volume}) / (4 \times \text{peak 15-minute volume}) \\
 &= 1800 / (4 \times 550) = 1800 / 2200 = 0.818
 \end{aligned}$$

Step 2 — comment

The flow rate during the busiest 15 minutes is $4 \times 550 = 2,200$ veh/h, against an average of 1,800 veh/h for the hour.

A PHF of 0.82 shows an appreciable degree of peaking within the hour; the facility must be checked against the rate of 2,200 veh/h, not 1,800 veh/h.

PHF = 0.82 ; the design flow rate within the peak hour = 2,200 veh/h

3.3.2 Spot speed studies

Spot speed — the instantaneous speed of a vehicle at a specified location.

Uses: to fix speed limits, to design geometric elements, to decide the location of signs and signals, to study accident-prone spots, and to check the effect of a traffic improvement measure.

Methods: the **enoscope** (mirror box) with a stop-watch, the **pressure-pad** and other electronic detectors, the **Doppler radar speedmeter**, and **photographic / videographic** methods.

Presentation of spot-speed study data

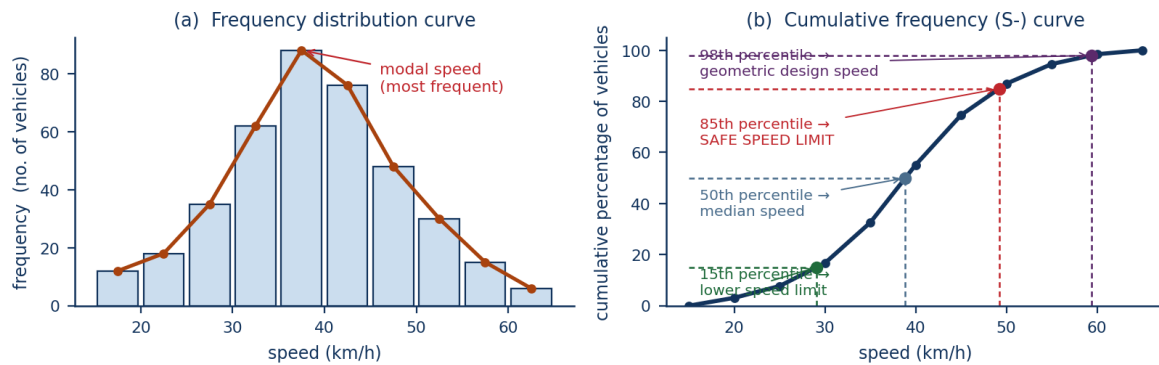


Fig. 3.1 Presentation of spot-speed data — the frequency distribution curve and the cumulative frequency (S-) curve from which the percentile speeds are read.

Percentile speed	What it is used for
85th percentile	The safe speed limit at the location. 85 % of the drivers travel at or below this speed; the remaining 15 % are considered to be driving unreasonably fast.
98th percentile	The design speed for geometric design — the elements of the road are designed so that even the fast 98th-percentile driver is safe.
15th percentile	The lower speed limit , to weed out the very slow drivers who obstruct the stream and cause accidents.
50th percentile	The median speed, used for general comparison.
Modal speed	The most frequently observed speed — the peak of the frequency distribution curve.

Table 3.5 Use of the percentile speeds — 85th, 98th and 15th are the ones asked in the exam.

SOLVED NUMERICAL — Percentile speeds from a spot-speed study

Q. A spot-speed study at a section gave the following data. Determine the 15th, 50th, 85th and 98th percentile speeds and state the use of each.

Speed range (km/h) : 15-20, 20-25, 25-30, 30-35, 35-40, 40-45, 45-50, 50-55, 55-60, 60-65

Number of vehicles : 12, 18, 35, 62, 88, 76, 48, 30, 15, 6

Solution:

Step 1 — cumulative frequency and cumulative percentage

Total number of observations = $12+18+35+62+88+76+48+30+15+6 = 390$

Upper limit 20 25 30 35 40 45 50 55 60 65

Cumulative 12 30 65 127 215 291 339 369 384 390

Cum. % 3.08 7.69 16.67 32.56 55.13 74.62 86.92 94.62 98.46 100.00

Step 2 — 15th percentile (between 25 and 30 km/h)

$$V_{15} = 25 + 5 \times (15 - 7.69) / (16.67 - 7.69) = 25 + 5 \times 7.31 / 8.98 = 29.1 \text{ km/h}$$

Step 3 — 50th percentile (between 35 and 40 km/h)

$$V_{50} = 35 + 5 \times (50 - 32.56) / (55.13 - 32.56) = 35 + 5 \times 17.44 / 22.57 = 38.9 \text{ km/h}$$

Step 4 — 85th percentile (between 45 and 50 km/h)

$$V_{85} = 45 + 5 \times (85 - 74.62) / (86.92 - 74.62) = 45 + 5 \times 10.38 / 12.30 = 49.2 \text{ km/h}$$

Step 5 — 98th percentile (between 55 and 60 km/h)

$$V_{98} = 55 + 5 \times (98 - 94.62) / (98.46 - 94.62) = 55 + 5 \times 3.38 / 3.84 = 59.4 \text{ km/h}$$

$V_{15} = 29.1 \text{ km/h}$ (lower speed limit) ; $V_{50} = 38.9 \text{ km/h}$ (median) ; $V_{85} = 49.2 \text{ km/h} \rightarrow$ adopt a speed limit of 50 km/h ; $V_{98} = 59.4 \text{ km/h} \rightarrow$ use 60 km/h as the design speed for the geometric elements

3.3.3 Speed and delay study — the moving observer method

A speed and delay study gives the **running speed**, **overall (journey) speed**, the **amount of delay** and the **causes of delay** over a stretch of road. The methods are the **floating car (moving observer) method**, the **licence-plate (vehicle number) method**, the **elevated observation method** and the **interview technique**.

Running speed — the length of the stretch divided by the time the vehicle was actually **in motion** (stopped time excluded).

Journey (overall) speed — the length of the stretch divided by the **total** time taken, including all stopped delays. Journey speed is therefore always less than or equal to the running speed.

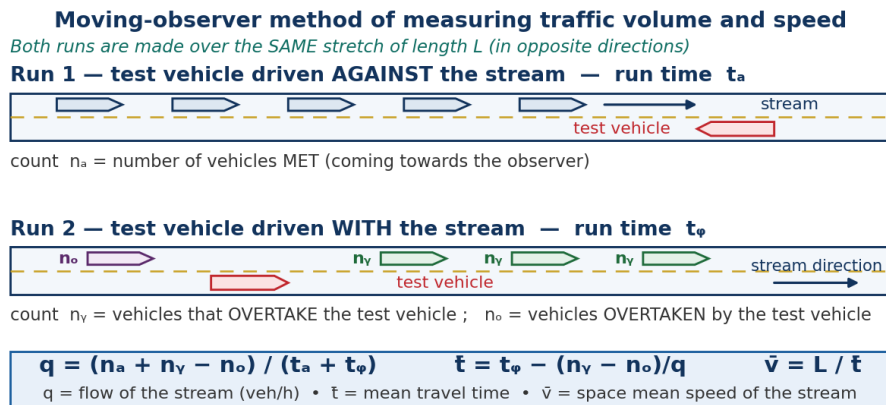


Fig. 3.2 The moving-observer method. Two runs are made over the same stretch — one against the stream and one with the stream.

MOVING OBSERVER METHOD

$$q = (n_a + n_v - n_o) / (t_a + t_\phi)$$

$$\bar{t} = t_\phi - (n_v - n_o) / q$$

$$\bar{v} = L / \bar{t} \text{ and } k = q / \bar{v}$$

- q = flow of the stream (veh per unit time)
- n_a = number of vehicles MET by the test vehicle while running AGAINST the stream, in time t_a
- n_v = number of vehicles that OVERTAKE the test vehicle while it runs WITH the stream, in time t_ϕ
- n_o = number of vehicles OVERTAKEN BY the test vehicle in the same run
- $\bar{t}$ = mean travel time of the stream over the stretch
- $\bar{v}$ = space mean speed of the stream ; k = density
- Keep the units consistent — if t is in minutes, q comes out in vehicles per minute

SOLVED NUMERICAL — Moving observer method

Q. In a moving-observer study over a stretch of road 1.0 km long, the test vehicle running **against** the stream took 1.5 minutes and met 80 vehicles. Running **with** the stream it took 2.0 minutes, during which 5 vehicles overtook it and it overtook 15 vehicles. Determine the flow, the mean travel time, the mean speed of the stream and the traffic density.

Solution:

Step 1 — list the data

$$n_a = 80, t_a = 1.5 \text{ min} ; n_v = 5, n_o = 15, t_\phi = 2.0 \text{ min} ; L = 1.0 \text{ km}$$

Step 2 — flow of the stream

$$q = (n_a + n_v - n_o)/(t_a + t_\phi) = (80 + 5 - 15)/(1.5 + 2.0) \\ = 70/3.5 = 20 \text{ vehicles per minute} = 20 \times 60 = \mathbf{1200 \text{ veh/h}}$$

Step 3 — mean travel time of the stream

$$\bar{t} = t_\phi - (n_v - n_o)/q = 2.0 - (5 - 15)/20 = 2.0 - (-10/20) \\ = 2.0 + 0.5 = 2.5 \text{ minutes}$$

Step 4 — mean (space mean) speed of the stream

$$\bar{v} = L/\bar{t} = 1.0/2.5 = 0.4 \text{ km/min} = 0.4 \times 60 = \mathbf{24 \text{ km/h}}$$

Step 5 — density

$$k = q/\bar{v} = 1200/24 = \mathbf{50 \text{ veh/km}}$$

The test vehicle itself took 2.0 min for 1 km, i.e. 30 km/h; since it overtook more vehicles (15) than overtook it (5), it was travelling faster than the stream — which is consistent with the stream speed of 24 km/h.

$$\mathbf{q = 1200 \text{ veh/h} ; \bar{t} = 2.5 \text{ min} ; \bar{v} = 24 \text{ km/h} ; k = 50 \text{ veh/km}}$$

3.3.4 Origin and destination (O-D) study

An O-D study finds out **where the trips begin and where they end**, and hence the **desire lines** of travel.

Uses: to plan the road network and to judge the need for a bypass or a new link; to locate terminals, parking areas and bus stops; to plan for improvement of an existing route; and to establish the desire lines of travel for transport planning.

Methods: roadside interview; licence-plate (registration number) method; return post-card method; tag-on-vehicle method; and the home-interview method.

Presentation: as an **O-D table (matrix)** giving the number of trips from each zone to every other zone, and as a **desire-line diagram**, in which straight lines are drawn between the origin and destination zones with the width of each line proportional to the number of trips. The desire-line diagram shows immediately where a new facility is most needed.

3.3.5 Accident studies

Objects: to study the causes of accidents and to suggest remedial measures; to evaluate the improvements already carried out; to support the design of the geometric and control elements; and to compute the accident cost.

The **causes** of accidents are grouped under the **road user** (excessive speed, overtaking violations, carelessness, fatigue, intoxication, jumping signals), the **vehicle** (brake or tyre failure, defective lights, steering failure), the **road** (poor alignment, inadequate sight distance, poor skid resistance, poor lighting, potholes), the **traffic conditions** (mixed traffic, encroachment, inadequate control) and the **environment** (rain, fog, glare).

Accident study — collision and condition diagrams

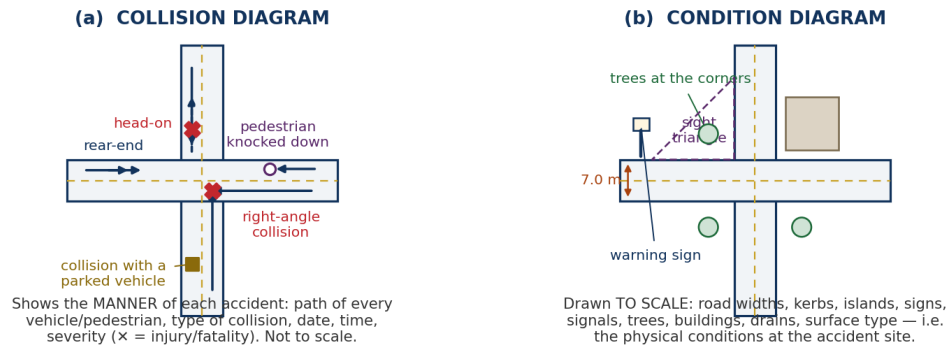


Fig. 3.3 The two accident diagrams. A **collision diagram** records the MANNER of each accident and is NOT to scale; a **condition diagram** records the physical conditions of the site and IS drawn to scale.

ACCIDENT RATES

Accident rate per 100 million vehicle-kilometres:

$$R = (A \times 10^8) / (365 \times \text{ADT} \times N \times L)$$

Accident rate per million entering vehicles (for an intersection):

$$R = (A \times 10^6) / (365 \times \text{ADT} \times N)$$

- A = total number of accidents in the period
- ADT = average daily traffic (vehicles per day)
- N = number of years of the study period
- L = length of the stretch in km

NOTE

The **3 E's** of accident prevention: **Engineering** (better geometric design, skid-resistant surfaces, road furniture, lighting), **Enforcement** (speed limits, licensing, the Motor Vehicles Act) and **Education** (road safety education and publicity). Some authors add a fourth E — **Environment** or **Emergency care**.

BEU / AKU REPEAT QUESTION

- ▶ Distinguish between a collision diagram and a condition diagram. [7 marks]
- ▶ Explain the moving observer method and derive/state its formulae; solve the given data. [14 marks — asked very often]
- ▶ Explain the significance of the 85th and 98th percentile speeds. [2 marks]
- ▶ What is a desire line diagram? What are the uses of an O-D study? [7 marks]
- ▶ State the causes of road accidents and the measures for their prevention. [7 marks]

3.4 Traffic flow characteristics and capacity

3.4.1 The three fundamental parameters

FUNDAMENTAL RELATIONS

$$q = k \times \bar{v}_s \text{ (the fundamental relation of traffic flow)}$$

$$\text{Time mean speed: } \bar{v}_t = (1/n) \sum v_i \text{ (arithmetic mean of the spot speeds)}$$

$$\text{Space mean speed: } \bar{v}_s = n / \sum (1/v_i) \text{ (harmonic mean of the spot speeds)}$$

$$\bar{v}_t = \bar{v}_s + \sigma_s^2 / \bar{v}_s \Rightarrow \text{time mean speed} \geq \text{space mean speed, always}$$

$$\text{Time headway } h = 3600 / q \text{ (s)} \quad \text{Space headway (spacing) } s = 1000 / k \text{ (m)}$$

- q = flow or volume in veh/h
- k = density in veh/km
- $\bar{v}_s$ = space mean speed in km/h — THIS is the speed that must be used in $q = k v$
- σ_s^2 = variance of the spot speeds about the space mean speed

SOLVED NUMERICAL — Time mean speed and space mean speed

Q. The spot speeds of five vehicles observed at a section are 30, 40, 50, 60 and 70 km/h. Compute the time mean speed and the space mean speed. If the density is 25 veh/km, find the flow.

Solution:

Step 1 — time mean speed (arithmetic mean)

$$\bar{v}_t = (30 + 40 + 50 + 60 + 70)/5 = 250/5 = 50.0 \text{ km/h}$$

Step 2 — space mean speed (harmonic mean)

$$\sum (1/v_i) = 1/30 + 1/40 + 1/50 + 1/60 + 1/70$$

$$= 0.03333 + 0.02500 + 0.02000 + 0.01667 + 0.01429 = 0.10929$$

$$\bar{v}_s = n / \sum (1/v_i) = 5 / 0.10929 = 45.75 \text{ km/h}$$

Step 3 — flow

$$q = k \times \bar{v}_s = 25 \times 45.75 = 1143.8 \approx 1144 \text{ veh/h}$$

$\bar{v}_t = 50.0 \text{ km/h}$; $\bar{v}_s = 45.75 \text{ km/h}$; $q \approx 1144 \text{ veh/h}$. Note that the space mean speed is the smaller of the two, as it always must be.

3.4.2 Greenshields' model and the fundamental diagrams

Greenshields assumed the simplest possible relation — a **linear** one — between speed and density. From it the flow-density and speed-flow relations follow as parabolas.

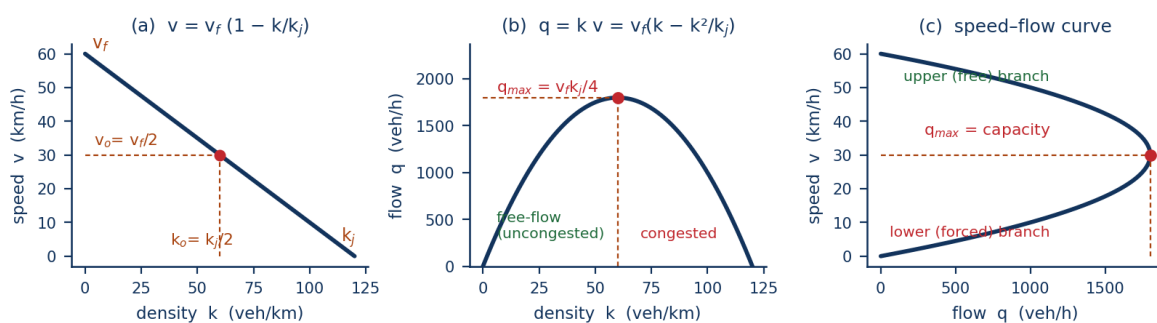
Fundamental diagrams of traffic flow (Greenshields' linear model)

Fig. 3.4 The three fundamental diagrams of traffic flow for Greenshields' linear speed-density model.

GREENSHIELDS' MODEL

$$v = v_f (1 - k / k_j) \text{ [linear speed-density]}$$

$$q = k v = v_f (k - k^2 / k_j) \text{ [parabolic flow-density]}$$

$$\text{Maximum flow (capacity): } q_{\max} = v_f k_j / 4$$

$$\text{which occurs at } k_o = k_j / 2 \text{ and } v_o = v_f / 2$$

- v_f = free (mean) speed, i.e. the speed as the density tends to zero
- k_j = jam density, i.e. the density when the vehicles are bumper to bumper and the speed is zero
- The upper branch of the speed-flow curve is free flow, the lower branch is congested (forced) flow; the same flow can occur at two different speeds

SOLVED NUMERICAL — Greenshields' model — capacity and the two possible speeds

Q. The free speed on a road is 60 km/h and the jam density is 120 veh/km. Assuming Greenshields' linear model, determine (a) the maximum flow and the speed and density at which it occurs, and (b) the two possible speeds and densities at which the flow is 1,500 veh/h.

Solution:

Step 1 — maximum flow

$$q_{\max} = v_f k_j / 4 = (60 \times 120) / 4 = 7200 / 4 = \mathbf{1800 \text{ veh/h}}$$

$$\text{It occurs at } k_o = k_j / 2 = 60 \text{ veh/km and } v_o = v_f / 2 = 30 \text{ km/h}$$

Step 2 — express the flow in terms of speed

$$\text{From } v = v_f (1 - k / k_j): k = k_j (1 - v / v_f) = 120 (1 - v / 60)$$

$$q = k v = 120 v (1 - v / 60) = 120 v - 2 v^2$$

Step 3 — put $q = 1500$ and solve the quadratic

$$120 v - 2 v^2 = 1500 \Rightarrow 2 v^2 - 120 v + 1500 = 0 \Rightarrow v^2 - 60 v + 750 = 0$$

$$v = [60 \pm \sqrt{(3600 - 3000)}] / 2 = [60 \pm \sqrt{600}] / 2 = (60 \pm 24.49) / 2$$

$$v = 42.25 \text{ km/h or } v = 17.75 \text{ km/h}$$

Step 4 — the corresponding densities

$$k = q / v = 1500 / 42.25 = 35.5 \text{ veh/km (free-flow branch)}$$

$$k = q / v = 1500 / 17.75 = 84.5 \text{ veh/km (congested branch)}$$

$q_{\max} = 1800 \text{ veh/h}$ at $v = 30 \text{ km/h}$ and $k = 60 \text{ veh/km}$. A flow of 1500 veh/h occurs either at 42.25 km/h with $k = 35.5 \text{ veh/km}$ (free flow) or at 17.75 km/h with $k = 84.5 \text{ veh/km}$ (congested flow).

3.4.3 Capacity and level of service

Term	Definition
Basic capacity	The maximum number of vehicles that can pass a point in unit time under the most ideal roadway and traffic conditions. It is a theoretical value and is the same for all roads of the same type.
Possible capacity	The maximum number of vehicles that can pass under the prevailing roadway and traffic conditions. It may be far lower than the basic capacity.
Practical (design) capacity	The maximum number of vehicles that can pass without unreasonable delay or restriction to the driver's freedom — the value actually used in design. It lies between the possible and the basic capacity.
Level of service (LOS)	A qualitative measure, on a scale A to F, of the operating conditions experienced by the driver. A = free flow, E = at capacity, F = forced or breakdown flow. It is measured by the volume-to-capacity ratio (v/c) and by the operating speed.

Table 3.6 Types of capacity and the level of service.

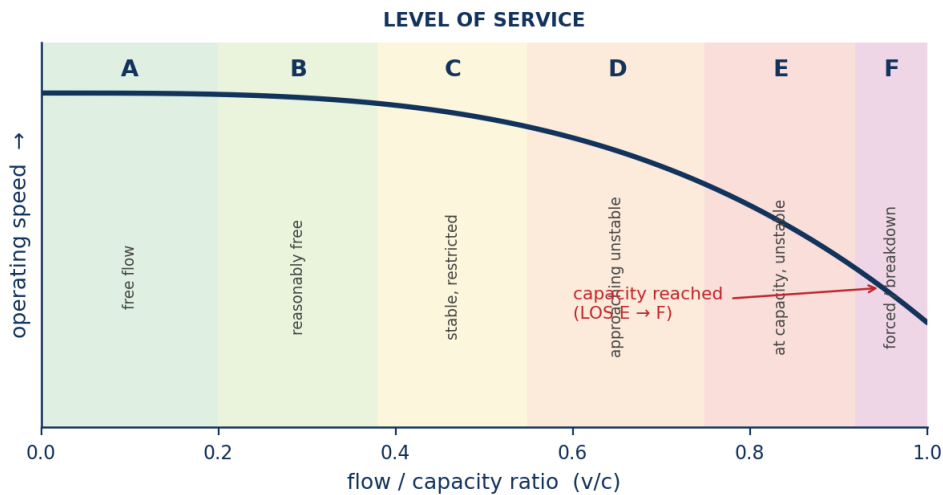


Fig. 3.5 Levels of service A to F against the volume-to-capacity ratio.

CAPACITY

Theoretical (basic) capacity from the spacing of vehicles:

$$C = 1000 V / S \text{ (vehicles per hour per lane)}$$

$$\text{Capacity from the time headway: } C = 3600 / H_t$$

$$\text{Minimum safe space headway: } S = SSD + \text{length of the vehicle}$$

- C = capacity in veh/h per lane
- V = speed in km/h
- S = average centre-to-centre spacing (space headway) of the vehicles in m
- H_t = average time headway in seconds

SOLVED NUMERICAL — Theoretical capacity of a lane from the safe stopping distance

Q. Calculate the theoretical maximum capacity of a single lane if the speed is 60 km/h, the average length of a vehicle is 6 m, the reaction time is 2.5 s and $f = 0.36$. Also find the capacity if the observed average time headway is 2.0 s.

Solution:

Step 1 — stopping sight distance at 60 km/h

$$\text{Lag distance} = 0.278 \times 60 \times 2.5 = 41.70 \text{ m}$$

$$\text{Braking distance} = 60^2 / (254 \times 0.36) = 3600 / 91.44 = 39.37 \text{ m}$$

$$SSD = 41.70 + 39.37 = 81.07 \text{ m}$$

Step 2 — minimum safe space headway

$$S = SSD + \text{length of the vehicle} = 81.07 + 6.0 = 87.07 \text{ m}$$

Step 3 — capacity from the spacing

$$C = 1000 V / S = (1000 \times 60) / 87.07 = 60000 / 87.07 = 689 \text{ veh/h per lane}$$

Step 4 — capacity from the observed time headway

$$C = 3600 / H_t = 3600 / 2.0 = 1800 \text{ veh/h per lane}$$

Theoretical capacity based on the safe stopping distance = 689 veh/h/lane; based on the observed headway of 2 s = 1800 veh/h/lane. The huge difference shows that drivers in practice keep far shorter headways than the safe-stopping criterion would allow.

IRC STANDARD VALUES

Design service volumes commonly quoted for rural roads (IRC): single-lane road 2,000 PCU/day; intermediate lane 6,000 PCU/day; two-lane road 15,000 PCU/day. For urban roads IRC 106-1990 gives the capacity of a two-lane one-way road as about 2,400 PCU/h and of a two-lane two-way road as about 1,500 PCU/h.

BEU / AKU REPEAT QUESTION

- ▶ Define basic, possible and practical capacity. [7 marks]
- ▶ Explain the terms level of service, time mean speed and space mean speed. [7 marks]
- ▶ Derive the relation $q = k v$ and explain the speed-flow-density diagrams. [7 marks]
- ▶ Using Greenshields' model, find the capacity for the given free speed and jam density. [7 marks]

▶ Reference: Kadiyali — *Traffic Engineering and Transport Planning*, chapters on traffic characteristics, traffic studies and traffic flow theory; Khanna, Justo & Veeraragavan, chapter 'Traffic Engineering'; IRC 9-1972 (traffic census) and IRC 106-1990 (capacity of urban roads).

UNIT 4 | Traffic Regulation and Control - Intersections • Parking • Highway Lighting

Syllabus allotment: 8 hours • about 19 % of the paper • heavy on numericals

4.1 Traffic regulation

Traffic regulations are the legal instruments by which traffic is controlled. In India the governing legislation is the **Motor Vehicles Act, 1988** (which replaced the Act of 1939) together with the Central Motor Vehicle Rules and the State rules. Regulations fall under three heads:

Regulation of	What is regulated
The driver	Minimum age; the driving licence and its renewal; medical fitness and eyesight; the professional driver's badge; the number of hours of continuous driving; prohibition of driving under the influence of drink or drugs; penalties and the suspension or cancellation of the licence.
The vehicle	Registration and the certificate of fitness; the maximum dimensions (width 2.5 m, height 3.8 m, length 11 m for a single unit); the maximum laden weight and axle load (single axle 10.2 t); brakes, lights, horn, silencer, mirror, wipers and other essential equipment; insurance; pollution-under-control certificate.
The flow of traffic	Speed limits (maximum and sometimes minimum, and separate limits for different classes of vehicle); one-way regulation; keep-left rule and lane discipline; the right of way at intersections; restrictions on overtaking , on turning and on the use of horns ; parking regulations; restriction of the entry of heavy vehicles to certain hours; regulation of pedestrians and of slow-moving traffic.

Table 4.1 The three heads of traffic regulation.

4.2 Traffic control devices

A traffic control device conveys a message to the road user. The four devices are **traffic signs**, **road markings**, **traffic islands** and **traffic signals**. Every device must fulfil four requirements: it must **attract attention**, convey a **simple and clear meaning**, command **respect**, and give **adequate time** for the response.

4.2.1 Traffic signs (IRC 67)

Classification of traffic signs (IRC 67) — shape identifies the class



Fig. 4.1 The three classes of traffic sign. The SHAPE identifies the class, so the driver knows the kind of message before he can read it.

Class	Shape and colour	Examples
Mandatory / regulatory signs	Circular, with a red border (some are blue circles); the octagonal STOP sign and the inverted triangle GIVE WAY sign belong to this class	Stop, Give way, No entry, One way, Speed limit, No parking, Compulsory turn left, Weight limit
Cautionary / warning signs	Equilateral triangle with the apex upwards and a red border	Right hand curve, Cross road, Narrow bridge, Steep descent, Pedestrian crossing, School ahead, Falling rocks

Class	Shape and colour	Examples
Informatory / guide signs	Rectangular	Direction and place identification signs, Destination and distance signs, Route markers, Public facility signs (petrol pump, hospital, parking, first aid)

Table 4.2 Classification of traffic signs. Disobeying a MANDATORY sign is a legal offence; a cautionary sign only warns.

4.2.2 Road markings (IRC 35)

- **Longitudinal markings** — **centre line** (a broken line on a two-lane road), **lane lines**, **no-passing zone markings** (a continuous line, or a continuous line with a broken line beside it, in which case the traffic on the side of the broken line may cross), **warning lines**, **edge lines** (marking the edge of the carriageway) and **carriageway-width transition markings**.
- **Transverse markings** — **stop lines**, **pedestrian crossings (zebra crossings)**, **give-way lines** and **parking space limits**.
- **Object markings** — markings on physical obstructions within or adjacent to the carriageway (piers, kerbs, islands), and hazard markers.
- **Word messages and arrows** — STOP, SLOW, BUS, and lane-use arrows.
- Markings are made in white, except no-parking and continuous-line prohibitions which use yellow.

4.2.3 Traffic islands

Type of island	Purpose
Divisional island	Separates opposing flows of traffic, or separates the traffic on a minor road at a junction, discouraging wrong turns.
Channelizing island	Guides the traffic into the proper channel through the intersection and controls the angle and point of merging.
Pedestrian refuge island	Gives the pedestrian a safe place to wait part way across a wide road.
Rotary island	The large central island of a rotary intersection around which all the traffic circulates.

Table 4.3 Types of traffic island.

4.3 Traffic signals

4.3.1 Types, advantages and limitations

- **Fixed-time (pre-timed) signals** — the cycle and the splits are fixed in advance; simple and cheap, but cannot adjust to a change in flow.
- **Vehicle-actuated signals** — detectors in the carriageway extend the green as long as vehicles keep arriving, up to a maximum; efficient but costly.
- **Semi-actuated signals** — detectors only on the minor road; the major road gets the green until a vehicle demands it on the minor road.
- **Manually operated signals** — operated by a police officer.

Advantages of signals	Limitations of signals
A definite and orderly movement is assured; the crossing conflicts are separated in TIME instead of in space	Rear-end collisions may increase
The capacity of the intersection is increased and the delay reduced compared with an uncontrolled junction	Delay is caused even when the cross traffic is absent (in a fixed-time signal)
Right-angled and pedestrian accidents are reduced	Improper design or unwarranted installation causes disrespect and violation
Cheaper than manual control or than grade separation; can give priority to the major road by co-ordination	Failure of the equipment or a power cut creates confusion

Table 4.4 Advantages and limitations of traffic signals.

4.3.2 Definitions — learn these exactly

Term	Definition
Cycle	One complete sequence of the signal indications.
Cycle length (C)	The time in seconds for one complete cycle.
Interval	Any part of the cycle during which the signal indication does not change.
Change interval (amber)	The yellow interval between the green and the red, warning the driver of the impending change; usually 2 to 4 s.
Clearance (all-red) interval	The interval during which all approaches get red, to clear the vehicles already in the intersection.
Green interval (G)	The actual duration of the green indication.
Effective green (g)	The time actually available for discharge at the saturation flow rate, after allowing for the lost time.
Phase	The part of the cycle allotted to one or more movements receiving the right of way simultaneously.
Lost time (L)	The time in a cycle that is not used for the effective discharge of traffic — the start-up delay of each phase (about 2 s) plus the all-red / inter-green intervals.
Saturation flow (s)	The maximum rate of discharge from an approach during the green, in PCU/h. It depends mainly on the approach width.
Offset	The time difference between the start of the green of one signal and that of the next signal in a co-ordinated system.

Table 4.5 Signal terminology.

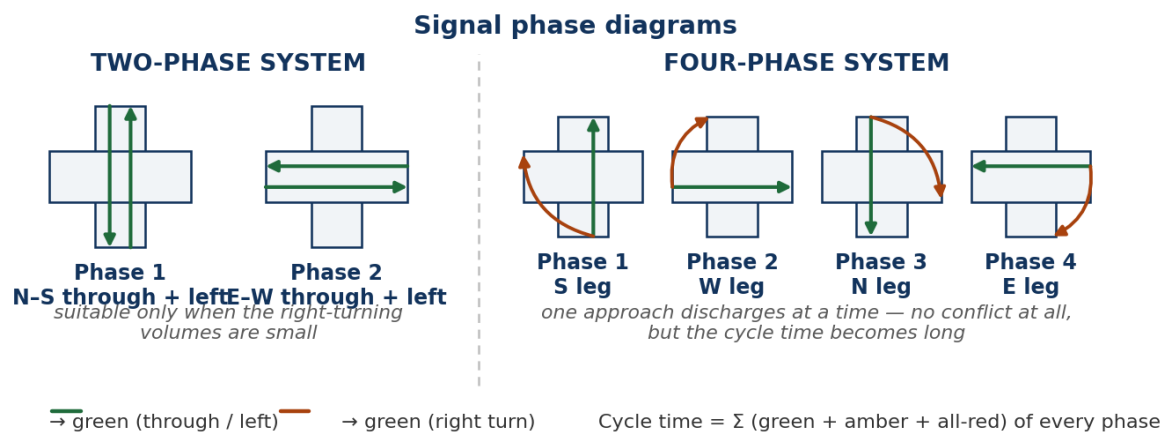


Fig. 4.2 Two-phase and four-phase signal systems. A two-phase system is used where the right-turning volumes are small; more phases mean fewer conflicts but a longer cycle and more lost time.

4.3.3 Webster's method — the optimum cycle time

Webster's method is a **rational method** that minimises the total delay to all the vehicles at the intersection. It is the method most often asked in the examination.

WEBSTER'S METHOD

$$y_i = q_i / s_i \quad Y = \Sigma y_i \text{ (summed over the phases)}$$

$$\text{Total lost time per cycle: } L = 2n + R$$

$$\text{OPTIMUM CYCLE TIME: } C_0 = (1.5L + 5) / (1 - Y)$$

$$\text{Effective green of phase } i: g_i = (y_i / Y) (C_0 - L)$$

- q_i = design (normal) flow on the critical approach of phase i , in PCU/h
- s_i = saturation flow of that approach, in PCU/h
- n = number of phases; the 2 s per phase is the start-up lost time
- R = total all-red (inter-green) time per cycle, in seconds
- C_0 is rounded off to the nearest 5 s for convenience of setting
- The method is valid only while $Y < 1$; if $Y \geq 1$ the intersection is oversaturated and signals alone cannot help

SOLVED NUMERICAL — Design of a two-phase signal by Webster's method

Q. A four-arm intersection is to be signalised with a two-phase fixed-time signal. The design hourly flow and the saturation flow on the critical approach of each phase are:

Phase 1 (north-south): $q_1 = 750$ PCU/h, $s_1 = 2,400$ PCU/h

Phase 2 (east-west): $q_2 = 600$ PCU/h, $s_2 = 2,000$ PCU/h

The amber (inter-green) time is 3 s per phase. Determine the optimum cycle time and the signal settings by Webster's method.

Solution:

Step 1 — the flow ratios

$$y_1 = q_1/s_1 = 750/2400 = 0.3125$$

$$y_2 = q_2/s_2 = 600/2000 = 0.3000$$

$$Y = y_1 + y_2 = 0.3125 + 0.3000 = 0.6125 \text{ (less than 1, so the design is feasible)}$$

Step 2 — total lost time per cycle

Number of phases $n = 2$; total inter-green (amber) $R = 2 \times 3 = 6$ s

$$L = 2n + R = 2 \times 2 + 6 = 10 \text{ s}$$

Step 3 — optimum cycle time

$$C_0 = (1.5L + 5)/(1 - Y) = (1.5 \times 10 + 5)/(1 - 0.6125)$$

$$= (15 + 5)/0.3875 = 20/0.3875 = 51.6 \text{ s}$$

Adopt a cycle time $C = 50$ s (rounded to the nearest 5 s)

Step 4 — apportion the effective green

Total effective green available = $C - L = 50 - 10 = 40$ s

$$g_1 = (y_1/Y)(C - L) = (0.3125/0.6125) \times 40 = 0.5102 \times 40 = 20.4 \text{ s} \rightarrow \text{say } 20 \text{ s}$$

$$g_2 = (y_2/Y)(C - L) = (0.3000/0.6125) \times 40 = 0.4898 \times 40 = 19.6 \text{ s} \rightarrow \text{say } 20 \text{ s}$$

Step 5 — the signal setting

Phase 1: green 20 s + amber 3 s = 23 s

Phase 2: green 20 s + amber 3 s = 23 s

Total = 23 + 23 = 46 s, plus the start-up lost time of $2 \times 2 = 4$ s, giving the cycle of 50 s

Red time for phase 1 = $C - (\text{green}_1 + \text{amber}_1) = 50 - 23 = 27$ s ; similarly for phase 2

Optimum cycle time ≈ 52 s, adopted as $C = 50$ s. Settings — Phase 1: green 20 s, amber 3 s, red 27 s ; Phase 2: green 20 s, amber 3 s, red 27 s.

4.3.4 IRC (approximate) method and the pedestrian requirement

The IRC approximate method fixes the cycle from the **pedestrian crossing requirement** and then apportions the green times to the approaches in the **ratio of the approach volumes**.

IRC APPROXIMATE METHOD

$$\text{Pedestrian green time: } G_p = 7 + W / 1.2$$

Green time apportioned to an approach:

$$G_1 = [q_1 / (q_1 + q_2)] \times (\text{total available green})$$

- W = width of the carriageway to be crossed, in m
- 1.2 m/s = the assumed walking speed of a pedestrian
- 7 s = the initial interval, for the pedestrian to start off
- q_1, q_2 = the approach volumes of the two phases

SOLVED NUMERICAL — Pedestrian green time

Q. Determine the pedestrian green time required to cross a road 15 m wide.

Solution:

Step 1 — walking time

$$= W / 1.2 = 15 / 1.2 = 12.5 \text{ s}$$

Step 2 — add the initial interval

$$G_p = 7 + 12.5 = 19.5 \text{ s}$$

Pedestrian green time ≈ 20 s. The cycle must be long enough to accommodate this in the appropriate phase.

4.3.5 Co-ordination of traffic signals

System	How it works	Comment
Simultaneous system	All the signals along the road show the same indication at the same time.	Simplest, but encourages speeding between the signals and gives poor progression
Alternate system	Alternate signals, or groups of signals, show opposite indications.	Better than the simultaneous system, but needs nearly equal spacing of the signals
Simple progressive system	The green periods are offset one from the next in accordance with the intended speed of progression, but the cycle and the splits are fixed.	Gives a genuine 'green wave' for the design speed
Flexible progressive system	The offsets, the cycle length and the splits can all be varied automatically by a master controller according to the traffic demand.	The best system; used on important urban arterials

Table 4.6 The four systems of signal co-ordination.

Time-space diagram of a linearly co-ordinated (progressive) signal system

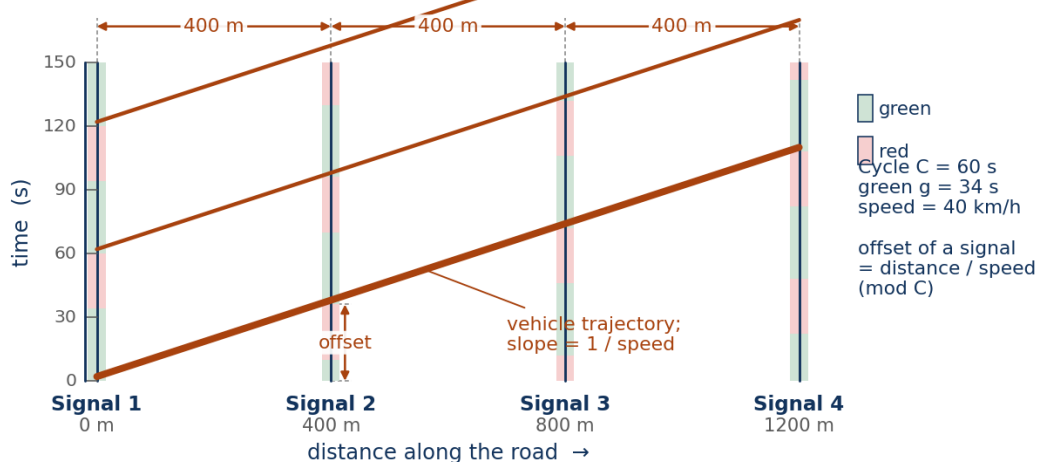


Fig. 4.3 Time-space diagram of a progressive (co-ordinated) signal system. The offset of each signal is set so that a vehicle travelling at the design speed meets a green at every signal.

SIGNAL CO-ORDINATION

Offset of a signal in a progressive system = (distance from the first signal) / (speed of progression)

Ideal spacing of signals for the alternate system: spacing = $v \times C / 2$

- v = speed of progression in m/s; C = cycle time in s
- The offset is taken modulo the cycle time

BEU / AKU REPEAT QUESTION

- Explain Webster's method of signal design and design a two-phase signal for the given data. [14 marks — appears in almost every paper]
- Define cycle, interval, phase, lost time and saturation flow. [7 marks]
- What are the different systems of signal co-ordination? Explain with a time-space diagram. [7 marks]
- State the advantages and limitations of traffic signals. [7 marks]
- Classify traffic signs with examples. [7 marks]

4.4 Design of road intersections

4.4.1 Types of intersection

Types of at-grade intersection

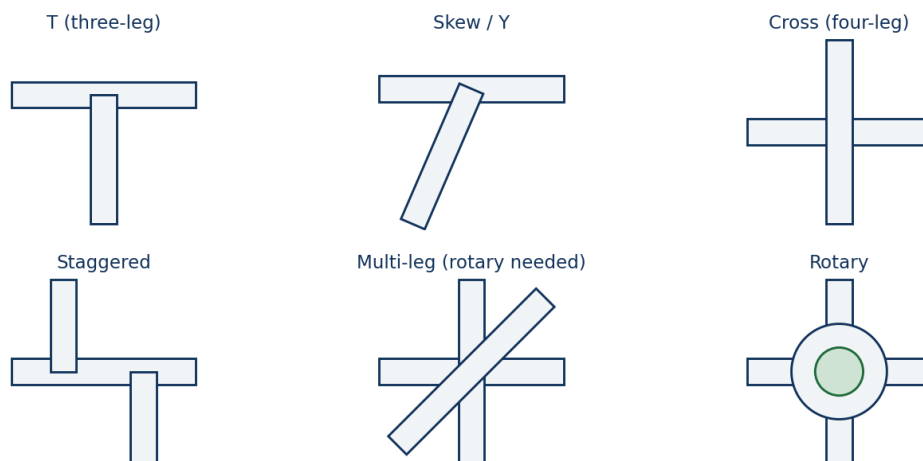
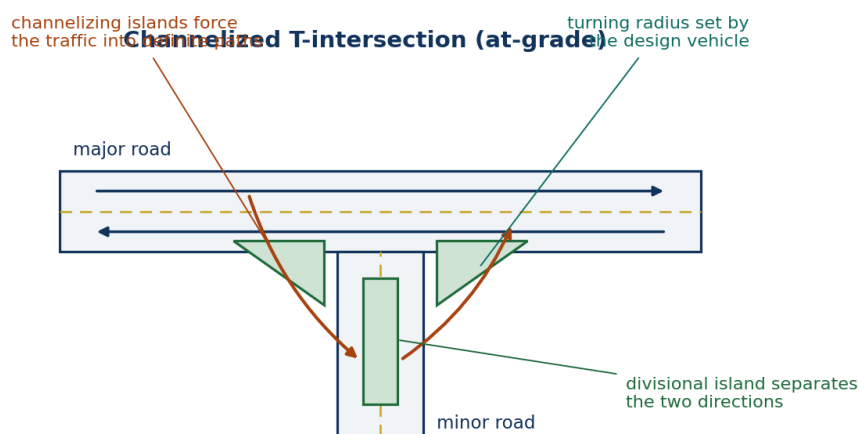


Fig. 4.4 Types of at-grade intersection. As the number of legs increases the number of conflict points rises steeply, and a rotary or a grade separation becomes necessary.

Conflicts at an intersection are of three kinds — **crossing, merging and diverging**. The crossing conflict is the most dangerous; the whole art of intersection design is to remove the crossing conflicts or to convert them into merging and diverging conflicts.

4.4.2 Channelization

Channelization is the separation or the regulation of the conflicting traffic movements into definite paths of travel by means of traffic islands, pavement markings and turning roadways.



CHANNELIZATION — regulating conflicting movements with islands, markings and turning roadways; it reduces the conflict area and fixes the angle and point of merging

Fig. 4.5 A channelized T-intersection.

Objects of channelization: to reduce the area of conflict; to fix the **angle and the point of merging or crossing**; to control the speed of the traffic entering the intersection; to protect the vehicles waiting to turn; to provide a refuge for pedestrians; to discourage the prohibited movements; and to provide a place for the signs and signals.

4.4.3 Rotary intersection

Rotary (roundabout) — an enlarged intersection in which the vehicles from the converging roads are compelled to move round a large central island in one direction (anticlockwise in India) before they can weave out into the road of their choice.

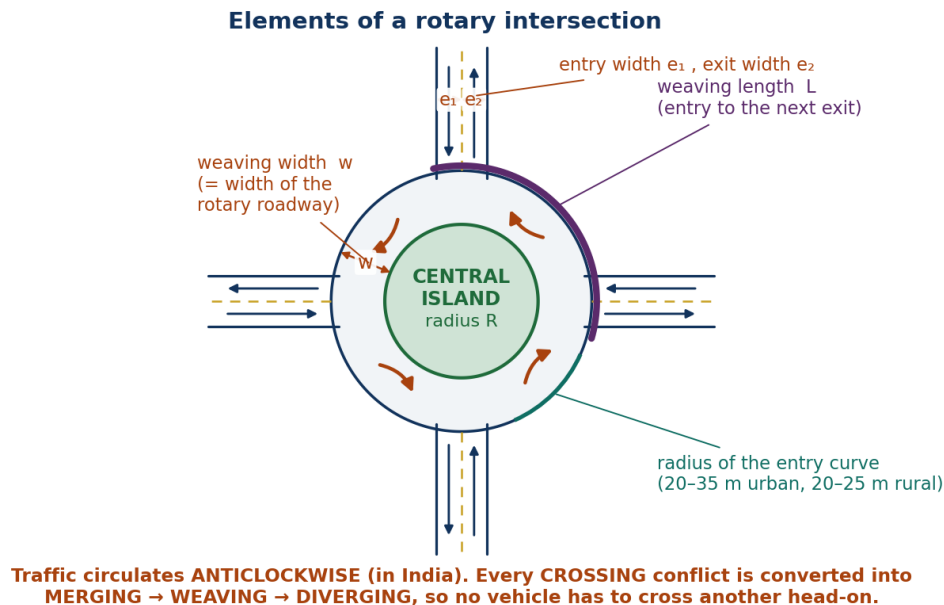


Fig. 4.6 Elements of a rotary intersection.

Advantages of a rotary	Disadvantages of a rotary
All the crossing conflicts are converted into merging, weaving and diverging — the most dangerous conflict is eliminated	Requires a very large area of land, so it is unsuitable in built-up areas
Self-governing; needs no police control or signals, and hence no recurring cost	Not suitable when the traffic volume is very high — the capacity is limited and long queues form
Traffic proceeds at a uniform low speed, so accidents are less severe	Not suitable when the pedestrian volume is high, or where any leg has a very heavy flow
Suitable when the number of legs is between 4 and 7 and the traffic is fairly evenly distributed	Unsuitable where a rail level crossing or a steep gradient is close by; needs a considerable amount of maintenance

Table 4.7 Rotary intersection — advantages and disadvantages.

Design elements of a rotary (IRC 65)

Element	Recommended value
Design speed	40 km/h for rural roads, 30 km/h for urban roads
Shape of the central island	Circular for four equal legs; elliptical, tangent or turbine shapes where the legs are unequal
Radius of the entry curve	20 to 35 m for urban roads; 20 to 25 m for rural roads
Radius of the exit curve	Usually 1.5 to 2.0 times the entry radius, so that the vehicles leave quickly
Radius of the central island	About 1.33 times the radius of the entry curve
Weaving angle	Should be small — of the order of 15° to 20°; a small weaving angle means a longer, safer weaving section
Weaving length	Minimum 45 m for a design speed of 40 km/h and 30 m for 30 km/h
Width of the carriageway at entry and exit	Entry 5 to 7 m for a two-lane road; the exit is usually made wider than the entry

Element	Recommended value
Width of the rotary roadway (weaving section)	The larger of (i) the width of the widest single entry + 3.5 m and (ii) the average of the entry widths + 3.5 m

Table 4.8 Design elements of a rotary as per IRC 65.

ROTARY CAPACITY (TRL / WARDROP FORMULA)

Practical capacity of a weaving section of a rotary:

$$Q_p = 280 w [1 + e/w] [1 - p/3] / [1 + w/L]$$

$$p = (b + c) / (a + b + c + d)$$

- Q_p = practical capacity of the weaving section, in PCU per hour
- w = weaving width in m (valid for $6 \leq w \leq 18$ m)
- e = average width of the entry and the exit = $(e_1 + e_2)/2$ (valid for $0.4 \leq e/w \leq 1.0$)
- L = length of the weaving section in m (valid for $0.12 \leq w/L \leq 0.4$)
- p = proportion of the weaving traffic (valid for $0.4 \leq p \leq 1.0$)
- a = traffic entering and leaving without weaving (left turning), b and c = the two weaving streams, d = the non-weaving stream

SOLVED NUMERICAL — Capacity of the weaving section of a rotary

Q. A rotary has a weaving section 12 m wide and 36 m long. The entry width is 8 m and the exit width 10 m. The traffic approaching the weaving section is: left-turning (non-weaving) $a = 400$ PCU/h, the two weaving streams $b = 600$ PCU/h and $c = 500$ PCU/h, and the non-weaving stream $d = 300$ PCU/h. Determine the practical capacity of the weaving section.

Solution:

Step 1 — the geometric parameters

$$w = 12 \text{ m} ; L = 36 \text{ m} ; e = (e_1 + e_2)/2 = (8 + 10)/2 = 9 \text{ m}$$

Step 2 — check that the formula is applicable

$$w = 12 \text{ m} \text{ — lies between 6 and 18 m (OK)}$$

$$e/w = 9/12 = 0.75 \text{ — lies between 0.4 and 1.0 (OK)}$$

$$w/L = 12/36 = 0.333 \text{ — lies between 0.12 and 0.4 (OK)}$$

Step 3 — proportion of weaving traffic

$$p = (b + c)/(a + b + c + d) = (600 + 500)/(400 + 600 + 500 + 300)$$

$$= 1100/1800 = 0.611 \text{ — lies between 0.4 and 1.0 (OK)}$$

Step 4 — substitute in the capacity formula

$$Q_p = 280 \times 12 \times (1 + 0.75) \times (1 - 0.611/3) / (1 + 0.333)$$

$$= 3360 \times 1.75 \times (1 - 0.2037) / 1.3333$$

$$= 3360 \times 1.75 \times 0.7963 / 1.3333$$

$$= 4682.2 / 1.3333 = 3511.7 \text{ PCU/h}$$

Step 5 — compare with the demand

Total traffic through the weaving section = 1800 PCU/h, which is well below the capacity of 3512 PCU/h, so the rotary is adequate.

Practical capacity of the weaving section $\approx 3,512$ PCU/h

4.4.4 Grade-separated intersections (interchanges)

Common types of grade-separated interchange

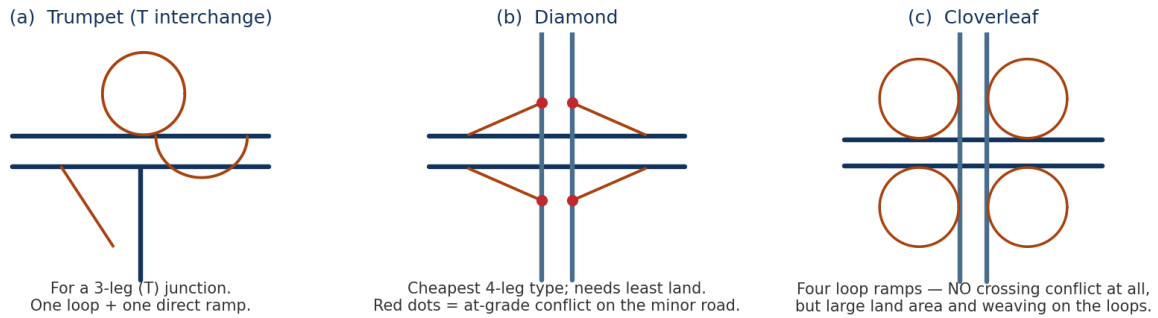


Fig. 4.7 Common types of grade-separated interchange.

Type	Where used
Trumpet	Three-leg (T) junctions; one loop ramp and one direct ramp
Diamond	Four-leg junctions where land is limited and the minor road flow is light; the cheapest four-leg type, but some at-grade conflict remains on the minor road
Cloverleaf	Four-leg junctions carrying heavy traffic; four loop ramps eliminate every crossing conflict, but a large area is needed and weaving occurs on the loops
Rotary interchange	Where several roads meet; a rotary is provided on one level with the through road grade-separated

Table 4.9 Types of interchange.

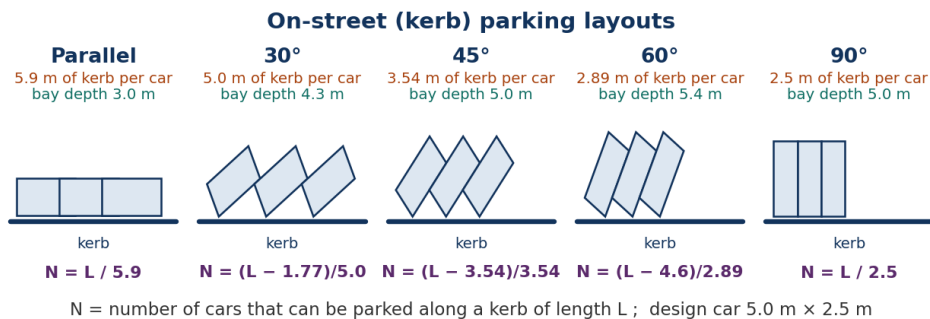
BEU / AKU REPEAT QUESTION

- What is a rotary intersection? State its advantages and disadvantages, and explain its design elements as per IRC. [14 marks — asked very often]
- Determine the capacity of the weaving section of a rotary for the given data. [7 marks]
- What is channelization? State its objects. [7 marks]
- Explain the different types of grade-separated interchange with sketches. [7 marks]
- What are the three types of conflict at an intersection? [2 marks]

4.5 Design of parking facilities

4.5.1 Types of parking

- **On-street (kerb) parking** — parallel parking, and angle parking at 30°, 45°, 60° or 90° to the kerb.
- **Off-street parking** — surface car parks, multi-storey parking garages, underground parking, mechanical (automated) parking.



As the parking angle increases, MORE cars are parked per metre of kerb, but a GREATER width of carriageway is lost

Fig. 4.8 On-street parking layouts, with the kerb length required per car and the formula for the number of cars N.

Layout	Kerb length per car	Depth of the bay	N for a kerb length L
Parallel	5.9 m	3.0 m	$N = L / 5.9$
30°	5.0 m	4.3 m	$N = (L - 1.77) / 5.0$

Layout	Kerb length per car	Depth of the bay	N for a kerb length L
45°	3.54 m	5.0 m	$N = (L - 3.54) / 3.54$
60°	2.89 m	5.4 m	$N = (L - 4.6) / 2.89$
90° (right angle)	2.5 m	5.0 m	$N = L / 2.5$

Table 4.10 Kerb parking — based on a design car 5.0 m × 2.5 m. Parallel parking accommodates the FEWEST cars per metre of kerb but obstructs the carriageway least; 90° parking accommodates the most but takes the greatest width.

SOLVED NUMERICAL — Number of cars that can be parked along a kerb

Q. A kerb length of 60 m is available for parking. Determine the number of cars that can be parked if the parking is (a) parallel, (b) at 45° and (c) at 90° to the kerb. Comment on the result.

Solution:

(a) Parallel parking

$$N = L/5.9 = 60/5.9 = 10.17 \rightarrow \mathbf{10 \text{ cars}}$$

(b) 45° angle parking

$$N = (L - 3.54)/3.54 = (60 - 3.54)/3.54 = 56.46/3.54 = 15.95 \rightarrow \mathbf{15 \text{ cars}}$$

(c) 90° (right-angle) parking

$$N = L/2.5 = 60/2.5 = \mathbf{24 \text{ cars}}$$

Comment

Right-angle parking accommodates 2.4 times as many cars as parallel parking in the same kerb length, but it needs a bay depth of 5.0 m against 3.0 m and the reversing manoeuvre obstructs the through traffic. Parallel parking is therefore preferred on narrow but busy streets.

(a) 10 cars (b) 15 cars (c) 24 cars

4.5.2 Parking statistics

PARKING STATISTICS

Parking accumulation = number of vehicles parked at a given instant

Parking volume = total number of vehicles parked during a given period

Parking load = Σ (accumulation × time interval) [in vehicle-hours]

Average parking duration = parking load / parking volume

Parking turnover = parking volume / number of parking bays

Parking index (occupancy) = (parking load / parking capacity) × 100 %

Parking capacity = (number of bays) × (duration of the study) [vehicle-hours]

SOLVED NUMERICAL — Parking load, duration, turnover and parking index

Q. An off-street parking facility with 20 bays was surveyed from 8 a.m. to 4 p.m. During the 8-hour period 160 vehicles used the facility and the parking load worked out to 120 vehicle-hours. Determine the average parking duration, the parking turnover, the parking capacity and the parking index.

Solution:

Step 1 — average parking duration

= parking load / parking volume = $120/160 = 0.75$ hour = 45 minutes

Step 2 — parking turnover

= parking volume / number of bays = $160/20 = 8$ vehicles per bay in the 8-hour period

= $8/8 = 1$ vehicle per bay per hour

Step 3 — parking capacity (the space-hours available)

= number of bays × duration of the study = $20 \times 8 = 160$ vehicle-hours

Step 4 — parking index

= (parking load / parking capacity) × 100 = $(120/160) \times 100 = 75\%$

Average duration = 45 min ; turnover = 8 vehicles per bay (1 per bay per hour) ; capacity = 160 vehicle-hours ; parking index = 75 %. An index of 75 % indicates the facility is well used but not saturated.

NOTE

Ill effects of on-street parking: it reduces the effective width of the carriageway and hence the capacity; the manoeuvring vehicles cause delay and accidents; opening doors endanger cyclists and pedestrians; it obstructs the sight distance at intersections; and it hinders fire-fighting and ambulance access.

4.6 Highway lighting

Need: about a quarter to a third of the accidents occur at night although the night traffic is much less; street lighting reduces night accidents, increases the night capacity, reduces crime and improves the appearance of the town. It is provided at intersections, on bridges, in tunnels, on curves, at pedestrian crossings, in built-up areas and at railway level crossings.

4.6.1 Terms used in lighting

Term	Meaning	Unit
Luminous flux	The light energy radiated per unit time by a source, measured in terms of its visual effect	lumen
Luminous intensity	The luminous flux radiated per unit solid angle in a given direction	candela (cd)
Illumination (illuminance)	The luminous flux received per unit area of the surface; $1 \text{ lux} = 1 \text{ lumen/m}^2$	lux
Luminance (brightness)	The luminous intensity per unit projected area of the surface, as seen by the eye	cd/m^2
Coefficient of utilization	The fraction of the luminous flux of the lamp that actually reaches the road surface	a ratio
Maintenance factor	Allows for the depreciation of the light output due to ageing and to dirt on the fittings; usually 0.8	a ratio

Table 4.11 Lighting terminology.

4.6.2 Design of the spacing of lamps

Layouts for highway lighting (S = spacing of lamps, h = mounting height, W = road width)

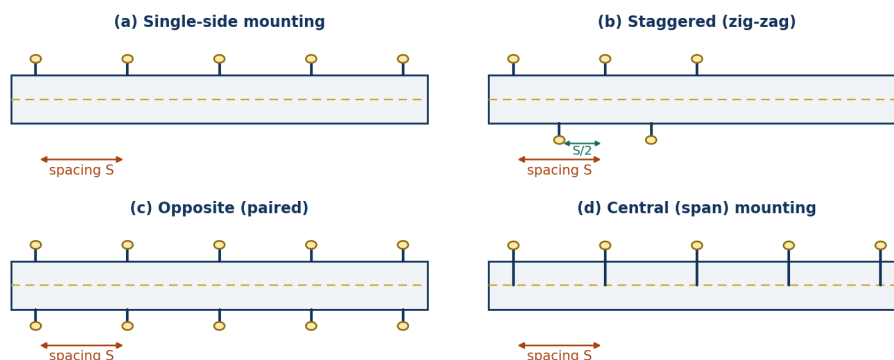


Fig. 4.9 Layouts for highway lighting — single-side, staggered, opposite (paired) and central (span) mounting.

LAMP SPACING

$$\text{Average illumination } E = (F \times CU \times MF) / (W \times S)$$

$$\Rightarrow \text{Spacing of the lamps } S = (F \times CU \times MF) / (E \times W)$$

- E = average illumination required on the road surface, in lux
- F = luminous flux output of one lamp, in lumens
- CU = coefficient of utilization
- MF = maintenance factor (about 0.8)
- W = width of the roadway to be lit, in m
- S = spacing of the lamp posts along the road, in m
- For the OPPOSITE (paired) layout there are two lamps per spacing, so the flux F is doubled; for the STAGGERED layout the successive lamps are $S/2$ apart measured along the road

Layout	When used
Single-side mounting	When the width of the road is less than the mounting height of the lamp
Staggered (zig-zag) mounting	When the road width is 1 to 1.5 times the mounting height
Opposite (paired) mounting	When the road width is more than 1.5 times the mounting height
Central (span) mounting	On divided highways, with the poles in the median; also used for its good appearance

Table 4.12 Choice of lighting layout. Typical mounting height 10 to 12 m and spacing 30 to 50 m on main roads.

SOLVED NUMERICAL — Spacing of street-lighting lamps

Q. Design the spacing of the lamps for a road 7.5 m wide, using lamps of 6,000 lumens mounted on one side only. The average illumination required is 15 lux, the coefficient of utilization is 0.35 and the maintenance factor is 0.8. What would the spacing be if lamps of 9,000 lumens were used instead?

Solution:

Step 1 — spacing with 6,000-lumen lamps

$$S = (F \times CU \times MF)/(E \times W)$$

$$= (6000 \times 0.35 \times 0.8)/(15 \times 7.5)$$

$$= 1680/112.5 = 14.93 \text{ m} \rightarrow \text{adopt } \mathbf{15 \text{ m}}$$

Step 2 — spacing with 9,000-lumen lamps

$$S = (9000 \times 0.35 \times 0.8)/(15 \times 7.5) = 2520/112.5 = 22.4 \text{ m} \rightarrow \text{adopt } \mathbf{22 \text{ m}}$$

Step 3 — comment

The spacing is directly proportional to the flux of the lamp. Using fewer, brighter lamps reduces the number of poles and the capital cost, but the uniformity of the illumination becomes poorer, so the spacing is usually limited to about 3 to 4 times the mounting height.

Spacing = 15 m with 6,000-lumen lamps and 22 m with 9,000-lumen lamps

BEU / AKU REPEAT QUESTION

- Explain the terms lumen, candela, lux and luminance. [2 marks each]
- Design the spacing of street lighting lamps for the given data. [7 marks]
- Explain the different layouts of highway lighting and state when each is used. [7 marks]
- Define parking accumulation, parking load, parking turnover and parking index. Solve the given parking survey data. [14 marks]
- Compare parallel and 90° parking. How many cars can be parked in ... m of kerb? [7 marks]

► *Reference: Kadiyali — Traffic Engineering and Transport Planning, chapters on traffic control devices, signals, intersections and parking; Khanna, Justo & Veeraragavan, chapter 'Traffic Engineering'; IRC 67 (signs), IRC 35 (markings), IRC 93-1985 (signals), IRC 65-2017 (rotaries), IRC SP 41 (intersections), IRC SP 12 / IS 1944 (lighting).*

UNIT 5 | Pavement Materials

Syllabus allotment: 5 hours • about 12 % of the paper

5.1 Materials used in highway construction

The four groups of materials used in a pavement are **soil** (the subgrade and the embankment), **stone aggregates** (the granular layers and the skeleton of every bound layer), **bituminous binders** (flexible pavements) and **cement and cement concrete** (rigid pavements and stabilised layers). For each material the examination asks the same three things: the **desirable properties**, the **tests** used to measure them, and the **specified limiting values**.

5.2 Soil as a highway material

5.2.1 Desirable properties of a subgrade soil

- **Stability** — adequate and sustained strength under the worst moisture condition it will meet in service.
- **Incompressibility** — freedom from excessive settlement and from consolidation under repeated loading.
- **Permanency of strength** — the strength should not change with changes in moisture and temperature.
- **Minimum change in volume** — low swelling and shrinkage; expansive clays are the worst subgrade material.
- **Good drainage** — the soil should drain freely, so that the moisture does not build up.
- **Ease of compaction** — it should be possible to compact it to a high density with the available plant.

5.2.2 Tests on subgrade soil

Test	What it measures	Where used
California Bearing Ratio (CBR)	The resistance of the soil to the penetration of a standard plunger, expressed as a percentage of the resistance of standard crushed stone	The design input for flexible pavements (IRC 37); the most important test in this subject
Plate bearing test	The pressure-settlement behaviour of the soil in the field; gives the modulus of subgrade reaction K (kg/cm ³ or MPa/m)	The design input for rigid pavements (IRC 58, Westergaard's analysis)
Dynamic cone penetration test	The number of blows required to drive a cone through a given depth	A rapid field test for the relative strength profile of the subgrade
Triaxial compression test	The shear strength parameters c and ϕ , and the modulus of elasticity	The triaxial method of flexible pavement design; research work
Standard Proctor / modified Proctor	The optimum moisture content and the maximum dry density	Field compaction control
Atterberg limits, sieve analysis	Liquid limit, plastic limit, plasticity index and the grain-size distribution	Classification of the soil and the group index
Field density test (sand replacement, core cutter)	The dry density actually achieved in the field	Quality control of compaction

Table 5.1 Tests on subgrade soil.

5.2.3 The CBR test

A cylindrical plunger of 50 mm diameter is made to penetrate the compacted soil specimen at a uniform rate of **1.25 mm per minute**, and the load is recorded against the penetration. The **load-penetration curve** is drawn; if its initial portion is **concave upwards** (which happens because of surface irregularities), a **tangent is drawn at the point of inflexion** and its intercept on the penetration axis is taken as the **corrected origin**. The loads at 2.5 mm and 5.0 mm penetration, measured from the corrected origin, are then read off.

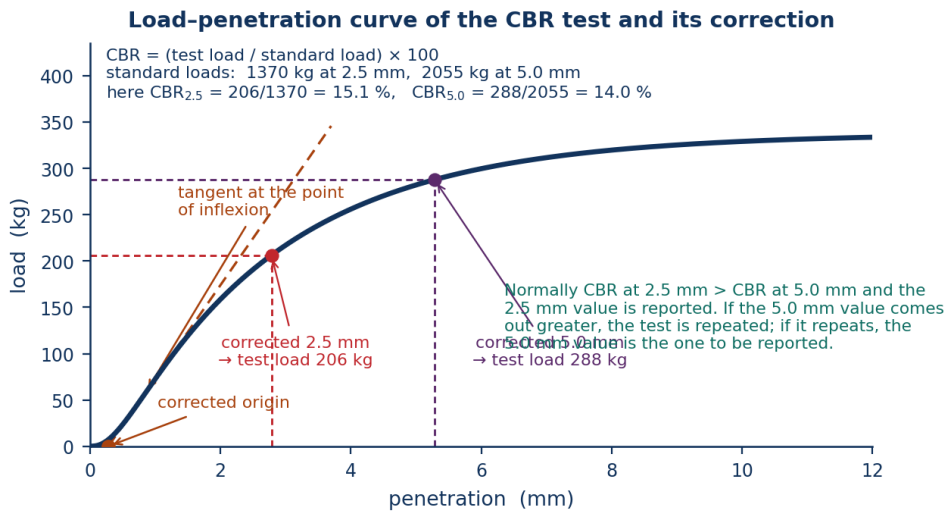


Fig. 5.1 The CBR load-penetration curve and its correction for the concave-upward initial portion.

CALIFORNIA BEARING RATIO

$\text{CBR} = (\text{load carried by the specimen} / \text{load carried by the standard aggregate}) \times 100$

Standard load at 2.5 mm penetration = 1370 kg (pressure 70 kg/cm²)

Standard load at 5.0 mm penetration = 2055 kg (pressure 105 kg/cm²)

- Normally CBR at 2.5 mm is greater than CBR at 5.0 mm and the **2.5 mm value is reported**
- If the 5.0 mm value comes out greater, the test is repeated; if it repeats, the 5.0 mm value is reported
- The specimen is soaked for 4 days before testing, to represent the worst moisture condition
- Rate of penetration of the plunger = 1.25 mm/min; plunger diameter = 50 mm

SOLVED NUMERICAL — Determination of the CBR value

Q. In a CBR test on a soaked subgrade soil specimen, the loads corresponding to 2.5 mm and 5.0 mm penetration (measured from the corrected origin) were 96 kg and 132 kg respectively. Determine the CBR value of the soil and comment on its suitability as a subgrade.

Solution:

Step 1 — CBR at 2.5 mm penetration

$$= (96/1370) \times 100 = 7.01 \%$$

Step 2 — CBR at 5.0 mm penetration

$$= (132/2055) \times 100 = 6.42 \%$$

Step 3 — which value to report

CBR at 2.5 mm (7.01 %) is greater than CBR at 5.0 mm (6.42 %), which is the normal behaviour. Hence the 2.5 mm value is reported.

CBR of the soil = 7.0 %. This is a fair to good subgrade; with a CBR of 7 % a reasonable crust thickness can be designed by IRC 37. A CBR below about 2 % would call for a capping layer or soil stabilisation.

5.2.4 Group index of a soil

The **group index** is a single number, from 0 to 20, that expresses the suitability of a soil as a subgrade — the **lower the group index the better the soil**. It is computed from the amount of fine material and the plasticity characteristics.

GROUP INDEX

$$GI = 0.2 a + 0.005 a c + 0.01 b d$$

- a = (percentage passing the 0.075 mm sieve) – 35, limited to the range 0 to 40
- b = (percentage passing the 0.075 mm sieve) – 15, limited to the range 0 to 40
- c = (liquid limit in %) – 40, limited to the range 0 to 20
- d = (plasticity index in %) – 10, limited to the range 0 to 20
- The group index is reported to the nearest whole number

SOLVED NUMERICAL — Group index of a soil

Q. A soil has 60 % passing the 0.075 mm sieve, a liquid limit of 50 % and a plasticity index of 25 %. Determine its group index.

Solution:

Step 1 — evaluate a, b, c and d with their limits

$$a = 60 - 35 = 25 \text{ (within 0-40, so } a = 25\text{)}$$

$$b = 60 - 15 = 45 \text{ (exceeds 40, so } b \text{ is taken as 40)}$$

$$c = 50 - 40 = 10 \text{ (within 0-20, so } c = 10\text{)}$$

$$d = 25 - 10 = 15 \text{ (within 0-20, so } d = 15\text{)}$$

Step 2 — substitute in the formula

$$GI = 0.2 a + 0.005 a c + 0.01 b d$$

$$= 0.2 \times 25 + 0.005 \times 25 \times 10 + 0.01 \times 40 \times 15$$

$$= 5.00 + 1.25 + 6.00 = 12.25$$

Group index = 12 (to the nearest whole number). Since the group index is high (the scale runs 0 to 20), this is a POOR subgrade soil and will need a thick pavement crust or improvement of the subgrade.

BEU / AKU REPEAT QUESTION

- Explain the CBR test with a sketch of the load-penetration curve. Why is a correction applied to the curve? [7 marks — asked very often]
- What are the desirable properties of a subgrade soil? [7 marks]
- Define group index and compute it for the given data. [4 marks]
- What is the modulus of subgrade reaction and how is it determined? [2 marks]

5.3 Stone aggregates

5.3.1 Desirable properties of road aggregates

Property	Why it is needed	Test that measures it
Strength	To resist the crushing action of the wheel loads, especially in a thin surfacing where the stresses are high	Aggregate crushing value; also the ten per cent fines value
Hardness	To resist the abrasive action of the tyres and of the grit trapped between the tyre and the surface	Los Angeles abrasion test; Deval attrition test
Toughness	To resist the sudden shock or impact of the wheel loads	Aggregate impact value
Durability	To resist weathering — alternate wetting and drying, freezing and thawing, and chemical attack	Soundness test (with sodium or magnesium sulphate)
Proper shape	Flaky and elongated particles break easily under load and give poor interlock and workability	Flakiness index, Elongation index, Angularity number
Adhesion to bitumen	The bitumen film must not be displaced by water, otherwise the aggregate strips and the mix fails	Stripping value test; static and dynamic immersion tests
Low porosity / water absorption	A porous aggregate absorbs bitumen and is less durable	Water absorption test (should be under 2 %)

Property	Why it is needed	Test that measures it
Suitable specific gravity and cleanliness	Needed for the mix design calculations; clay coatings prevent bonding	Specific gravity test; deleterious material test

Table 5.2 Desirable properties of road aggregates and the corresponding tests — a standard 7-mark answer.

Standard tests on road aggregates

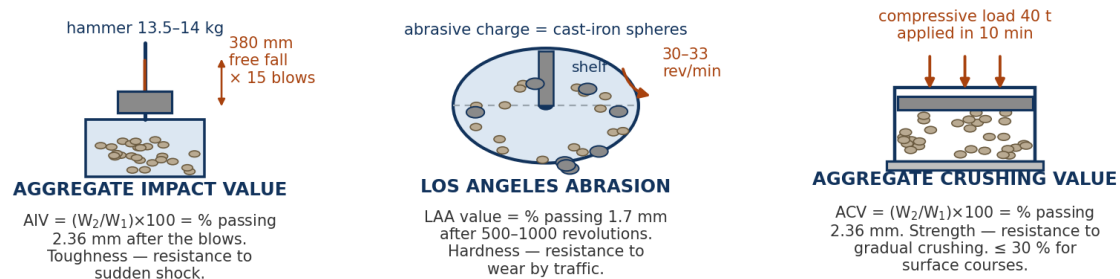


Fig. 5.2 Three of the standard aggregate tests — impact, Los Angeles abrasion and crushing.

5.3.2 The tests and their limiting values

Test	How it is expressed	Commonly specified maximum
Aggregate crushing value (ACV)	Percentage of the sample passing the 2.36 mm sieve after a compressive load of 40 tonnes applied gradually in 10 minutes	30 % for surface (wearing) courses; 45 % for base courses
Aggregate impact value (AIV)	Percentage passing the 2.36 mm sieve after 15 blows of a 13.5–14 kg hammer falling 380 mm	30 % for wearing courses; 35 % for bituminous macadam; 40 % for WBM base courses
Los Angeles abrasion value	Percentage passing the 1.7 mm sieve after 500 to 1000 revolutions of the drum with the abrasive charge of steel spheres	30 % for wearing courses; 40–50 % for base courses
Flakiness index (FI)	Percentage by weight of particles whose LEAST dimension is less than 0.6 times their mean dimension	25 % (15 % for bituminous concrete)
Elongation index (EI)	Percentage by weight of particles whose GREATEST dimension is more than 1.8 times their mean dimension	15 % (combined FI + EI $\leq$ 30 %)
Water absorption	Percentage increase in weight on soaking for 24 hours	2 %
Soundness	Loss in weight after 5 cycles of immersion in sodium or magnesium sulphate solution	12 % (sodium) / 18 % (magnesium)
Stripping value	Percentage of the aggregate surface from which the bitumen film is displaced after 24 h immersion in water at 40 °C	25 %

Table 5.3 Aggregate tests and the values commonly specified. The exact limit depends on the layer and on the MoRTH / IRC specification being used, so quote the value given in the question if one is supplied.

NOTE

Interpretation of the impact value: below 10 % exceptionally strong, 10–20 % strong, 20–30 % satisfactory for surfacing, and above 35 % weak and unsuitable for surfacing.

SOLVED NUMERICAL — Aggregate impact value and crushing value

Q. (a) In an aggregate impact test the total weight of the oven-dried sample was 350 g and the weight of the material passing the 2.36 mm sieve after the test was 42 g. (b) In an aggregate crushing test on another sample of 2,500 g, the weight passing the 2.36 mm sieve after the test was 425 g. Determine both values and comment.

Solution:

(a) Aggregate impact value

$$AIV = (W_2/W_1) \times 100 = (42/350) \times 100 = 12.0 \%$$

Being between 10 % and 20 %, the aggregate is classified as **strong** and is satisfactory for a wearing course (limit 30 %).

(b) Aggregate crushing value

$$ACV = (W_2/W_1) \times 100 = (425/2500) \times 100 = 17.0 \%$$

This is well below the limit of 30 % for a surface course, so the aggregate is acceptable.

AIV = 12 % (strong) ; ACV = 17 % — both are within the specified limits and the aggregate is suitable even for a wearing course.

SOLVED NUMERICAL — Flakiness index, elongation index and the combined index

Q. A 2,000 g sample of aggregate was tested for shape. The weight of the flaky particles separated by the thickness gauge was 260 g. From the remaining non-flaky material, the elongated particles separated by the length gauge weighed 210 g. Determine the flakiness index, the elongation index and the combined index.

Solution:

Step 1 — flakiness index

$$\begin{aligned} FI &= (\text{weight of flaky particles} / \text{total weight of the sample}) \times 100 \\ &= (260/2000) \times 100 = 13.0 \% \end{aligned}$$

Step 2 — elongation index

The elongation index is determined on the material left after the flaky particles have been removed:

$$\begin{aligned} \text{Weight of the non-flaky material} &= 2000 - 260 = 1740 \text{ g} \\ EI &= (210/1740) \times 100 = 12.07 \% \end{aligned}$$

Step 3 — combined index

$$= [(260 + 210)/2000] \times 100 = (470/2000) \times 100 = 23.5 \%$$

FI = 13.0 % ; EI = 12.1 % ; combined index = 23.5 %. All three are within the usual limits (FI 25 %, EI 15 %, combined 30 %), so the shape of the aggregate is acceptable.

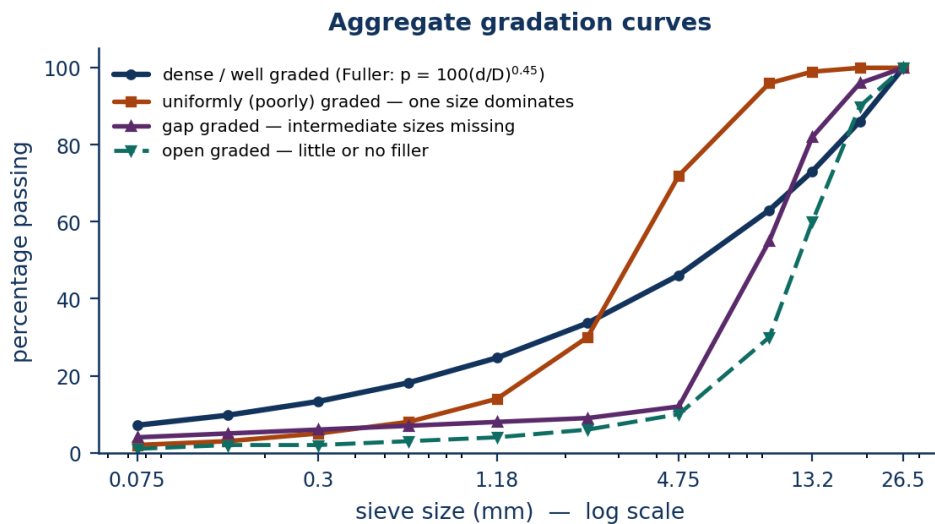


Fig. 5.3 Aggregate gradation curves. A DENSE (well) graded aggregate gives the lowest voids and the highest strength; an OPEN graded aggregate has little filler and high voids; a GAP graded aggregate lacks the intermediate sizes.

BEU / AKU REPEAT QUESTION

- ▶ What are the desirable properties of road aggregates? Name the test for each. [7 marks — asked very often]
- ▶ Explain the aggregate impact test and state the IRC limiting values. [7 marks]
- ▶ Define flakiness index and elongation index. Why are flaky and elongated particles undesirable? [4 marks]
- ▶ Compute the AIV / ACV / FI for the given data. [4 marks]

5.4 Bituminous binders

5.4.1 Types of bituminous binder

Binder	Description and use
Bitumen (paving bitumen)	A residue of the fractional distillation of crude petroleum; a thermoplastic material, soluble in carbon disulphide. Now graded by viscosity as VG-10, VG-20, VG-30 and VG-40 (IS 73), the higher number being the harder and more viscous grade. VG-30 is the general-purpose paving grade; VG-40 is used for heavily trafficked and high-stress locations. The older system graded it by penetration (30/40, 60/70, 80/100).
Road tar	Obtained by the destructive distillation of coal or wood; graded RT-1 to RT-5. It has better adhesion to wet aggregate than bitumen but is more temperature-susceptible and is now little used because of health concerns.
Cutback bitumen	Bitumen whose viscosity has been reduced by a volatile diluent, so that it can be used without heating. RC (rapid curing, cut back with gasoline), MC (medium curing, with kerosene) and SC (slow curing, with diesel or light oil). Used for surface dressing and patch repair in cold weather.
Bitumen emulsion	Bitumen dispersed as fine globules in water with an emulsifying agent. RS (rapid setting), MS (medium setting) and SS (slow setting). Used for surface dressing, patch repair, tack coat and for maintenance in wet or cold weather, because no heating is needed.
Modified bitumen	Polymer modified bitumen (PMB) and crumb rubber modified bitumen (CRMB) — the polymer or rubber improves the resistance to rutting at high temperature and to cracking at low temperature, and widens the useful temperature range. Used on heavily trafficked highways.

Table 5.4 Types of bituminous binder.

5.4.2 Tests on bitumen

Standard tests on bituminous binders

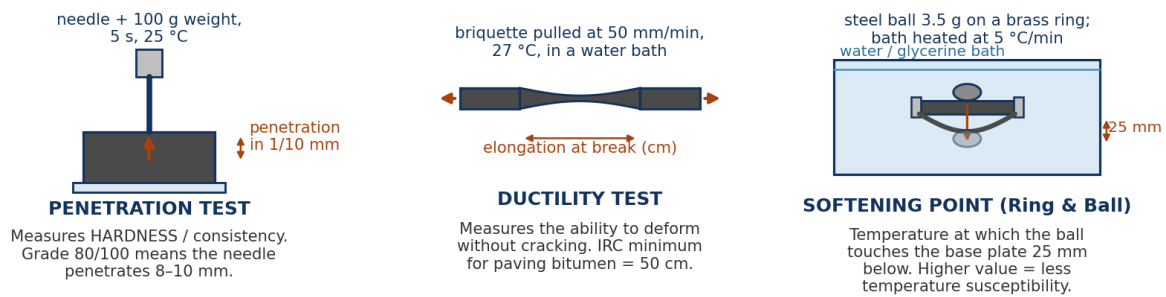


Fig. 5.4 Three standard tests on bituminous binders — penetration, ductility and softening point.

Test	What is done	What it tells us
Penetration test	The distance, in units of 0.1 mm, that a standard needle penetrates the sample in 5 s under a load of 100 g at 25 °C	The hardness or consistency of the binder. A grade of 80/100 means the needle penetrates 8 to 10 mm
Ductility test	A briquette of standard shape is pulled apart at 50 mm/min at 27 °C in a water bath; the elongation at break is measured in cm	The ability to deform without cracking — essential because the pavement flexes under every wheel load. Minimum 50 cm for paving bitumen
Softening point (Ring and Ball)	A steel ball of 3.5 g rests on the sample held in a brass ring; the bath is heated at 5 °C/min; the temperature at which the sample softens enough for the ball to fall 25 mm is recorded	The temperature susceptibility . A higher softening point means less susceptibility to temperature and less risk of bleeding in hot weather
Viscosity test	Absolute viscosity at 60 °C (in poise) and kinematic viscosity at 135 °C (in centistokes)	The resistance to flow ; it is now the basis of the grading system (VG grades) and decides the mixing and rolling temperatures
Flash and fire point	The lowest temperatures at which the vapour flashes momentarily and at which the material burns continuously	The safe heating temperature . Minimum flash point 220 °C for paving grades
Specific gravity	By the pycnometer or the balance method at 27 °C	Needed for the mix design calculations ; usually 0.97 to 1.02
Solubility in trichloroethylene	The percentage soluble in the solvent	The purity — the insoluble part is inert matter; minimum 99 %
Loss on heating / thin film oven test	The loss in weight and the change in penetration after heating	The volatility and the ageing (hardening) of the binder in service

Table 5.5 Tests on bituminous binders.

EXAM TIP

The three tests asked most often are **penetration (hardness)**, **ductility (ability to deform)** and **softening point (temperature susceptibility)**. Remember for each: the temperature, the load or the rate, and what the result signifies. Penetration — 25 °C, 100 g, 5 s. Ductility — 27 °C, 50 mm/min, minimum 50 cm. Softening point — ball 3.5 g, heating 5 °C/min, fall of 25 mm.

5.5 Bituminous paving mixes

5.5.1 Types of bituminous construction

Type	Where used	Character
Surface dressing	A thin wearing surface or a renewal coat	A layer of binder covered with a single layer of stone chips; not a structural layer
Premix carpet (PC) with seal coat	Thin wearing course, 20 mm	Open graded, needs a seal coat to close the surface
Bituminous macadam (BM)	Base or binder course	Open graded, high voids, low bitumen content
Dense bituminous macadam (DBM)	Binder course of heavy-duty pavements	Dense graded, designed by the Marshall method

Type	Where used	Character
Semi-dense bituminous concrete (SDBC)	Wearing course, 25–40 mm	Between BM and BC in density
Bituminous concrete (BC) / asphaltic concrete	The highest quality wearing course , 30–50 mm	Dense graded, impervious, designed by the Marshall method
Mastic asphalt	Bridge decks, heavily trafficked junctions, toll plazas	Very high binder content, void-less, laid hot and hand tamped

Table 5.6 Types of bituminous construction, in increasing order of quality.

5.5.2 Requirements of a good bituminous mix

- **Stability** — resistance to permanent deformation (rutting and shoving) under repeated loading, obtained from the aggregate interlock and the binder.
- **Durability** — resistance to weathering and to the ageing of the binder, obtained by a sufficient binder film thickness and low voids.
- **Flexibility** — the ability to deform with the underlying layers without cracking.
- **Adequate skid resistance** — a rough, harsh microtexture, particularly when wet.
- **Impermeability** — to keep water out of the pavement layers.
- **Workability** — the mix must be capable of being laid and compacted with the available plant.
- **Economy** — the minimum binder content that satisfies all the above.

5.5.3 Marshall method of mix design

Marshall's method is the standard method of designing a dense bituminous mix. Cylindrical specimens 101.6 mm in diameter and 63.5 mm high are prepared at several bitumen contents, compacted with **75 blows on each face** for heavy traffic, and then tested at **60 °C** at a deformation rate of **50 mm per minute**. Two results are obtained from the test:

- **Marshall stability** — the maximum load the specimen carries before failure, in kg or kN.
- **Flow value** — the vertical deformation at the instant of failure, in units of 0.25 mm.

Marshall mix design — the five plots; OBC = mean of (max stability, max unit weight, 4 % air voids) = 5.41 %

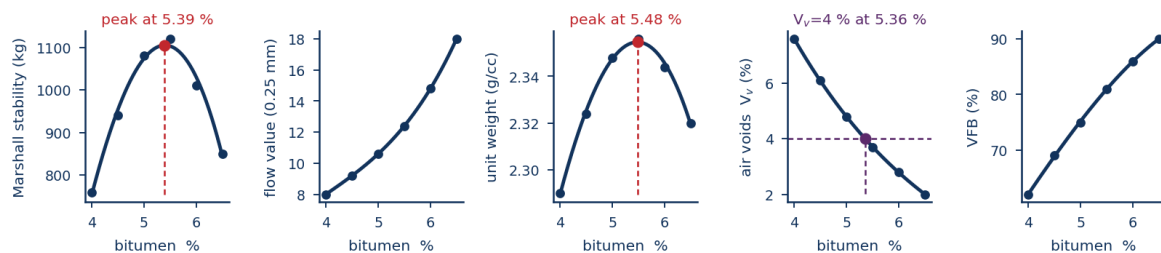


Fig. 5.5 The five Marshall plots. The optimum bitumen content is the average of the bitumen contents at (i) maximum stability, (ii) maximum unit weight and (iii) 4 % air voids.

MARSHALL MIX DESIGN COMPUTATIONS

Bulk specific gravity of the compacted mix:

$$G_m = W_a / (W_a - W_u)$$

Theoretical maximum specific gravity (no air voids):

$$G_t = 100 / [(W_1/G_1) + (W_2/G_2) + \dots + (W^b/G^b)]$$

$$\text{Air voids: } V_v = [(G_t - G_m) / G_t] \times 100 \%$$

$$\text{Volume of bitumen: } V^b = (W^b \times G_m) / G^b \%$$

$$\text{Voids in the mineral aggregate: } VMA = V_v + V^b \%$$

$$\text{Voids filled with bitumen: } VFB = (V^b / VMA) \times 100 \%$$

- W_a = weight of the specimen in air, W_u = weight in water
- W_i , G_i = percentage by weight and the specific gravity of each ingredient (aggregate fractions, filler, bitumen)
- W^b , G^b = percentage and specific gravity of the bitumen
- Typical criteria: Marshall stability minimum 9 kN, flow 2–4 mm (8–16 units), air voids 3–5 %, VFB 65–75 %

SOLVED NUMERICAL — Marshall mix design computations

Q. A compacted Marshall specimen of a bituminous mix weighs 1,200 g in air and 690 g in water. The bitumen content is 5.0 % by total weight of the mix. The specific gravity of the aggregate is 2.70 and that of the bitumen is 1.02. Determine the bulk specific gravity, the theoretical maximum specific gravity, the percentage air voids, the VMA and the VFB, and comment on the mix.

Solution:

Step 1 — bulk specific gravity of the mix

$$G_m = W_a / (W_a - W_u) = 1200 / (1200 - 690) = 1200 / 510 = 2.353$$

Step 2 — theoretical maximum specific gravity

Percentage of aggregate = $100 - 5.0 = 95.0 \%$

$$G_t = 100 / [(95/2.70) + (5/1.02)] = 100 / (35.185 + 4.902) \\ = 100 / 40.087 = 2.495$$

Step 3 — percentage air voids

$$V_v = [(G_t - G_m) / G_t] \times 100 = [(2.495 - 2.353) / 2.495] \times 100 \\ = (0.142 / 2.495) \times 100 = 5.68 \%$$

Step 4 — volume of bitumen

$$V^b = (W^b \times G_m) / G^b = (5.0 \times 2.353) / 1.02 = 11.765 / 1.02 = 11.53 \%$$

Step 5 — VMA and VFB

$$VMA = V_v + V^b = 5.68 + 11.53 = 17.21 \%$$

$$VFB = (V^b / VMA) \times 100 = (11.53 / 17.21) \times 100 = 67.0 \%$$

Step 6 — comment

The VFB of 67 % lies within the usual range of 65–75 %, but the air voids of 5.68 % are slightly above the desirable range of 3–5 % for a bituminous concrete. The mix would be improved either by a small increase in the bitumen content or by heavier compaction.

$$G_m = 2.353 ; G_t = 2.495 ; V_v = 5.68 \% ; V^b = 11.53 \% ; VMA = 17.21 \% ; VFB = 67.0 \%$$

BEU / AKU REPEAT QUESTION

- ▶ Explain the Marshall method of bituminous mix design. How is the optimum bitumen content determined? [14 marks — asked very often]
- ▶ Compute the air voids, VMA and VFB for the given Marshall specimen data. [7 marks]
- ▶ What are the requirements of a good bituminous mix? [7 marks]
- ▶ Differentiate between BM, DBM and BC. [4 marks]
- ▶ What is cutback bitumen and bitumen emulsion? When is each used? [7 marks]

5.6 Cement and cement concrete for pavements

5.6.1 Desirable properties and tests on cement

Test on cement	What it checks
Fineness (sieve or Blaine air permeability)	The rate of hydration and the strength development
Consistency (Vicat apparatus)	The water required for the standard paste, needed for the other tests
Setting time — initial and final (Vicat)	Initial setting time not less than 30 minutes and final setting time not more than 600 minutes for OPC — governs the time available for placing and finishing
Soundness (Le Chatelier, autoclave)	Freedom from excessive expansion after setting, due to free lime or magnesia
Compressive strength (mortar cubes at 3, 7 and 28 days)	The grade of the cement (33, 43 or 53)
Heat of hydration	Important for thick slabs, where the temperature rise causes cracking

Table 5.7 Tests on cement.

5.6.2 Requirements of concrete for a cement concrete pavement

The critical property of pavement quality concrete (PQC) is **flexural strength**, not compressive strength, because a rigid pavement slab carries the load essentially in **bending**.

Requirement	Value normally specified
Characteristic 28-day flexural strength	4.5 MPa minimum — the design input in IRC 58
Grade of concrete	M 40 (28-day characteristic compressive strength 40 MPa) for PQC
Minimum cement content	About 350 kg/m ³
Maximum water-cement ratio	About 0.45 to 0.50
Maximum size of coarse aggregate	25 mm to 31.5 mm
Workability (slump)	Low — 25 to 50 mm for slip-form or fixed-form paving, since a stiff mix is needed to hold the shape
Air content, where freezing occurs	4 to 6 % entrained air

Table 5.8 Requirements for pavement quality concrete (IRC 15 / IRC 58). The relation between flexural strength f_r and compressive strength f_{ck} is often taken as $f_r \approx 0.7 \sqrt{f_{ck}}$ (in MPa).

Other uses of cement in road works: cement-stabilised soil sub-bases, **dry lean concrete (DLC)** sub-bases beneath the PQC slab, cement-treated granular bases, and roller-compacted concrete.

BEU / AKU REPEAT QUESTION

- ▶ Why is flexural strength, and not compressive strength, the design criterion for a cement concrete pavement? [2 marks]
- ▶ State the requirements of concrete used in a cement concrete pavement. [7 marks]
- ▶ Enumerate the tests on cement and state the purpose of each. [7 marks]

▶ *Reference: Khanna, Justo & Veeraragavan, chapter 'Highway Materials'; Partha Chakraborty & Animesh Das — Principles of Transportation Engineering for the mix-design volumetrics; IS 2386 (tests on aggregates), IS 73 (paving bitumen), IS 1203-1220 (tests on bitumen), IRC 15 and IRC 58 (concrete pavements), MoRTH Specifications for Road and Bridge Works.*

UNIT 6 | Design of Pavements

Syllabus allotment: 10 hours • about 24 % of the paper — the highest weightage

6.1 Types of pavement

A pavement is a structure that **distributes the concentrated wheel load over a large area of the subgrade**, so that the pressure reaching the soil is within its safe bearing capacity, and provides a smooth, dust-free, skid-resistant riding surface. Pavements are of two kinds, distinguished by **the way they carry the load**.

How a flexible and a rigid pavement transmit the wheel load

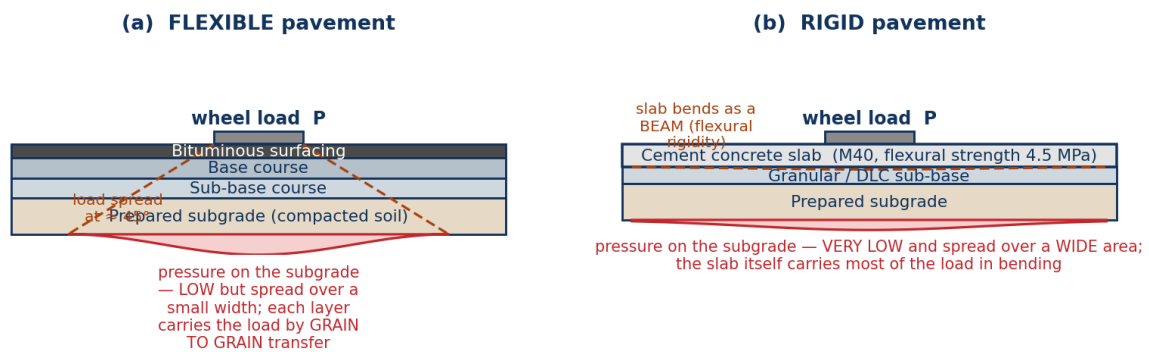


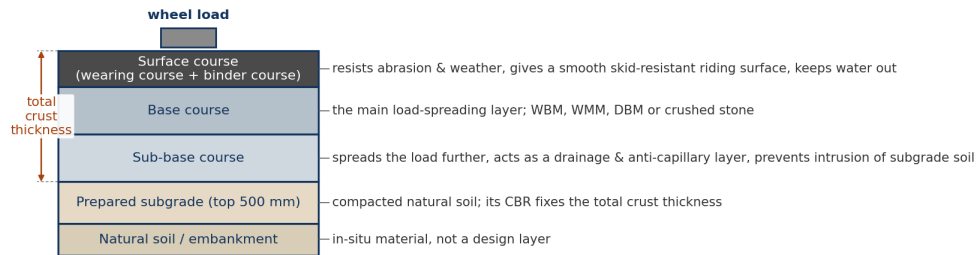
Fig. 6.1 How a flexible pavement and a rigid pavement transmit the wheel load to the subgrade. **This diagram, and the comparison table below, together make a complete 7-mark answer.**

Point of comparison	Flexible pavement	Rigid pavement
Load transfer mechanism	By grain-to-grain transfer through the points of contact of the aggregate, layer by layer	By the flexural (beam) action of the slab, which has considerable flexural rigidity
Design is based on	The strength of the subgrade (CBR)	The flexural strength of the concrete and the modulus of subgrade reaction K
Stress on the subgrade	Higher, spread over a small area	Very low, spread over a wide area
Effect of a weak spot in the subgrade	Reflects at the surface as a depression — the pavement deforms	The slab bridges over it; no surface depression, but the slab may crack
Design life	Usually 10 to 15 years	Usually 30 to 40 years
Initial cost	Low	High
Maintenance cost	High — frequent renewal of the surfacing	Very low
Joints	No joints required	Expansion, contraction and longitudinal joints are essential
Temperature stresses	Negligible	Significant — warping and frictional stresses must be computed
Night visibility, skid resistance	Darker surface, poorer night visibility	Lighter surface, better night visibility
Suitability for stage construction	Very suitable	Not suitable
Behaviour under water logging	Loses strength badly	Not much affected

Table 6.1 Flexible pavement compared with rigid pavement.

6.2 Components of a flexible pavement and their functions

Components of a flexible pavement and the function of each layer



Stress decreases with depth $\Rightarrow$ the strongest and most expensive material is placed at the TOP

Fig. 6.2 Components of a flexible pavement. Since the stress decreases with depth, the strongest and costliest material is placed at the top.

Layer	Functions
Surface course (wearing course + binder course)	Provides a smooth, comfortable and skid-resistant riding surface; resists the abrasive action of traffic; is impervious , so it keeps surface water out of the lower layers; and takes the highest wheel-load stresses.
Base course	The main load-spreading layer . It takes up the stress transmitted from the surfacing and distributes it to the sub-base; it also provides a stable platform on which the surfacing is laid.
Sub-base course	Distributes the load further to the subgrade; acts as a drainage layer and as an anti-capillary (filter) layer that prevents the intrusion of the fine subgrade soil into the base course; and provides a working platform for construction traffic.
Prepared subgrade (top 500 mm)	The foundation of the pavement. Its strength, measured by the CBR, decides the total thickness of the crust. It must be compacted to the specified density and protected from moisture.

Table 6.2 Functions of each layer of a flexible pavement.

A **rigid pavement** has fewer layers: the **cement concrete slab** (the pavement quality concrete or PQC), a **dry lean concrete (DLC) or granular sub-base**, and the **prepared subgrade**. The slab itself performs the functions of the surface, base and sub-base of a flexible pavement.

6.3 Factors affecting the design of a flexible pavement

1. **Design wheel load** — the magnitude and repetition of the wheel load, the contact pressure, the load on a dual or multiple wheel assembly, and the equivalent single wheel load.
2. **Subgrade soil** — its strength (CBR), its variation with moisture, and its volume stability.
3. **Climatic factors** — rainfall (which raises the water table and the subgrade moisture), temperature (which affects the stiffness of the bituminous layers), and frost action.
4. **Pavement component materials** — the strength and stiffness of each layer.
5. **Environmental factors** — the height of the embankment, the depth of the water table, and the drainage conditions.
6. **Special factors** — the behaviour of the pavement under repeated loading (fatigue) and the possibility of stage construction.

6.3.1 Contact pressure and rigidity factor

CONTACT PRESSURE

$$\text{Contact pressure} = (\text{load on the wheel}) / (\text{contact area})$$

$$\text{Rigidity factor} = (\text{contact pressure}) / (\text{tyre pressure})$$

$$\text{Radius of the equivalent circular contact area: } a = \sqrt{(P / (\pi p))}$$

- The rigidity factor is **1.0** when the tyre pressure is 7 kg/cm²
- It is **greater than 1** for tyre pressures LOWER than 7 kg/cm² and **less than 1** for HIGHER pressures
- P = load on the wheel, p = tyre pressure, a = radius of the contact area

6.3.2 Equivalent single wheel load (ESWL)

Heavy vehicles carry the load on a **dual wheel assembly**. Near the surface each wheel acts independently, while at a great depth the two overlap completely and act as one. The **ESWL** is the load on a single wheel that would produce the same magnitude of the chosen stress or deflection, at the given depth, as the dual wheel assembly does.

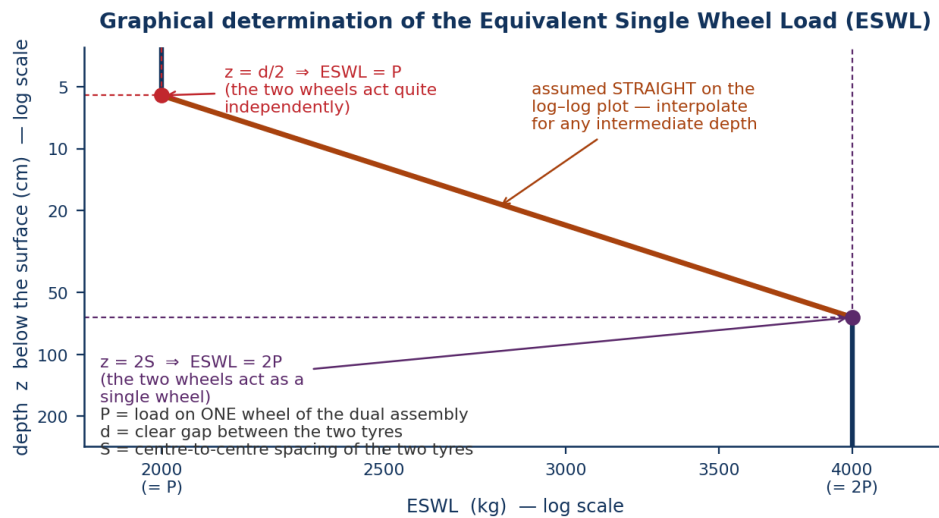


Fig. 6.3 Graphical determination of the ESWL on a log-log plot.

EQUIVALENT SINGLE WHEEL LOAD

$\text{ESWL} = P$ for $z \leq d/2$ (the wheels act independently)

$\text{ESWL} = 2P$ for $z \geq 2S$ (the wheels act as a single wheel)

For any intermediate depth, by linear interpolation on the log-log plot:

$$\log_{10}(\text{ESWL}) = \log_{10} P + [0.301 \times \log_{10}(2z/d)] / [\log_{10}(4S/d)]$$

- P = load on ONE wheel of the dual assembly
- S = centre-to-centre spacing of the two tyres
- d = clear gap between the two tyres = $S - 2a$
- z = depth below the surface at which the ESWL is required
- $0.301 = \log_{10} 2$, because the load doubles from P to $2P$

SOLVED NUMERICAL — Equivalent single wheel load at various depths

Q. A dual wheel assembly carries 2,000 kg on each wheel. The centre-to-centre spacing of the tyres is 30 cm and the tyre pressure is 7 kg/cm². Determine the ESWL at depths of 5 cm, 15 cm, 30 cm and 60 cm.

Solution:

Step 1 — radius of the contact area

$$a = \sqrt{P/\pi p} = \sqrt{(2000/(\pi \times 7))} = \sqrt{(2000/21.99)} = \sqrt{90.94} = 9.54 \text{ cm}$$

Step 2 — clear gap between the tyres

$$d = S - 2a = 30 - 2 \times 9.54 = 30 - 19.07 = 10.93 \text{ cm}$$

$$d/2 = 5.46 \text{ cm and } 2S = 2 \times 30 = 60 \text{ cm}$$

Step 3 — the two limiting depths

At $z = 5 \text{ cm}$, which is less than $d/2 = 5.46 \text{ cm}$: **ESWL = P = 2000 kg**

At $z = 60 \text{ cm}$, which equals $2S$: **ESWL = 2P = 4000 kg**

Step 4 — interpolate for $z = 15 \text{ cm}$

$$\log(2z/d) = \log(2 \times 15/10.93) = \log 2.744 = 0.4382$$

$$\log(4S/d) = \log(4 \times 30/10.93) = \log 10.979 = 1.0406$$

$$\log(\text{ESWL}) = \log 2000 + 0.301 \times (0.4382/1.0406) = 3.3010 + 0.301 \times 0.4212$$

$$= 3.3010 + 0.1268 = 3.4278 \Rightarrow \text{ESWL} = 10^{3.4278} = \mathbf{2678 \text{ kg}}$$

Step 5 — interpolate for $z = 30 \text{ cm}$

$$\log(2z/d) = \log(60/10.93) = \log 5.4895 = 0.7391$$

$$\log(\text{ESWL}) = 3.3010 + 0.301 \times (0.7391/1.0406) = 3.3010 + 0.301 \times 0.7103$$

$$= 3.3010 + 0.2138 = 3.5148 \Rightarrow \text{ESWL} = \mathbf{3273 \text{ kg}}$$

ESWL = 2000 kg at 5 cm ; 2678 kg at 15 cm ; 3273 kg at 30 cm ; 4000 kg at 60 cm. The ESWL rises from P to 2P as the depth increases from $d/2$ to $2S$.

6.3.3 Design traffic — cumulative standard axles (msa)

A flexible pavement fails by **fatigue** under repeated loading, so the design traffic is expressed as the **cumulative number of standard axles** (of 8.16 tonnes or 80 kN) expected during the design life, in **million standard axles (msa)**.

DESIGN TRAFFIC (IRC 37)

$$N = 365 \times A \times [(1 + r)^n - 1] / r \times D \times F$$

$$A = P (1 + r)^x$$

- N = cumulative number of standard axles during the design life (convert to msa by dividing by 10^6)
- A = number of commercial vehicles per day (CVPD) in the **year of completion** of construction
- P = present number of CVPD, x = number of years between the last count and the year of completion
- r = annual rate of growth of commercial vehicles (as a decimal; IRC suggests 5 % minimum, commonly 7.5 %)
- n = design life in years (15 years for NH and SH; 20 years for expressways)
- D = lane distribution factor — 1.0 for a single lane; 0.75 for a two-lane single carriageway; 0.40 for a four-lane single carriageway; 0.75 of the directional traffic for a dual two-lane carriageway
- F = vehicle damage factor (VDF) — the number of standard axles per commercial vehicle; it depends on the terrain and the traffic volume (indicative values 1.5 to 4.5)

SOLVED NUMERICAL — Cumulative standard axles for pavement design

Q. A two-lane single-carriageway national highway is to be strengthened. The present traffic is 1,500 commercial vehicles per day (two-way). The construction will be completed in 2 years, the annual growth rate of commercial traffic is 7.5 % and the design life is 15 years. Take the vehicle damage factor as 4.5. Determine the design traffic in msa.

Solution:

Step 1 — traffic in the year of completion

$$A = P (1 + r)^x = 1500 \times (1.075)^2 = 1500 \times 1.1556 = 1733.4 \text{ CVPD}$$

Step 2 — the growth factor for the design life

$$(1.075)^{15} = 2.9591$$

$$[(1+r)^n - 1]/r = (2.9591 - 1)/0.075 = 1.9591/0.075 = 26.12$$

Step 3 — lane distribution factor

For a two-lane single carriageway, $D = 0.75$

Step 4 — cumulative standard axles

$$N = 365 \times 1733.4 \times 26.12 \times 0.75 \times 4.5$$

$$= 365 \times 1733.4 = 632,691$$

$$632,691 \times 26.12 = 1.6527 \times 10^7$$

$$\times 0.75 = 1.2395 \times 10^7$$

$$\times 4.5 = 5.578 \times 10^7 \text{ standard axles}$$

Step 5 — express in msa

$$N = 5.578 \times 10^7 / 10^6 = 55.8 \text{ msa}$$

Design traffic = 55.8 msa. The pavement would be designed for the next standard step, i.e. 60 msa, using the IRC 37 catalogue or IITPAVE.

6.4 Stresses in flexible pavements

Theory	Assumptions and result
Boussinesq's theory (single layer)	The pavement and the subgrade are treated as ONE homogeneous, isotropic, elastic half-space. For a point load P the vertical stress at depth z on the axis is $\sigma_z = 3P/(2\pi z^2)$. For a circular load of radius a and pressure p the stress on the axis is $\sigma_z = p [1 - z^3/(a^2 + z^2)^{1.5}]$. Simple, but the assumption of a single layer is far from the truth.
Burmister's two-layer theory	The pavement is an elastic layer of modulus E_1 and thickness h resting on a subgrade of modulus E_2 , with $E_1 > E_2$. The deflection is expressed as $\Delta = 1.5 p a F_2 / E_2$, where the deflection factor F_2 depends on E_2/E_1 and h/a and is read from Burmister's charts. It also gives the interface stresses. Extended by Burmister to three layers.
Odemark's method of equivalent thickness	A layered system is transformed into an equivalent single layer of the subgrade material, so that Boussinesq's equations can then be applied.
Mechanistic-empirical analysis (IRC 37-2018)	The pavement is analysed as a multi-layer elastic system (using the program IITPAVE) to obtain the horizontal tensile strain at the bottom of the bituminous layer (which controls fatigue cracking) and the vertical compressive strain at the top of the subgrade (which controls rutting). These strains are then limited by empirical performance equations.

Table 6.3 Theories for the analysis of stresses in flexible pavements.

SOLVED NUMERICAL — Vertical stress under a circular load by Boussinesq's theory

Q. A wheel load transmits a pressure of 7 kg/cm² over a circular contact area of radius 15 cm. Determine the vertical stress on the axis of loading at depths of 30 cm and 60 cm, and comment.

Solution:

Formula

$$\sigma_z = p \left[1 - \frac{z^3}{(a^2 + z^2)^{1.5}} \right]$$

At z = 30 cm

$$a^2 + z^2 = 15^2 + 30^2 = 225 + 900 = 1125$$

$$(1125)^{1.5} = 1125 \times \sqrt{1125} = 1125 \times 33.54 = 37,733$$

$$z^3 = 27,000 \Rightarrow \text{ratio} = 27000/37733 = 0.7155$$

$$\sigma_z = 7 (1 - 0.7155) = 7 \times 0.2845 = 1.99 \text{ kg/cm}^2$$

At z = 60 cm

$$a^2 + z^2 = 225 + 3600 = 3825$$

$$(3825)^{1.5} = 3825 \times 61.85 = 236,565$$

$$z^3 = 216,000 \Rightarrow \text{ratio} = 216000/236565 = 0.9131$$

$$\sigma_z = 7 (1 - 0.9131) = 7 \times 0.0869 = 0.61 \text{ kg/cm}^2$$

$\sigma_z = 1.99 \text{ kg/cm}^2$ at 30 cm depth and 0.61 kg/cm^2 at 60 cm depth. The stress falls to about 28 % of the contact pressure at a depth of 2a and to about 9 % at a depth of 4a — which is exactly why the granular layers below can be of progressively poorer quality.

6.5 Design methods for flexible pavements

Method	Basis
Group index method	Empirical. The total thickness is read from a chart against the group index of the subgrade soil and the traffic volume class. Does not use the strength of the soil directly, and is now obsolete.
CBR method (IRC 37)	Semi-empirical. The thickness is obtained from design curves plotted between the CBR of the subgrade and the cumulative standard axles. This is the method used in India.
Triaxial test method	Uses the modulus of elasticity from a triaxial test in an elastic-theory based thickness equation, with corrections for traffic and saturation.
McLeod method	Based on plate bearing test results and the logarithm of the number of load repetitions.
Burmister's method	Based on the two-layer elastic theory, limiting the surface deflection.

Table 6.4 Methods of flexible pavement design.

6.5.1 The IRC 37 design procedure

1. Determine the **subgrade CBR** from soaked specimens compacted at the field density; take the design value as the lower of the values along the stretch.
2. Compute the **effective resilient modulus** of the subgrade: $M_R = 10 \times \text{CBR}$ for $\text{CBR} \leq 5 \%$, and $M_R = 17.6 \times \text{CBR}^{0.64}$ for $\text{CBR} > 5 \%$ (M_R in MPa).
3. Estimate the **design traffic in msa** from the axle-load survey and the traffic count (§ 6.3.3).
4. Select the **pavement composition** — the thickness of the bituminous surfacing, the bituminous base (DBM), the granular base (WMM) and the granular sub-base (GSB) — from the IRC 37 design catalogue, or analyse a trial section with **IITPAVE**.
5. Check the trial section against the two performance criteria — the **fatigue criterion** (horizontal tensile strain at the bottom of the bituminous layer) and the **rutting criterion** (vertical compressive strain at the top of the subgrade) — for the chosen reliability.
6. Provide the specified **minimum thicknesses** of each layer and check the drainage of the granular sub-base.

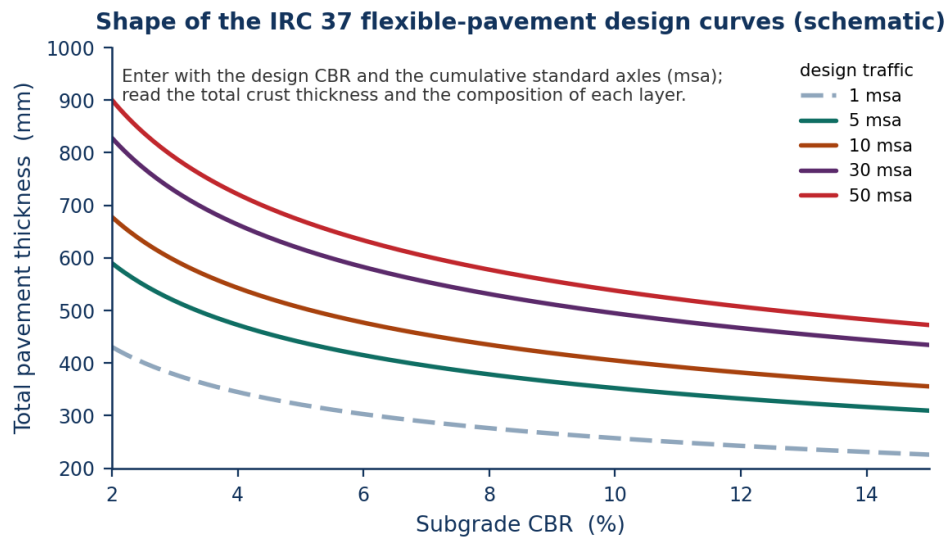


Fig. 6.4 The shape of the IRC 37 design curves — the total thickness increases with the design traffic and decreases as the subgrade CBR improves.

EXAM TIP

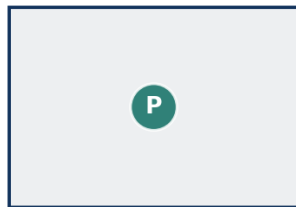
For a numerical in the examination you will normally be asked only for the **design traffic in msa** (as in Example 6.2) and sometimes to read a thickness from a supplied chart. Learn the msa formula thoroughly — it is the single most likely numerical from this section.

6.6 Stresses in rigid pavements — Westergaard's analysis

Westergaard treated the concrete slab as a **thin elastic plate resting on a dense liquid subgrade**, i.e. the reaction of the subgrade at any point is proportional to the deflection at that point ($p = K\Delta$). The three critical load positions are the **interior**, the **edge** and the **corner**.

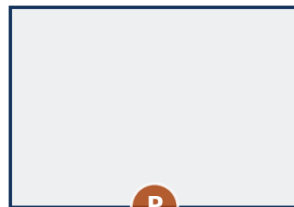
Westergaard's three critical load positions on a rigid pavement slab

INTERIOR (centre) loading



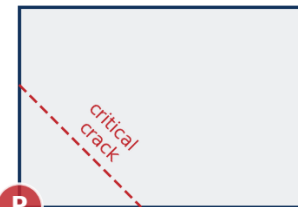
load well inside the slab

EDGE loading



load on the edge, away from a corner

CORNER loading



load at the corner of the slab

Critical stress for a corner load acts at the TOP of the slab; for edge and interior loads at the BOTTOM

l = radius of relative stiffness = $\left[\frac{E h^3}{12 (1-\mu^2) K} \right]^{1/4}$ b = equivalent radius of the resisting section

Fig. 6.5 Westergaard's three critical load positions.

WESTERGAARD'S PARAMETERS

Radius of relative stiffness:

$$l = [E h^3 / (12 (1 - \mu^2) K)]^{1/4}$$

Equivalent radius of the resisting section:

$$b = \sqrt{(1.6 a^2 + h^2)} - 0.675 h \text{ when } a < 1.724 h$$

$$b = a \text{ when } a \geq 1.724 h$$

- E = modulus of elasticity of the concrete (about 3×10^5 kg/cm²)
- h = thickness of the slab (cm)
- μ = Poisson's ratio of concrete (0.15)
- K = modulus of subgrade reaction (kg/cm³), from the plate bearing test
- a = radius of the contact area of the wheel (cm)

WESTERGAARD'S STRESS EQUATIONS (AS MODIFIED)

$$\text{Interior loading: } \sigma_i = (0.316 P / h^2) [4 \log_{10}(l/b) + 1.069]$$

$$\text{Edge loading: } \sigma_e = (0.572 P / h^2) [4 \log_{10}(l/b) + 0.359]$$

$$\text{Corner loading: } \sigma_c = (3 P / h^2) [1 - (a\sqrt{2} / l)^{0.6}]$$

- P = design wheel load (kg); h = slab thickness (cm)
- The stresses come out in kg/cm²
- For the INTERIOR and EDGE loads the critical tensile stress is at the **BOTTOM** of the slab
- For the CORNER load the critical tensile stress is at the **TOP** of the slab

SOLVED NUMERICAL — Westergaard's stresses at the interior, edge and corner

Q. A cement concrete pavement slab is 25 cm thick and rests on a subgrade of modulus of reaction $K = 8 \text{ kg/cm}^3$. The design wheel load is 5,100 kg with a contact radius of 15 cm. Take $E = 3 \times 10^5 \text{ kg/cm}^2$ and $\mu = 0.15$. Compute the load stresses at the interior, the edge and the corner, and state which governs.

Solution:

Step 1 — radius of relative stiffness

$$l = [E h^3 / (12(1 - \mu^2) K)]^{1/4}$$

$$h^3 = 25^3 = 15,625 ; 1 - \mu^2 = 1 - 0.0225 = 0.9775$$

$$12 \times 0.9775 \times 8 = 93.84$$

$$E h^3 = 3 \times 10^5 \times 15,625 = 4.6875 \times 10^9$$

$$l = (4.6875 \times 10^9 / 93.84)^{1/4} = (4.995 \times 10^7)^{1/4} = 84.07 \text{ cm}$$

Step 2 — equivalent radius of the resisting section

Check: $1.724 h = 1.724 \times 25 = 43.1 \text{ cm}$; $a = 15 \text{ cm} < 43.1 \text{ cm}$, so use the first expression

$$b = \sqrt{(1.6 a^2 + h^2)} - 0.675 h = \sqrt{(1.6 \times 225 + 625)} - 0.675 \times 25$$

$$= \sqrt{(360 + 625)} - 16.875 = \sqrt{985} - 16.875 = 31.38 - 16.88 = 14.51 \text{ cm}$$

Step 3 — the logarithmic term

$$l/b = 84.07/14.51 = 5.794 \Rightarrow \log_{10}(l/b) = 0.7630$$

$$4 \log_{10}(l/b) = 3.052$$

$$P/h^2 = 5100/625 = 8.16$$

Step 4 — interior load stress

$$\sigma_i = 0.316 \times 8.16 \times (3.052 + 1.069) = 2.579 \times 4.121 = 10.63 \text{ kg/cm}^2$$

Step 5 — edge load stress

$$\sigma_e = 0.572 \times 8.16 \times (3.052 + 0.359) = 4.667 \times 3.411 = 15.92 \text{ kg/cm}^2$$

Step 6 — corner load stress

$$a\sqrt{2}/l = 15 \times 1.4142/84.07 = 21.21/84.07 = 0.2523$$

$$(0.2523)^{0.6} = 0.4376$$

$$\sigma_c = 3 \times 8.16 \times (1 - 0.4376) = 24.48 \times 0.5624 = 13.77 \text{ kg/cm}^2$$

$\sigma_i = 10.63$, $\sigma_e = 15.92$ and $\sigma_c = 13.77 \text{ kg/cm}^2$. The edge load stress governs here. All three are well below the flexural strength of 45 kg/cm^2 (4.5 MPa), but the warping and frictional stresses must still be added in the critical combinations.

6.6.1 Temperature (warping) stresses

The top and bottom of the slab are at different temperatures, so the slab tries to curl. Its own weight and the friction of the subgrade restrain the curling, and this restraint induces **warping stresses**.

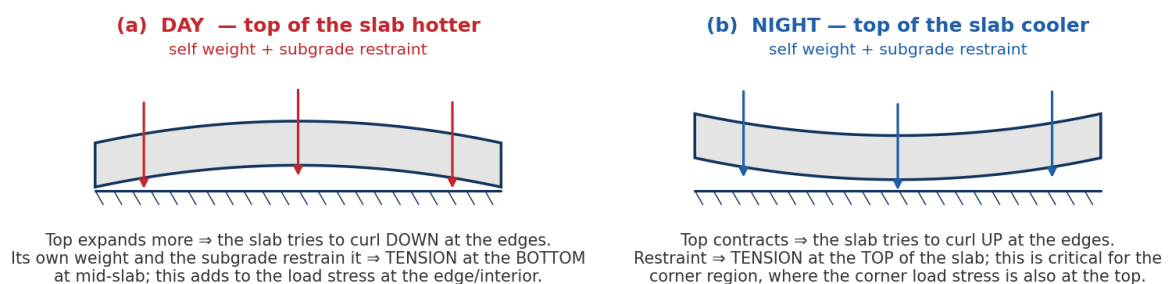
Warping (curling) stresses in a cement concrete slab caused by the temperature gradient

Fig. 6.6 Warping of a slab by day and by night, and the resulting location of the tensile stress.

WARPING STRESS (BRADBURY'S COEFFICIENTS)

$$\text{Interior: } \sigma_{ti} = (E \alpha t / 2) \times [(C_x + \mu C_y) / (1 - \mu^2)]$$

$$\text{Edge: } \sigma_{te} = C E \alpha t / 2$$

$$\text{Corner: } \sigma_{tc} = [E \alpha t / (3 (1 - \mu))] \times \sqrt{(a / l)}$$

- α = coefficient of thermal expansion of concrete $\approx 10 \times 10^{-6}$ per $^{\circ}\text{C}$
- t = temperature differential between the top and the bottom of the slab
- C, C_x, C_y = Bradbury's coefficients, read from a chart against L_x/l and L_y/l
- L_x, L_y = the two dimensions of the slab panel

6.6.2 Frictional stresses

As the slab contracts on cooling or on drying shrinkage, the friction of the subgrade resists the movement and induces a **tensile frictional stress** in the slab. It is a maximum at the mid-length of the slab and zero at the free ends.

FRICTIONAL STRESS

$$\sigma_e = W L f / (2 \times 10^4)$$

- σ_e = frictional stress in kg/cm^2
- W = unit weight of concrete $\approx 2400 \text{ kg/m}^3$
- L = length of the slab in metres
- f = coefficient of friction between the slab and the sub-base ≈ 1.5
- Note that the slab thickness cancels out — the frictional stress does not depend on h

6.6.3 Critical combination of stresses

Condition	Stresses that combine	Critical location
Summer, mid-day	Load stress + warping stress (the slab curls DOWN at the edges, so the warping tension is at the bottom) – frictional stress (the slab is expanding, so friction gives compression)	Bottom of the slab at the EDGE or the INTERIOR
Winter, mid-day	Load stress + warping stress + frictional stress (the slab is contracting, so friction gives tension) — the worst combination	Bottom of the slab at the EDGE
Mid-nights	Load stress + warping stress (the slab curls UP at the edges, so the warping tension is at the top)	Top of the slab at the CORNER

Table 6.5 Critical combinations of stress in a rigid pavement.

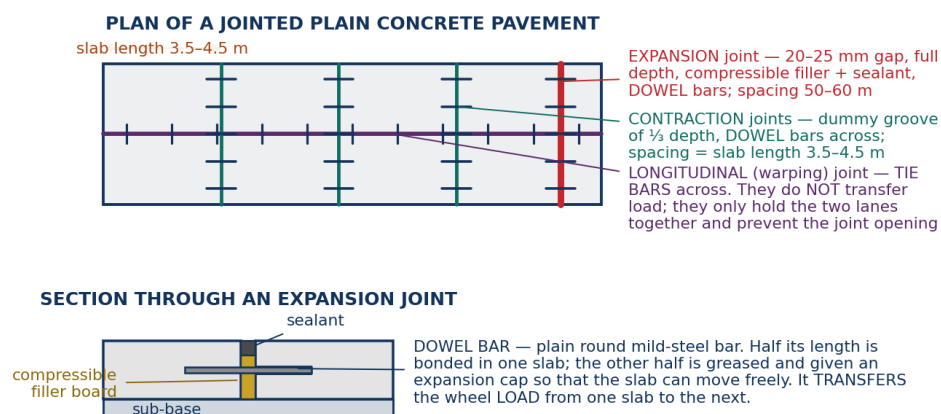
6.7 Design of joints in cement concrete pavements

Fig. 6.7 Joints in a jointed plain concrete pavement, and a section through an expansion joint showing the dowel bar, the filler and the sealant.

Joint	Purpose	Load transfer
Expansion joint	To allow the slab to EXPAND in hot weather without buckling; a gap of 20–25 mm through the full depth, filled with a compressible filler board and sealed at the top. Spacing 50 to 60 m.	By dowel bars — plain round bars, half the length bonded and half greased with an expansion cap
Contraction joint	To relieve the tensile stress caused by CONTRACTION and shrinkage, and so to control the location of the cracks; usually a dummy groove of one-third the slab depth. Spacing 3.5 to 4.5 m.	By dowel bars , or by aggregate interlock in thin slabs
Longitudinal (warping) joint	Provided between the lanes to relieve the warping stress and to prevent longitudinal cracking; it also allows for a slight differential settlement.	By tie bars — deformed bars, fully bonded, which do NOT transfer load but hold the two slabs together
Construction joint	Provided wherever the day's work stops, or at an obstruction.	As for a contraction joint

Table 6.6 Joints in a cement concrete pavement. Note the key difference: DOWEL bars transfer LOAD, TIE bars only hold the slabs together.

DESIGN OF JOINTS

Spacing of expansion joints: $L_e = \delta / (2 \alpha \Delta T)$

Spacing of contraction joints (plain slab):

$$L_c = 2 \times 10^4 \times S_c / (W \times f)$$

Area of tie steel per unit length of the longitudinal joint:

$$A_s = b f h W / S_s$$

Length of the tie bar: $L_t = S_s d / (2 S^b)$

- δ = width of the expansion joint gap (cm); $\alpha = 10 \times 10^{-6}/^\circ\text{C}$; ΔT = maximum temperature rise
- S_c = allowable tensile stress in the concrete during curing (0.8 kg/cm^2); $W = 2400 \text{ kg/m}^3$; $f = 1.5$
- b = half the width of the slab (i.e. the lane width) in cm, h = slab thickness in cm, W = unit weight in kg/cm^3
- S_s = allowable tensile stress in the steel; S^b = allowable bond stress; d = diameter of the tie bar

SOLVED NUMERICAL — Spacing of expansion and contraction joints

Q. A cement concrete pavement is to be laid. (a) Determine the spacing of the expansion joints if the joint gap is 2.5 cm, the maximum temperature rise is 25°C and $\alpha = 10 \times 10^{-6}$ per $^\circ\text{C}$. (b) Determine the spacing of the contraction joints for a plain slab, taking the allowable tensile stress in the concrete during curing as 0.8 kg/cm^2 , the unit weight of concrete as $2,400 \text{ kg/m}^3$ and the coefficient of friction as 1.5.

Solution:

(a) Spacing of the expansion joints

Each of the two slab ends meeting at the joint can move by $\delta/2$, so

$$L_e \times \alpha \times \Delta T = \delta/2$$

$$L_e = \delta / (2 \alpha \Delta T) = 2.5 / (2 \times 10 \times 10^{-6} \times 25)$$

$$= 2.5 / (5.0 \times 10^{-4}) = 5000 \text{ cm} = \mathbf{50 \text{ m}}$$

(b) Spacing of the contraction joints

The frictional stress at mid-length must not exceed S_c :

$$\sigma_e = W L f / (2 \times 10^4) \leq S_c$$

$$L_c = 2 \times 10^4 \times S_c / (W f) = 2 \times 10^4 \times 0.8 / (2400 \times 1.5)$$

$$= 16,000 / 3,600 = 4.44 \text{ m}$$

Expansion joints at 50 m centres; contraction joints at 4.44 m, adopted as 4.5 m — which is exactly why concrete slab panels are normally about 4.5 m long. Note that the slab thickness cancels out of the contraction-joint formula.

SOLVED NUMERICAL — Design of tie bars for a longitudinal joint

Q. Design the tie bars for the longitudinal joint of a cement concrete pavement 25 cm thick with a lane width of 3.5 m. Take the coefficient of friction as 1.5, the unit weight of concrete as 2,400 kg/m³, the allowable tensile stress in steel as 1,250 kg/cm² and the allowable bond stress as 24.6 kg/cm². Use 12 mm diameter deformed bars.

Solution:

Step 1 — frictional force to be resisted, per cm length of joint

The tie steel must hold the frictional force developed under one lane width:

$$W = 2400 \text{ kg/m}^3 = 2400/10^6 = 2.4 \times 10^{-3} \text{ kg/cm}^3$$

$$\text{Force per cm length} = f \times W \times b \times h = 1.5 \times 2.4 \times 10^{-3} \times 350 \times 25$$

$$= 1.5 \times 2.4 \times 10^{-3} = 3.6 \times 10^{-3}; \times 350 = 1.26; \times 25 = 31.5 \text{ kg per cm length}$$

Step 2 — area of steel required

$$A_s = 31.5/1250 = 0.0252 \text{ cm}^2 \text{ per cm length}$$

$$= 2.52 \text{ cm}^2 \text{ per metre length of joint}$$

Step 3 — spacing of 12 mm bars

$$\text{Area of one 12 mm bar} = (\pi/4)(1.2)^2 = 1.131 \text{ cm}^2$$

$$\text{Number of bars per metre} = 2.52/1.131 = 2.23$$

$$\text{Spacing} = 1000/2.23 = 448 \text{ mm} \rightarrow \text{adopt } \mathbf{440 \text{ mm centres}}$$

Step 4 — length of the tie bar

$$L_t = S_s d / (2 S_b) = 1250 \times 1.2 / (2 \times 24.6) = 1500/49.2 = 30.5 \text{ cm}$$

Adding an allowance of about 50-100 mm for inaccurate placement, adopt a length of **400 mm**.

Provide 12 mm diameter deformed tie bars, 400 mm long, at 440 mm centres (IRC 58 in practice specifies 12 mm bars, 480-640 mm long, at 300-600 mm centres, so this design is of the right order).

DOWEL BAR CAPACITY

Load transfer capacity of ONE dowel bar:

$$\text{Shear: } P_s = 0.785 d^2 F_s$$

$$\text{Bending: } P_x = 2 d^3 F_x / (L^d + 8.8 \delta)$$

$$\text{Bearing: } P_b = F_b L^d d / [12.5 (L^d + 1.5 \delta)]$$

- d = diameter of the dowel bar (cm); L^d = total embedded length; δ = width of the joint
- F_s , F_x , F_b = permissible shear, flexural and bearing stresses in the dowel
- The LEAST of the three capacities governs
- The load assumed to be transferred across the joint is taken as 40 % of the design wheel load, distributed over a length of 1.8 l on either side of the load

6.8 Design of cement concrete pavements as per IRC 58

1. Fix the **design period** (30 years) and compute the **cumulative repetitions of each axle-load group** from the axle-load spectrum and the traffic growth rate.
2. Determine the **effective modulus of subgrade reaction** K of the subgrade with the sub-base (DLC or granular) in place.
3. Assume a **trial thickness** of the slab.
4. Compute the **flexural stress** for each axle-load group for the two critical cases — **bottom-up cracking** (axle load at the edge in the day, with a positive temperature differential) and **top-down cracking** (axles at the corner at night, with a negative temperature differential).
5. Compute the **stress ratio** $SR = (\text{flexural stress}) / (\text{modulus of rupture, } 4.5 \text{ MPa})$ and obtain the allowable number of repetitions from the **fatigue equation**.

6. Sum the **cumulative fatigue damage**, $CFD = \sum (n_i/N_i)$. The trial thickness is safe if **CFD $\leq$ 1.0** for both cracking modes; otherwise increase the thickness and repeat.
7. Design the **joints, the dowel bars and the tie bars**, and check the sub-base and the drainage layer.

NOTE

Dry lean concrete (DLC) sub-base. Modern IRC 58 designs place the PQC slab on a 100–150 mm DLC sub-base with a debonding layer between them. The DLC greatly increases the effective K value and therefore reduces the required slab thickness.

BEU / AKU REPEAT QUESTION

- Compare flexible and rigid pavements. Explain how each transmits the wheel load. [7 marks — asked very often]
- Enumerate the components of a flexible pavement and state the function of each layer. [7 marks]
- What is ESWL? Determine the ESWL at the given depths for a dual wheel assembly. [14 marks — asked very often]
- Explain the factors affecting the design of a flexible pavement. [7 marks]
- Determine the design traffic in msa for the given data. [7 marks]
- State Westergaard's assumptions. Compute the interior, edge and corner stresses for the given data. [14 marks — asked very often]
- Define radius of relative stiffness and equivalent radius of the resisting section. [2 marks each]
- What are warping and frictional stresses? State the critical combinations of stress. [7 marks]
- Explain the different types of joint in a cement concrete pavement. Distinguish between dowel bars and tie bars. [7 marks]
- Design the spacing of expansion and contraction joints / design the tie bars for the given data. [7 marks]

► *Reference: Khanna, Justo & Veeraragavan, chapter 'Design of Pavements'; Partha Chakraborty & Animesh Das — Principles of Transportation Engineering for the analysis of layered systems; **IRC 37-2018** (flexible pavements), **IRC 58-2015** (rigid pavements), IRC 15 (construction of concrete roads), IRC SP 62 (rural roads).*

Formula Sheet and IRC Standard Values

read this last, on the morning of the paper

UNIT 1 — PLANNING

Road density = (road length / area) $\times$ 100
km per 100 km²

Plan targets: Nagpur 16, Bombay 32,
Lucknow 82 km per 100 km²

Saturation system:
utility per km = total utility units / length
(highest value gets first priority)

UNIT 2 — CAMBER AND WIDTH

Parabolic camber: $y = 2x^2 / (n W)$
Rise of crown above edge = $W / (2n)$
Camber 1 in $n \equiv (100/n) \%$
Lane widths: 3.75 single • 7.0 two-lane
5.5 intermediate • 3.5 per lane multi

SIGHT DISTANCE

$SSD = 0.278 V t + V^2 / (254 f)$
 $SSD = v t + v^2 / (2 g f)$ [v in m/s]
On a grade: $V^2 / [254(f \pm 0.01n)]$
+ for ASCENDING, – for DESCENDING
Single-lane two-way: $SSD_{1-lane} = 2 \times SSD$
 $ISD = 2 \times SSD$
 $t = 2.5 \text{ s (SSD)}, 2 \text{ s (OSD)}$; $f = 0.35-0.40$
 $OSD = d_1 + d_2 + d_3$
 $d_1 = 0.278 V^b t$
 $d_2 = 0.278 V^b T + 2s$
 $d_3 = 0.278 V T$
 $s = 0.7 v^b + 6$ [v^b in m/s]
 $T = \sqrt{(4s/a)}$ or $\sqrt{(14.4 s/A)}$
 $V^b = V - 16$ if not given
Overtaking zone: min $3 \times OSD$, desirable $5 \times OSD$
Divided highway: $OSD = d_1 + d_2$ only

HORIZONTAL ALIGNMENT

$e + f = v^2 / (gR) = V^2 / (127 R)$
Mixed traffic: $e = V^2 / (225 R)$
[$= (0.75V)^2 / (127 R)$]
Rise of outer edge $E = e \times B$
 $R_{min} = V^2 / [127(e + f)]$
 $e_{max} = 7 \%$ plain/rolling, 10% hill,
 4% urban; $f = 0.15$
IRC steps: (1) $e = V^2 / 225R$; (2) if $e > 7 \%$
take $e = 7 \%$; (3) check $f = V^2 / 127R - 0.07$
 ≤ 0.15 ; (4) else $V_a = \sqrt{(27.94 R)}$
 $W_m = n l^2 / (2R)$ [$l = 6.1 \text{ m}$]
 $W_{ps} = V / (9.5 \sqrt{R})$
 $W_e = W_m + W_{ps}$

TRANSITION CURVE

L = greatest of the three:
(i) $L = 0.0215 V^3 / (C R)$, $C = 80 / (75 + V)$
 $0.5 \leq C \leq 0.8$
(ii) $L = e N (W + W_e)$ [about inner edge]
 $L = e N (W + W_e) / 2$ [about centre line]
empirical $2.7V^2 / R$ plain, V^2 / R hilly
(iii) IRC minimum for the road class
Shift $S = L^2 / (24 R)$
Tangent length = $(R + S) \tan(\Delta/2) + L/2$

SET-BACK DISTANCE

$L^c \geq S$: $m = R - (R - d) \cos[S / (2(R - d))]$
Single lane: $m = R - R \cos(S/2R)$
 $L^c < S$: $m = R - (R - d) \cos(\alpha/2)$
+ $[(S - L^c) / 2] \sin(\alpha/2)$
 $\alpha/2 = L^c / [2(R - d)]$
 $d = 1.75 \text{ m}$ for a 7 m two-lane road
ANGLE IN RADIANS

VERTICAL ALIGNMENT

N = algebraic difference of grades
Grade compensation = $(30 + R) / R \%$,
max $75 / R \%$; none below 4%
SUMMIT (parabola), sight distance governs:
 $L > S$: $L = N S^2 / [2(\sqrt{H} + \sqrt{h})^2]$
 $= N S^2 / 4.4$ (SSD), $N S^2 / 9.6$ (OSD)
 $L < S$: $L = 2S - 2(\sqrt{H} + \sqrt{h})^2 / N$
 $= 2S - 4.4 / N$, $2S - 9.6 / N$
 $H = 1.2 \text{ m}$, $h = 0.15 \text{ m}$ (SSD); $H = h = 1.2$ (OSD)
VALLEY — take the GREATER of:
comfort: $L = 0.38 (N V^3)^{1/2}$ [$c = 0.6 \text{ m/s}^3$]
head-light, $L > S$: $L = N S^2 / (1.5 + 0.035 S)$
head-light, $L < S$: $L = 2S - (1.5 + 0.035 S) / N$
 $h_1 = 0.75 \text{ m}$, $\beta = 1^\circ$

UNIT 3 — TRAFFIC FLOW

$q = k \times \bar{v}_s$ (use SPACE mean speed)
 $\bar{v}_t = (1/n) \sum v_i$ (arithmetic mean)
 $\bar{v}_s = n / \sum (1/v_i)$ (harmonic mean)
 $\bar{v}_t = \bar{v}_s + \sigma_s^2 / \bar{v}_s \Rightarrow \bar{v}_t \geq \bar{v}_s$
 Time headway $h = 3600/q$ (s)
 Space headway $s = 1000/k$ (m)
 Greenshields: $v = v_f (1 - k/k_j)$
 $q = v_f (k - k^2/k_j)$
 $q_{\max} = v_f k_j / 4$ at $k_j/2$ and $v_f/2$

MOVING OBSERVER METHOD

$q = (n_a + n_v - n_o) / (t_a + t_\phi)$
 $\bar{t} = t_\phi - (n_v - n_o) / q$
 $\bar{v} = L / \bar{t}$ $k = q / \bar{v}$
 n_a met against stream in t_a
 n_v overtake the test vehicle
 n_o overtaken by the test vehicle

CAPACITY AND VOLUME

$C = 1000 V / S$ (veh/h per lane)
 $C = 3600 / H_t$
 $S_{\min} = SSD + \text{length of vehicle}$
 PHF = peak hour vol / (4 × peak 15-min vol)
 Design volume: 30th highest hourly volume

SPEEDS AND ACCIDENTS

85th percentile → safe speed LIMIT
 98th percentile → geometric DESIGN speed
 15th percentile → lower speed limit
 R per 100 million veh-km
 $= A \times 10^8 / (365 \times ADT \times N \times L)$
 R per million entering vehicles
 $= A \times 10^6 / (365 \times ADT \times N)$

UNIT 4 — SIGNAL DESIGN

WEBSTER:
 $y_i = q_i / S_i$ $Y = \sum y_i$ (need $Y < 1$)
 $L = 2n + R$ (n = phases, R = all-red)
 $C_0 = (1.5 L + 5) / (1 - Y)$
 $g_i = (y_i / Y) (C_0 - L)$
 Round C_0 to the nearest 5 s
 IRC / pedestrian: $G_p = 7 + W/1.2$
 $G_i = [q_i / \sum q] \times \text{available green}$
 Offset = distance / speed (mod C)
 Alternate system spacing = $v C/2$

PARKING

Parking load = $\sum (\text{accumulation} \times \text{interval})$
 [vehicle-hours]
 Average duration = load / volume
 Turnover = volume / number of bays
 Capacity = bays × duration of study
 Index = (load / capacity) × 100 %
 Kerb length per car and N for kerb L :
 parallel 5.9 m $N = L/5.9$
 30° 5.0 m $N = (L - 1.77)/5.0$
 45° 3.54 m $N = (L - 3.54)/3.54$
 60° 2.89 m $N = (L - 4.6)/2.89$
 90° 2.5 m $N = L/2.5$

ROTARY INTERSECTION

$Q_p = 280 w (1 + e/w) (1 - p/3) / (1 + w/L)$
 $p = (b + c) / (a + b + c + d)$
 Valid: $6 \leq w \leq 18$ m
 $0.4 \leq e/w \leq 1.0$
 $0.12 \leq w/L \leq 0.4$
 $0.4 \leq p \leq 1.0$
 $e = (e_1 + e_2) / 2$
 $R_{\text{islan}}^d \approx 1.33 \times R_{\text{entry}}$
 Weaving length ≥ 45 m at 40 km/h

HIGHWAY LIGHTING

$E = (F \times CU \times MF)/(W \times S)$
 $S = (F \times CU \times MF)/(E \times W)$
 $E \text{ lux} \cdot F \text{ lumens} \cdot MF \approx 0.8$
 $1 \text{ lux} = 1 \text{ lumen/m}^2$
 Layout: single side if $W < h$;
 staggered if $W = 1\text{--}1.5 h$;
 opposite if $W > 1.5 h$
 Mounting height 10–12 m, spacing 30–50 m

UNIT 5 — SOIL AND AGGREGATE

$CBR = (\text{test load/standard load}) \times 100$
 standard 1370 kg at 2.5 mm
 2055 kg at 5.0 mm
 report the 2.5 mm value normally
 $GI = 0.2a + 0.005ac + 0.01bd$
 $a = (\% \text{ passing } 0.075) - 35 \text{ [0–40]}$
 $b = (\% \text{ passing } 0.075) - 15 \text{ [0–40]}$
 $c = LL - 40 \text{ [0–20]}$
 $d = PI - 10 \text{ [0–20]}$
 $AIV, ACV, LAA = (W_2/W_1) \times 100$
 $FI = \text{flaky wt} / \text{total wt} \times 100$
 $EI = \text{elongated wt} / \text{non-flaky wt} \times 100$

MARSHALL MIX DESIGN

$G_m = W_a/(W_a - W_u)$
 $G_t = 100/\Sigma(W_i/G_i)$
 $V_v = [(G_t - G_m)/G_t] \times 100 \%$
 $V^b = (W^b \times G_m)/G^b \%$
 $VMA = V_v + V^b$
 $VFB = (V^b/VMA) \times 100 \%$
 OBC = mean of the bitumen contents at
 max stability, max unit weight, 4 % V_v
 Criteria: stability $\geq 9 \text{ kN}$, flow 2–4 mm,
 V_v 3–5 %, VFB 65–75 %
 Test at 60 °C, 50 mm/min, 75 blows/face

UNIT 6 — FLEXIBLE PAVEMENT

$a = \sqrt{(P/\pi p)} \quad d = S - 2a$
 $ESWL = P \text{ for } z \leq d/2; 2P \text{ for } z \geq 2S$
 $\log ESWL = \log P$
 $+ 0.301 \log(2z/d)/\log(4S/d)$
 Rigidity factor = contact/tyre pressure
 (= 1.0 at a tyre pressure of 7 kg/cm²)
 $N = 365 A [(1+r)^n - 1]/r \times D \times F$
 $A = P(1+r)^x$
 $D = 1.0 \text{ single lane, } 0.75 \text{ two-lane,}$
 $0.40 \text{ four-lane single carriageway}$
 Standard axle 8.16 t (80 kN); N in msa
 $M_r = 10 \text{ CBR (CBR} \leq 5)$
 $M_r = 17.6 \text{ CBR}^{0.64} \text{ (CBR} > 5) \text{ [MPa]}$
 Boussinesq, circular load, on the axis:
 $\sigma_z = p[1 - z^3/(a^2 + z^2)^{1.5}]$
 point load: $\sigma_z = 3P/(2\pi z^2)$
 Burmister: $\Delta = 1.5 p \text{ a } F_z/E_z$

RIGID PAVEMENT — WESTERGAARD

$l = [E h^3/(12(1-\mu^2)K)]^{1/4}$
 $b = \sqrt{(1.6a^2 + h^2)} - 0.675h, a < 1.724h$
 $b = a, a \geq 1.724h$
 $\sigma_i = (0.316P/h^2)[4\log_{10}(l/b) + 1.069]$
 $\sigma_e = (0.572P/h^2)[4\log_{10}(l/b) + 0.359]$
 $\sigma_c = (3P/h^2)[1 - (av^2/l)^{0.6}]$
 interior, edge → tension at the BOTTOM
 corner → tension at the TOP
 $E = 3 \times 10^5 \text{ kg/cm}^2, \mu = 0.15$

TEMPERATURE, FRICTION, JOINTS

$\sigma_{ti} = (Eat/2)[(C_x + \mu C_y)/(1-\mu^2)]$
 $\sigma_{te} = C Eat/2$
 $\sigma_{tc} = [Eat/(3(1-\mu))]\sqrt{(a/l)}$
 $\alpha = 10 \times 10^{-6}/^\circ\text{C}$
 $\sigma_e = W L f/(2 \times 10^4) \text{ [h cancels]}$
 $L_e = \delta/(2\alpha\Delta T) \text{ (expansion joint)}$
 $L_c = 2 \times 10^4 Sc/(W f) \text{ (contraction)}$
 $A_s = b f h W/S_s \text{ (tie steel)}$
 $L_t = S_s d/(2 S^b) \text{ (tie bar length)}$
 Dowel: $P_s = 0.785 d^2 F_s$
 $P_x = 2d^3 F_x/(L^d + 8.86)$
 $P^b = F^b L^d/[12.5(L^d + 1.56)]$
 load transferred = 40 % of wheel load

Standard values worth memorising

Quantity	Value	Quantity	Value
Reaction time for SSD	2.5 s	Max superelevation, plain/rolling	7 %
Reaction time for OSD	2.0 s	Max superelevation, hill roads	10 %
Coefficient of longitudinal friction f	0.35-0.40	Max superelevation, urban roads	4 %
Coefficient of lateral friction f	0.15	Coefficient of subgrade friction	1.5
Driver's eye height H	1.2 m	Wheel base of a design vehicle l	6.1 m
Object height for SSD h	0.15 m	Standard axle load	8.16 t (80 kN)
Head-light height h_1	0.75 m	Legal max single axle load	10.2 t
Head-light beam divergence β	1°	Max vehicle width / height	2.5 m / 3.8 m
Rate of change of centrifugal accn (valley)	0.6 m/s ³	Flexural strength of PQC	4.5 MPa
Rate of introduction of e , plain terrain	1 in 150	Modulus of elasticity of concrete	3×10^5 kg/cm ²
Rate of introduction of e , hilly terrain	1 in 60	Poisson's ratio of concrete	0.15
Ruling gradient, plain terrain	3.3 % (1 in 30)	Thermal expansion of concrete α	$10 \times 10^{-6}/^\circ\text{C}$
Limiting gradient, plain terrain	5 % (1 in 20)	Unit weight of concrete	2400 kg/m ³
Exceptional gradient, plain terrain	6.7 % (1 in 15)	Expansion joint gap / spacing	20-25 mm / 50-60 m
Camber, CC and high-type bituminous	1.7-2.0 %	Contraction joint spacing	3.5-4.5 m
Camber, thin bituminous	2.0-2.5 %	Design life, flexible pavement	15 years
Camber, WBM and gravel	2.5-3.0 %	Design life, rigid pavement	30 years
Camber, earth roads	3.0-4.0 %	Growth rate of commercial traffic	5 % min, 7.5 % typical
Minimum shoulder width (NH, SH)	2.5 m	Marshall stability, minimum	9 kN
Minimum median width	1.2 m	Marshall air voids	3-5 %
Minimum footpath width	1.5 m	Marshall VFB	65-75 %
ROW, NH/SH plain, open area	45 m	Aggregate impact value, wearing course	30 % max
Standard CBR load at 2.5 / 5.0 mm	1370 / 2055 kg	Flakiness index	25 % max

Table F.1 The numbers that carry the 2-mark questions.

EXAM TIP

If you remember nothing else, remember these five: **SSD** = $0.278Vt + V^2/254f$, **$e + f$** = $V^2/127R$, **L** = $NS^2/4.4$ for a summit curve, **C_o** = $(1.5L + 5)/(1 - Y)$ for a signal, and **N** = $365 A[(1+r)^n - 1]/r \times D \times F$ for the design traffic. Between them they cover a numerical in almost every paper.

Question Bank and Model Question Paper

compiled from the recurring pattern of BEU / AKU papers

NOTE

The questions below are **compiled and reworded from the pattern that recurs in BEU and AKU papers for this subject**; they are not verbatim reproductions of any one paper. Every question is cross-referenced to the section, figure or worked example in these notes where the answer is to be found. Work through the two-mark bank first — it covers Q.1, which is compulsory.

Part A — Two-mark question bank (for the compulsory Q.1)

Unit 1

Question	Where to revise
State the four modes of transportation. Give two advantages of road transport over rail transport.	§ 1.1, Table 1.1
What was the most important contribution of Macadam to road construction?	§ 1.2.5
How did Telford obtain the camber in his cross-section?	§ 1.2.4, Fig. 1.2
State any two recommendations of the Jayakar Committee and their outcome.	Table 1.3
Write the full form of IRC and CRRI, and the year each was set up.	Table 1.4
State the target road density of each of the three twenty-year road plans.	Table 1.5
What road pattern was assumed in the Nagpur Road Plan?	§ 1.4.1, Fig. 1.4
Name the five classes of road in the Nagpur classification.	Table 1.6
What are rural roads? Which two classes constitute them?	Table 1.6
Differentiate between an arterial street and a collector street.	§ 1.5.4
What is the length of the Golden Quadrilateral? Which four cities does it connect?	Table 1.8
State the objective and the population criterion of PMGSY.	Table 1.8
What are obligatory points? Give one example of each kind.	Table 1.9, Fig. 1.6
Name the four stages of engineering survey for a highway.	Table 1.10, Fig. 1.7
What is the saturation system of road planning?	§ 1.10.2
Define road density.	Example 1.1

Unit 1 — two-mark questions.

Unit 2

Question	Where to revise
Define camber. State any two of its objects.	§ 2.2.2
State the IRC recommended camber for a cement concrete surface and for an earth road.	Table 2.4
Why is a carriageway width between 3.75 m and 5.5 m never provided?	Table 2.5
Differentiate between mountable, semi-barrier and barrier kerbs.	Table 2.6
How is terrain classified for geometric design?	Table 2.1
What is design speed? State the ruling design speed for a NH in plain terrain.	Table 2.2
What is PIEV theory?	§ 2.3.1, Fig. 2.4
State the criteria for measuring SSD and OSD.	Table 2.7
Why is the SSD greater on a descending gradient than on an ascending gradient?	§ 2.3.2
What is intermediate sight distance? How is it obtained?	§ 2.3
State the minimum and desirable length of an overtaking zone.	§ 2.3.3
Define superelevation. State its maximum permissible value for plain terrain and for hill roads.	§ 2.4.1

Question	Where to revise
Why is extra widening provided at a horizontal curve? Name its two components.	§ 2.4.2
What is the ideal shape of a transition curve and why?	§ 2.4.3
Define shift of a curve. Write its expression.	Fig. 2.10
Define set-back distance.	§ 2.4.4
Differentiate between ruling, limiting and exceptional gradient.	Table 2.8
What is grade compensation? Write its expression.	§ 2.5.2
Why is a minimum gradient necessary?	Table 2.8
Why is riding comfort a design criterion for a valley curve but not for a summit curve?	§ 2.5.3
Which shape of curve is used for a summit curve, and why?	§ 2.5.3

Unit 2 — two-mark questions.

Unit 3

Question	Where to revise
What is PCU? State the PCU value of a truck, a cycle and a cycle rickshaw.	Table 3.3
Differentiate between ADT and AADT.	Table 3.4
Define peak hour factor.	Table 3.4, Example 3.2
What is the 30th highest hourly volume and why is it used?	Table 3.4
State the significance of the 85th and the 98th percentile speed.	Table 3.5
Differentiate between running speed and journey speed.	§ 3.3.3
Differentiate between time mean speed and space mean speed. Which is larger?	§ 3.4.1
Write the fundamental relation between flow, density and speed.	§ 3.4.1
What is jam density and free speed?	§ 3.4.2
State Greenshields' expression for the maximum flow.	§ 3.4.2
Differentiate between basic, possible and practical capacity.	Table 3.6
What is level of service? Name the LOS at capacity.	Table 3.6, Fig. 3.5
What is a desire line diagram?	§ 3.3.4
Differentiate between a collision diagram and a condition diagram.	Fig. 3.3
State the 3 E's of accident prevention.	§ 3.3.5
Name any four methods of an origin-destination study.	§ 3.3.4

Unit 3 — two-mark questions.

Unit 4

Question	Where to revise
Name the four traffic control devices.	§ 4.2
What shape identifies a mandatory sign, a cautionary sign and an informative sign?	Table 4.2, Fig. 4.1
Give two examples each of a cautionary and an informative sign.	Table 4.2
Name the four types of traffic island.	Table 4.3
Define cycle, cycle length and phase of a signal.	Table 4.5
What is lost time in a signal cycle? How is it estimated?	Table 4.5
What is saturation flow?	Table 4.5
Write Webster's expression for the optimum cycle time.	§ 4.3.3
What is offset in a co-ordinated signal system?	Table 4.5, § 4.3.5
Name the four systems of signal co-ordination.	Table 4.6
State the three types of conflict at an intersection.	§ 4.4.1

Question	Where to revise
What is channelization? State any two of its objects.	§ 4.4.2
Define weaving length and weaving angle of a rotary.	Table 4.8, Fig. 4.6
State any two advantages and two disadvantages of a rotary.	Table 4.7
Name any three types of grade-separated interchange.	Table 4.9, Fig. 4.7
Define parking accumulation and parking load.	§ 4.5.2
Define parking turnover and parking index.	§ 4.5.2, Example 4.5
How many cars can be parked in 30 m of kerb at 90° and at parallel parking?	Table 4.10
Define lumen, candela and lux.	Table 4.11
What is the maintenance factor in lighting design? What is its usual value?	Table 4.11
When is staggered lighting layout preferred?	Table 4.12

Unit 4 — two-mark questions.

Unit 5

Question	Where to revise
State any four desirable properties of a subgrade soil.	§ 5.2.1
What is the modulus of subgrade reaction and how is it determined?	Table 5.1
State the standard loads used in the CBR test at 2.5 mm and 5.0 mm penetration.	§ 5.2.3
Why is the CBR load-penetration curve corrected? How?	§ 5.2.3, Fig. 5.1
Which CBR value is normally reported and why?	§ 5.2.3
Define group index. What is its range?	§ 5.2.4
Name the test used to measure the toughness, the hardness and the strength of an aggregate.	Table 5.2
Define flakiness index and elongation index.	Table 5.3, Example 5.4
Why are flaky and elongated particles undesirable in a road aggregate?	Table 5.2
What is the stripping value of an aggregate?	Table 5.3
What is the maximum permissible aggregate impact value for a wearing course?	Table 5.3
What are VG grades of bitumen? Name them.	Table 5.4
What is cutback bitumen? Name its three types.	Table 5.4
What is a bitumen emulsion and when is it used?	Table 5.4
What does the penetration test measure? State its standard conditions.	Table 5.5
What does the softening point indicate?	Table 5.5
Why is the ductility test necessary? State the minimum value for paving bitumen.	Table 5.5
Differentiate between BM, DBM and BC.	Table 5.6
What is the flow value in the Marshall test?	§ 5.5.3
How is the optimum bitumen content determined in Marshall's method?	Fig. 5.5
Why is flexural strength the design criterion for a concrete pavement?	§ 5.6.2
State the minimum 28-day flexural strength of pavement quality concrete.	Table 5.8

Unit 5 — two-mark questions.

Unit 6

Question	Where to revise
State two points of difference between a flexible and a rigid pavement.	Table 6.1
How does a flexible pavement transmit the wheel load to the subgrade?	§ 6.1, Fig. 6.1
State any two functions of a sub-base course.	Table 6.2
Why is the strongest material placed at the top of a flexible pavement?	Fig. 6.2

Question	Where to revise
Define ESWL.	§ 6.3.2
What is rigidity factor? What is its value at a tyre pressure of 7 kg/cm ² ?	§ 6.3.1
What is a standard axle? State its magnitude.	§ 6.3.3
Define vehicle damage factor and lane distribution factor.	§ 6.3.3
What is msa?	§ 6.3.3
State Boussinesq's expression for the vertical stress under a point load.	Table 6.3
Name any three methods of flexible pavement design.	Table 6.4
State Westergaard's assumption regarding the subgrade.	§ 6.6
Define radius of relative stiffness. Write its expression.	§ 6.6
Define equivalent radius of the resisting section.	§ 6.6
For a corner load, where is the critical tensile stress located?	§ 6.6
What causes warping stress in a concrete slab?	§ 6.6.1, Fig. 6.6
Why does the slab thickness not appear in the frictional stress formula?	§ 6.6.2
State the most critical combination of stresses in a rigid pavement.	Table 6.5
Differentiate between a dowel bar and a tie bar.	Table 6.6
What is the usual spacing of an expansion joint and of a contraction joint?	Table 6.6
What is the function of the filler board and the sealant in an expansion joint?	Fig. 6.7
What is DLC and why is it used?	§ 6.8

Unit 6 — two-mark questions.

Part B — Long-question bank (for Q.2 to Q.8)

Each long question in the paper carries 14 marks and is normally split into two or three parts — typically one theory part and one numerical or one diagram-based part. The questions marked » are the ones that recur most often.

Question	Where to revise
» Describe with neat sketches the Tresaguet, Telford and Macadam methods of road construction, and compare them in a table.	§ 1.2, Table 1.2
Discuss the recommendations of the Jayakar Committee and the organisations that resulted from them. Add the chronology of highway development in India.	§ 1.3
Explain the three twenty-year road development plans, with their target road densities, and solve a road-density numerical.	§ 1.4, Example 1.1
Classify the roads of India as per the Nagpur plan and as per the modified classification; give the function and the responsible authority of each class.	§ 1.5
Write notes on NHDP and its phases, Bharatmala and PMGSY.	§ 1.6
» What are the requirements of an ideal alignment? Discuss the factors controlling it, and the special considerations for hill roads.	§ 1.7
» Explain the four engineering surveys for highway location, and enumerate the drawings and reports of a highway project.	§ 1.8, § 1.9
Explain the saturation system and select between two given alignments.	§ 1.10, Example 1.2
» Draw a typical cross-section of a two-lane highway in embankment and in cutting, and explain every element with its IRC value.	§ 2.2
What is camber? Give its objects, shapes, expressions and IRC values. Also explain the width of the carriageway and of the shoulder.	§ 2.2.2, 2.2.3
» Derive the expression for stopping sight distance and explain the effect of gradient. Solve the given numerical.	§ 2.3.2, Examples 2.1, 2.2
» Explain overtaking sight distance with a sketch, derive the expression, and compute the OSD and the length of the overtaking zone.	§ 2.3.3, Example 2.3

Question	Where to revise
» Derive $e + f = V^2/127R$. Explain the IRC procedure for the design of superelevation and design it for the given data.	§ 2.4.1, Example 2.4
Why is extra widening required at a curve? Derive/state the two components and compute the total width at the curve.	§ 2.4.2, Example 2.6
» What is a transition curve? State its functions and the three criteria for its length; compute the length and the shift for the given data.	§ 2.4.3, Example 2.7
Explain set-back distance with a sketch and compute it for the given data.	§ 2.4.4, Example 2.8
Define the types of gradient and give the IRC values. Explain grade compensation and compute the compensated gradient.	§ 2.5.1, 2.5.2, Example 2.9
» Design the length of a summit curve for the given grades and sight distance, explaining both cases ($L > S$ and $L < S$).	§ 2.5.3, Example 2.10
» Explain the two criteria for the design of a valley curve and design its length for the given data.	§ 2.5.3, Example 2.11
Discuss the road user and vehicular characteristics that affect traffic operation. What is PCU and why is it needed? Convert the given count into PCU.	§ 3.2, Example 3.1
Explain the traffic volume study — the terms, the methods of counting and the methods of presenting the data.	§ 3.3.1
Explain the spot-speed study, its uses and the methods, and determine the percentile speeds from the given data.	§ 3.3.2, Example 3.3
» Explain the moving observer method, state its formulae and solve the given data.	§ 3.3.3, Example 3.4
Explain the origin-destination study — uses, methods and presentation.	§ 3.3.4
Discuss the causes of road accidents and the measures for prevention. Distinguish between a collision diagram and a condition diagram.	§ 3.3.5
» Derive $q = k v$. Explain the speed-flow-density diagrams and Greenshields' model; compute the capacity and the two possible speeds.	§ 3.4, Example 3.6
Define basic, possible and practical capacity and level of service; compute the theoretical capacity of a lane.	§ 3.4.3, Example 3.7
Explain traffic regulation under the three heads, and classify traffic signs and road markings with examples.	§ 4.1, § 4.2
» Explain Webster's method of signal design and design a two-phase signal for the given data.	§ 4.3.3, Example 4.1
Define all the signal terms, state the advantages and limitations of signals, and explain the four systems of co-ordination with a time-space diagram.	§ 4.3.2, 4.3.5
» What is a rotary? Give its advantages, disadvantages and design elements as per IRC, and compute the capacity of the weaving section.	§ 4.4.3, Example 4.3
Explain channelization and the types of at-grade and grade-separated intersections with sketches.	§ 4.4.2, 4.4.4
» Explain the types of parking and the parking statistics; solve the given parking survey and kerb-length problems.	§ 4.5, Examples 4.4, 4.5
Explain the terms used in lighting and the layouts; design the spacing of the lamps for the given data.	§ 4.6, Example 4.6
» State the desirable properties of road aggregates and the test for each; compute the AIV, ACV, FI and EI for the given data.	§ 5.3, Examples 5.3, 5.4
Explain the CBR test with the load-penetration curve and its correction; compute the CBR and the group index.	§ 5.2, Examples 5.1, 5.2
Explain the types of bituminous binder and the tests on bitumen with their significance.	§ 5.4
» Explain the Marshall method of mix design and the determination of the OBC; compute the air voids, VMA and VFB.	§ 5.5.3, Example 5.5
State the requirements of concrete for a CC pavement and the tests on cement.	§ 5.6
» Compare flexible and rigid pavements and explain how each transmits the load. Enumerate the components of a flexible pavement and their functions.	§ 6.1, § 6.2
» What is ESWL and why is it needed? Determine the ESWL at the given depths for a dual wheel assembly.	§ 6.3.2, Example 6.1

Question	Where to revise
Explain the factors affecting the design of a flexible pavement and compute the design traffic in msa.	§ 6.3, Example 6.2
Explain Boussinesq's and Burmister's theories and the IRC 37 design procedure; compute the vertical stress at the given depths.	§ 6.4, § 6.5, Example 6.3
» State Westergaard's assumptions, define l and b , and compute the interior, edge and corner stresses for the given data.	§ 6.6, Example 6.4
Explain warping and frictional stresses and the critical combinations of stress in a rigid pavement.	§ 6.6.1, 6.6.2, 6.6.3
» Explain the types of joint in a CC pavement; distinguish dowel bars from tie bars and design the joint spacing / the tie bars for the given data.	§ 6.7, Examples 6.5, 6.6

Part B — long-question bank. » marks the most frequently repeated questions.

Part C — Model question paper

B.TECH. SEMESTER EXAMINATION

TRANSPORTATION ENGINEERING • Course Code 101401 • Civil Engineering

Time : 3 hours | Full marks : 70

Instructions: Question No. 1 is compulsory. Answer any FOUR questions from the remaining. Assume any missing data suitably and state it clearly. Use of IS/IRC codes is permitted where indicated.

Q.1 Answer any SEVEN of the following. [7 × 2 = 14]

1. Define camber and state its objects.
2. What is PIEV theory?
3. Differentiate between ruling gradient and limiting gradient.
4. What is PCU? Give the PCU value of a truck and of a cycle.
5. State the criteria for measuring stopping sight distance.
6. Distinguish between a dowel bar and a tie bar.
7. Define equivalent single wheel load.
8. What is the group index of a soil? State its range.
9. State the target road density of the third twenty-year road plan.
10. Define parking turnover and parking index.

Q.2

- (a) Describe, with neat labelled sketches, the Tresaguet, Telford and Macadam methods of road construction. [8]
- (b) Discuss the recommendations of the Jayakar Committee and state the organisation that resulted from each. [6]

Q.3

- (a) Derive an expression for the stopping sight distance and explain the effect of gradient on it. [7]
- (b) Calculate the SSD for a design speed of 80 km/h on a descending gradient of 3 %. Take $t = 2.5$ s and $f = 0.35$. [7]

Q.4

- (a) Explain the IRC step-by-step procedure for the design of superelevation. [6]
- (b) A two-lane highway of 7.0 m width has a horizontal curve of radius 250 m and a design speed of 80 km/h in plain terrain. Design the superelevation, the extra widening and the length of the transition curve, and find the shift. [8]

Q.5

- (a) Explain the two criteria used for the design of a valley curve. [6]
- (b) A summit curve is formed by an ascending gradient of 1 in 40 meeting a descending gradient of 1 in 30. Design the length of the summit curve for a stopping sight distance of 180 m. [8]

Q.6

- (a) Explain the moving observer method of traffic study and state its formulae. [6]
- (b) In a moving observer study over a 2 km stretch, the test vehicle running against the stream took 3 minutes and met 120 vehicles; running with the stream it took 4 minutes, 8 vehicles overtook it and it overtook 20 vehicles. Find the flow, the mean speed of the stream and the density. [8]

Q.7

- (a) Explain Webster's method of traffic signal design. [6]
- (b) Design a two-phase fixed-time signal by Webster's method for an intersection where the critical approach flows are 900 and 700 PCU/h and the corresponding saturation flows are 2,700 and 2,500 PCU/h. Take the amber period as 3 s per phase. [8]

Q.8

- (a) State Westergaard's assumptions, define the radius of relative stiffness and the equivalent radius of the resisting section, and state the three critical load positions. [6]
- (b) A concrete slab 20 cm thick rests on a subgrade of $K = 6 \text{ kg/cm}^3$. The design wheel load is 4,100 kg with a contact radius of 12 cm. Taking $E = 3 \times 10^5 \text{ kg/cm}^2$ and $\mu = 0.15$, compute the interior, edge and corner load stresses. [8]

Answers to the numerical parts of the model paper

Q	Answer	Method
3 (b)	SSD = 134.3 m $\approx$ 134 m	$0.278 \times 80 \times 2.5 = 55.6$; $6400/[254(0.35-0.03)] = 78.7$; sum = 134.3
4 (b)	$e = 7\%$ (curve safe, f needed = 0.132) ; $W_e = 0.68 \text{ m}$; $L = 85.3 \text{ m} \rightarrow$ adopt 85 m ; shift $S = 1.20 \text{ m}$	$e = V^2/225R = 11.4\% > 7\% = e = 7\%$, check $f = V^2/127R - 0.07 = 0.132 < 0.15$; $W_m = 0.149$, $W_{ps} = 0.533$; $L = 0.0215V^3/CR$ with $C = 0.516$; $S = L^2/24R$
5 (b)	$N = 0.05833$; $L = 429.5 \text{ m} \rightarrow$ adopt 430 m	Assume $L > S$: $L = N S^2/4.4 = 0.05833 \times 180^2/4.4$; $429.5 > 180$, so the assumption holds
6 (b)	$q = 926 \text{ veh/h}$; $\bar{t} = 4.78 \text{ min}$; $\bar{v} = 25.1 \text{ km/h}$; $k = 36.9 \text{ veh/km}$	$q = (120+8-20)/(3+4) = 15.43/\text{min}$; $\bar{t} = 4 - (8-20)/15.43 = 4.78$; $\bar{v} = 2/4.78$; $k = q/\bar{v}$
7 (b)	$Y = 0.6133$; $L = 10 \text{ s}$; $C_0 = 51.7 \text{ s} \rightarrow$ adopt 50 s ; $g_1 = 21.7 \text{ s}$, $g_2 = 18.3 \text{ s}$	$y_1 = 900/2700 = 0.3333$, $y_2 = 700/2500 = 0.28$; $C_0 = (1.5 \times 10 + 5)/(1 - 0.6133)$; $g_i = (y_i/Y)(C - L)$
8 (b)	$l = 76.42 \text{ cm}$; $b = 11.61 \text{ cm}$; $\sigma_i = 14.06$, $\sigma_e = 21.29$, $\sigma_c = 18.28 \text{ kg/cm}^2$ — the EDGE stress governs	$l = [Eh^3/(12(1-\mu^2)K)]^{1/4}$; $b = \sqrt{(1.6a^2+h^2)} - 0.675h$; $4\log_{10}(l/b) = 3.273$; $P/h^2 = 10.25$

Check your working against these. Full worked solutions of the same types are in Examples 2.2, 2.4, 2.6, 2.7, 2.10, 3.4, 4.1 and 6.4.

Part D — Revision strategy

If you have...	Do this
7 days	Two days on Unit 2 and Unit 6 (all the numericals, worked twice). One day each on Unit 4, Unit 3 and Unit 1. Half a day on Unit 5. The last day: the formula sheet, the two-mark bank and the model paper against the clock.
3 days	Day 1 — Unit 2 completely (sight distance, superelevation, transition, summit and valley curves). Day 2 — Unit 6 (comparison table, ESWL, msa, Westergaard, joints) and Webster's method from Unit 4. Day 3 — the two-mark bank for all six units, plus the formula sheet. This covers four long questions and Q.1.
1 day	The formula sheet, Table 6.1 (flexible vs rigid), Table 1.2 (historic methods), the two-mark bank, and the six worked numericals: SSD, superelevation, summit curve, moving observer, Webster and Westergaard.

Table Q.1 What to study when time is short.

EXAM TIP

How to write the answers. (1) Draw the labelled sketch FIRST — it earns marks on its own and organises the rest of the answer. (2) Write the formula, then the substitution, then the answer with its UNIT, and box the answer. (3) In a design problem always state the code value you have assumed ($t = 2.5$ s, $f = 0.35$, $e_{\max} = 7\%$) — examiners give credit for it. (4) Where two cases exist ($L > S$ or $L < S$, ascending or descending gradient), always show the CHECK. (5) Answer in points and tables rather than long paragraphs.